

Identification of the trail-following pheromone receptor in termites

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Souleymane Diallo, Kateřina Kašparová, Josef Šulc, Jibin Johnny, Jan Křivánek, Jana Nebesářová, David Sillam-Dussès, Pavlína Kyjaková, Jiří Vondrášek, Aleš Machara, Ondřej Lukšan , Ewald Grosse-Wilde, Robert Hanus 

Institute of Organic Chemistry and Biochemistry of the Czech Academy of Sciences, Prague, Czech Republic • Czech University of Life Sciences, Prague, Czech Republic • Faculty of Science, Charles University, Prague, Czech Republic • Biology Centre of the Czech Academy of Sciences, České Budějovice, Czech Republic • University Sorbonne Paris Nord, Villetaneuse, France

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eLife Assessment

This **important** work by Diallo et al. substantially advances our understanding of the chemosensory system of a non-hymenopteran eusocial insect by identifying the first olfactory receptor for the trail pheromone in termites. The evidence supporting the conclusions that the receptor PsimOR14 is very narrowly tuned for the pheromone neocembrene is **compelling**. The work will be of broad interest to entomologists, chemical ecologists, neuroscientists, and molecular biologists.

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Abstract

Chemical communication is the cornerstone of eusocial insect societies since it mediates the social hierarchy, division of labor, and concerted activities of colony members. The chemistry of social insect pheromones received considerable attention in both major groups of social insects, the eusocial Hymenoptera and termites. By contrast, current knowledge on molecular mechanisms of social insect pheromone detection by odorant receptors (ORs) is limited to hymenopteran social insects and no OR was yet functionally characterized in termites, the oldest eusocial insect clade. Here, we present the first OR deorphanization in termites. We selected four OR sequences from the previously annotated antennal transcriptome of the termite *Prorehinotermes simplex* (Rhinotermitidae), expressed them in Empty Neuron *Drosophila*, and functionally characterized using single sensillum recording (SSR) and a panel of termite semiochemicals. In one of the selected ORs, PsimOR14, we succeeded in obtaining strong and reliable responses to the main component of *P. simplex* trail-following pheromone, the monocyclic diterpene neocembrene. PsimOR14 showed a narrow tuning to neocembrene; only one additional compound out of 67 tested (geranylgeraniol) generated non-negligible responses. Subsequently, we used SSR in *P. simplex* workers and identified the olfactory sensillum specifically responding to neocembrene, thus likely expressing *PsimOR14*. We report on homology-based modelling of neocembrene binding by PsimOR14 and show how different ligands impact the receptor dynamicity using molecular dynamics simulations.

Finally, we demonstrate that *PsimOR14* is significantly more expressed in worker antennae compared to soldiers, which correlates with higher sensitivity of workers to neocembrene.

Introduction

Chemical communication is the cornerstone of eusocial insect societies. It mediates the social hierarchy and division of labor, social cohesion and concerted activities of colony members. Different clades of eusocial insects have independently evolved chemical signals acting in similar social contexts, such as trail, alarm and queen pheromones, colony and caste recognition cues, and other signals. The convergent evolution of chemical signaling is the best manifested by many similarities between ants and termites, in spite of the great evolutionary distance between these two dominant groups of social insects with great ecological significance (Leonhardt et al., 2016; Tuma et al. 2020).

Since the identification of the multigene family of insect odorant receptors (ORs) expressed in antennae and maxillary palps of *Drosophila melanogaster* (Clyne et al., 1999), neurophysiology of insect olfaction has seen great progress in terms of phylogenetic reconstructions of OR evolution across taxa, functional characterizations (deorphanisations) of multiple ORs and ultimately their recent structural characterizations (Butterwick et al., 2018; del Marmol et al., 2021; Wang et al., 2024; Zhao et al. 2024). Even though the insect ORs have seven transmembrane domains like the ORs known from vertebrates, they do not share many tangible sequence similarities. Insect ORs have an inverted membrane topology in the dendrites of olfactory sensory neurons compared to vertebrate ORs (Benton et al., 2006; Clyne et al., 1999) and unlike the vertebrate ORs known to act as GPCR receptors, insect ORs function as odorant-gated ion channels (Sato et al., 2008; Wicher et al., 2008). While the ORs from the basal wingless insect lineage Archaeognatha are homotetramers (del Marmol et al., 2021; Brand et al., 2018), ORs of all other Insecta form heteromeric complexes with a highly conserved coreceptor protein (ORCo) (Brand et al., 2018; Butterwick et al., 2018; Larsson et al., 2004; Sato et al., 2008). As recently shown in mosquitoes and aphids, these complexes consist of one OR and three ORCo subunits (Wang et al., 2024; Zhao et al. 2024).

The OR repertoire is greatly variable across Insecta and ranges from units in the wingless Archaeognatha and basal winged order Odonata to tens or hundreds ORs identified in most other flying insects (Robertson, 2019; Yan et al., 2020). Insect ORs often lack a clear orthology pattern across phylogeny, because the OR family evolved via rapid birth-and-death process, accompanied by multiple gene duplications, pseudogenizations, and losses. Lineage-specific expansions, together with a considerable variability in OR ligand specificities (from broad to narrow tuning) allow for a rapid response of olfactory system to ecological and life history changes (Andersson et al., 2015; Benton, 2015; Nei & Rooney, 2005; Robertson, 2019). Ecology-driven plasticity in OR evolution has been convincingly demonstrated by comparisons of specialist vs. generalist insects, the former often having much lower repertoires of ORs and other chemosensory proteins (Robertson, 2019).

The knowledge on OR function and ligand specificities has historically been obtained mainly from deorphanisation studies on *Drosophila* and other holometabolous insects using various heterologous expression systems. Recently, this bias was in part compensated by OR deorphanisations in more basal taxa, e.g., the wingless Archaeognatha (del Marmol et al., 2021), or the hemimetabolous aphids (Zhang et al., 2017, 2019a) and locusts (Guo et al., 2020; Chang et al., 2023).

Within social insects, eusocial Hymenoptera received considerable attention both in terms of OR repertoire reconstructions and functional characterizations with multiple ORs being deorphanized in ants (Pask et al., 2017; Slone et al., 2017) and the honey bee (e.g., Gomez Ramirez et al.,

2023 [↗](#); Wanner et al., 2007 [↗](#)). The amassed knowledge suggests that the complex communication and orientation capabilities in the colonies of eusocial Hymenoptera are facilitated by the greatly expanded repertoire of ORs, especially that of the 9-exon subfamily in ants and paper wasps participating in the detection of cuticular hydrocarbons (CHCs) as important cues in contact chemoreception of colony and caste identity and fertility status in eusocial insects (Engsontia et al., 2015 [↗](#); Legan et al., 2021 [↗](#); McKenzie et al., 2016 [↗](#); Pask et al., 2017 [↗](#); Zhou et al., 2015 [↗](#)). The 9-exon subfamily and the overall OR richness have been inherited by the extant eusocial Hymenoptera from the ancestor of Aculeata (McKenzie et al., 2016 [↗](#)), and also solitary aculeate taxa display large OR repertoires (Obiero et al., 2021 [↗](#)); this preadaptation might have been important for the repeated emergence of eusociality and the related complex communication. This is supported by the reduction of OR array (including 9-exon genes) in parasitic ant taxa, along with the simplification of their behavioral repertoire (Jongepier et al., 2022 [↗](#)).

The multiple convergences in biology and life histories between termites and ants call for comparison of olfactory detection of chemical signals and environmental cues in the two groups. Yet, despite relatively good knowledge on chemistry of termite pheromones and recognition cues (reviewed in Bagnères & Hanus, 2015 [↗](#); Bordereau & Pasteels, 2011 [↗](#); Mitaka & Akino, 2021 [↗](#)), molecular aspects of olfaction remain largely understudied in termites. Termite ORs were so far only addressed with respect to their diversity, inferred from genome assemblies (Harrison et al., 2018 [↗](#); Terrapon et al., 2014 [↗](#)) or whole-body transcriptomes (Mitaka et al., 2016 [↗](#)), recently complemented by comprehensive search for chemosensory protein repertoire using the antennal transcriptomes of three termite species (Johnny et al., 2023 [↗](#)). Termite ORs are organized in relatively conserved, highly orthologous pattern and their total numbers range from 28 to 69 (Johnny et al., 2023 [↗](#); Terrapon et al., 2014 [↗](#)). These numbers are lower than in their solitary cockroach relative *Blattella germanica* (Harrison et al., 2018 [↗](#)), and dramatically lower than in ants, having up to over 400 ORs (Engsontia et al., 2015 [↗](#); Legan et al., 2021 [↗](#); McKenzie et al., 2016 [↗](#); Pask et al., 2017 [↗](#); Zhou et al., 2015 [↗](#)). Thus, termites clearly contradict the paradigm on eusociality as a driver of OR richness, even though their chemical communication is by far more complex than in solitary insects and includes pheromone components from a variety of chemical classes, such as fatty-acyl derived alcohols, aldehydes and ketones, terpenoids, and, last, but not least, the CHCs. Independently of eusocial Hymenoptera, termites evolved an intricate communication system using CHCs as kin- and nestmate discrimination cues and as indicators of caste identity and fertility status (reviewed in Bagnères & Hanus, 2015 [↗](#); Mitaka & Akino, 2021 [↗](#)). Therefore, compared to ants, the termite social evolution seemingly adopted a different trajectory to accommodate the needs of chemical communication, including CHC detection.

One of alternative hypotheses springs from observations that termites possess an extraordinarily rich set of ionotropic receptors (IRs), reaching up to more than one hundred in some species (Harrison et al., 2018 [↗](#); Johnny et al., 2023 [↗](#); Terrapon et al., 2014 [↗](#)). Even though the richness of IRs is shared by termites and their cockroach relatives, with *B. germanica* having the largest IR repertoire ever identified in insects (Robertson et al., 2018 [↗](#)), IRs also underwent termite-specific expansions (Harrison et al., 2018 [↗](#); Johnny et al., 2023 [↗](#)). Since insect IRs have been shown to respond to volatile ligands (Benton et al., 2009 [↗](#)), we cannot rule out that they also participate in the complex chemical communication in termites as has been previously speculated (Harrison et al., 2018 [↗](#)). Nevertheless, since no chemosensory proteins have yet been functionally characterized in termites, ORs remain the prime candidates for pheromone detection.

Here, we report on the first OR deorphanization in termites. We build on the knowledge about the chemical ecology of the Cuban subterranean termite *Prorethitermes simplex* (Rhinotermitidae) (Hanus et al., 2006 [↗](#), 2009 [↗](#); Jirošová et al., 2017 [↗](#); Piskorski et al., 2007 [↗](#)), on the annotated repertoire of 50 ORs from *P. simplex* antennal transcriptome (Johnny et al., 2023 [↗](#)), on additional *P. simplex* sequencing data (caste-specific head transcriptomes, draft genome assembly), and on laboratory culture of the species. We select four *P. simplex* OR sequences, study their function by means of the Empty Neuron *Drosophila* expression system and single-sensillum recording (SSR)

using panels of biologically relevant ligands including the *P. simplex* pheromones and chemically related compounds. We identify PsimOR14 as the pheromone receptor narrowly tuned to the monocyclic diterpene neocembrene, known as the major component of the trail-following pheromone (TFP) (Sillam-Dussès et al., 2009). We demonstrate strong and selective response of PsimOR14 to neocembrene, with only one additional ligand, geranylgeraniol, generating non-negligible receptor response. We further report on homology-based modelling of neocembrene binding by PsimOR14 and perform molecular dynamics simulations to estimate the impact of ligand binding on PsimOR14 dynamicity. Finally, we identify the *P. simplex* olfactory sensillum specifically responding to neocembrene and document worker caste-biased expression of PsimOR14 accompanied by significantly higher sensitivity to neocembrene compared to *P. simplex* soldiers.

Results

Phylogenetic reconstruction and candidate OR selection

In the first step, we reconstructed the phylogeny of termite ORs and ORCoS using published protein sequences from two species in combination with our antennal transcriptome data on three species and the bristletail *Lepisma saccharina* as basal insect outgroup. In the resulting tree, all ORCo sequences and ORs from *L. saccharina* were basally situated, while majority of termite OR sequences were organized into two large sister clusters, both of which were further split into several sub-branches mostly containing one sequences from all five termite species (Fig. 1A and Supplementary Fig. S1). Only a few exceptions to this highly orthologous pattern were spotted, such as isolated sequences or species-specific expansions with a maximum of four paralogs.

Out of the 50 ORs identified in *P. simplex*, 26 sequences represented full open reading frames with at least 6 undisputed transmembrane domains predicted using TMHMM-2.0. From these, we selected four sequences (PsimOR9, 14, 30 and 31) situated in different parts (subbranches) of the tree (Fig. 1) and used them for transgenic *D. melanogaster* generation and SSR screening.

Functional characterization of *P. simplex*

ORs in *D. melanogaster* ab3 sensillum

We expressed the four selected *P. simplex* ORs in the recently improved version of the *Drosophila melanogaster* Empty Neuron system (Chahda et al., 2019); the crossing scheme for fly generation adapted from Gonzales et al. (Gonzalez et al., 2016) is shown in Supplementary Fig. S2. Spontaneous SSR firing rates of the four transgenic lines showed an expected pattern with no abnormal bursts, indicating that the ORs were functional. The flies were first subjected to SSR screening with Panel 1, consisting of 11 semiochemicals relevant to termite chemical communication and structurally related compounds. As shown in Fig. 2A, PsimOR9 and PsimOR30 did not provide any strong response to any of the tested compounds, and PsimOR31 broadly and weakly responded to several compounds. By contrast, PsimOR14 systematically and strongly responded to stimulations by the monocyclic diterpene hydrocarbon neocembrene, which is the main component of the trail-following pheromone in the genus *Prorethitermes* (Sillam-Dussès et al., 2009). Additionally, a moderate PsimOR14 response was also recorded for the linear diterpene alcohol geranylgeraniol, while all other compounds in the panel, including two other terpenoids, only elicited weak or no responses (Fig. 2A, Supplementary Tables S1–S4).

We then compared the responses of *Drosophila* ab3 sensillum in PsimOR14 expressing flies with those of *W*¹¹¹⁸ flies. As evidenced in Fig. 2B, *W*¹¹¹⁸ ab3 sensillum did not show any significant neuronal response to Panel 1 compounds, while the transgenic PsimOR14 line generated an average Δ spike number of >50 spikes/s for neocembrene and a minor secondary response of ~25

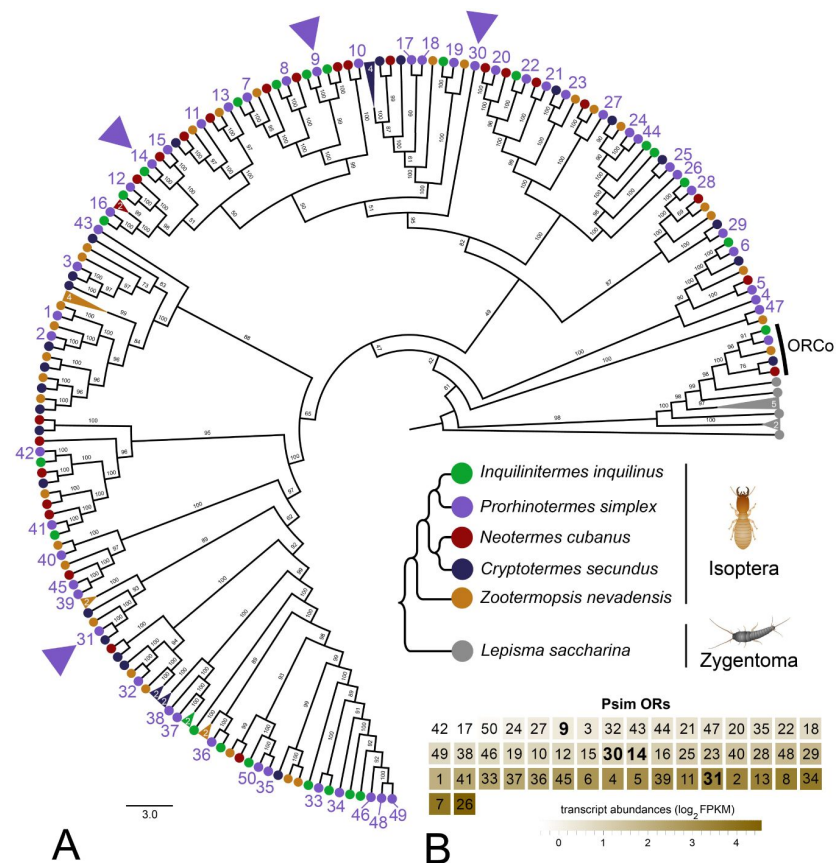


Fig. 1.

Phylogenetic reconstruction of termite ORs and their transcript abundances in *P. simplex* workers.

A. Phylogenetic tree is based on 182 protein sequences from five species of termites and the bristletail *Lepisma saccharina* as a basal insect outgroup, and also includes the sequences of ORCo. The topology and branching supports were inferred using the IQ-TREE maximum likelihood algorithm with the JTT+F+R8 model and supported by 10,000 iterations of ultrafast bootstrap approximation. Protein sequences of termite ORs can be found under the same labeling in [Johnny et al. \(2023\)](#). *Lepisma saccharina* sequences are listed in [Thoma et al. \(2019\)](#). Arrowheads highlight the four ORs from *Prorhinotermes simplex* selected for functional characterization. Fully annotated version of the tree is provided as **Supplementary Fig. S1**. **B.** Heatmap shows the transcript abundances of 50 ORs identified in the RNAseq data from *P. simplex* worker antennae available in NCBI SRA archive under accession SRX17749141.

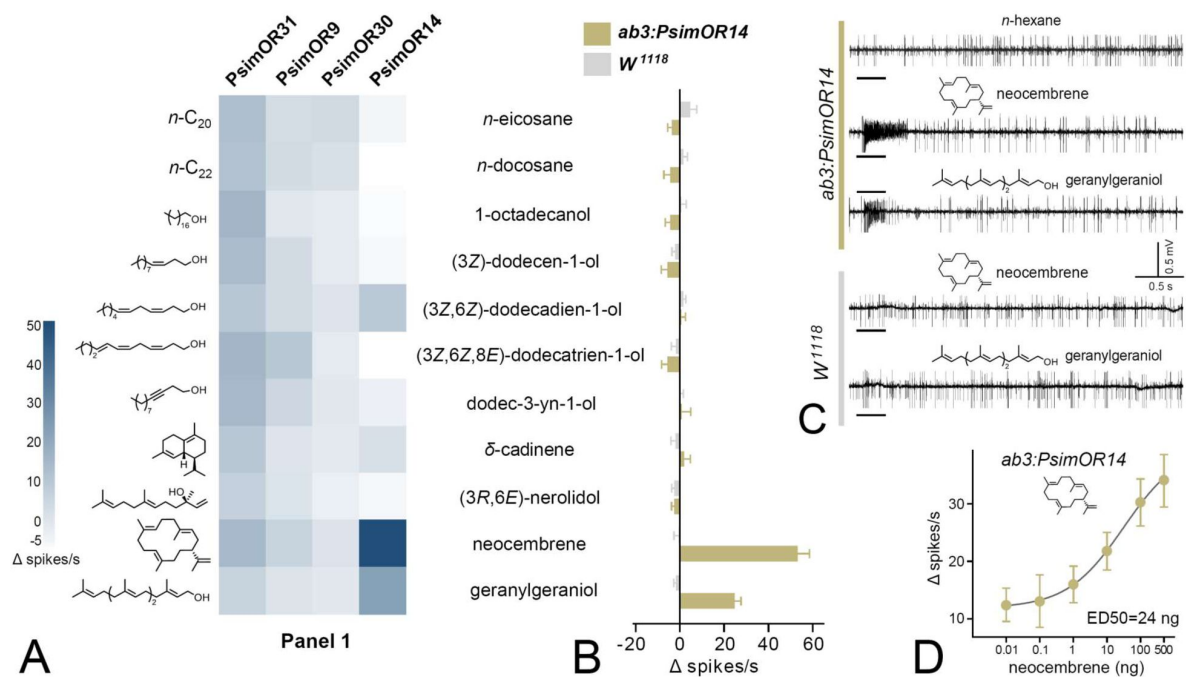


Fig. 2.

SSR responses of transgenic *D. melanogaster* ab3 sensillum expressing PsimOR9, 14, 30 and 31 to the initial screening of 11 volatiles with biological relevance for termites.

A. Heatmap showing the average responses of the four ORs as Δ spikes/s from 3–6 independent replicates. **B.** Comparison of SSR responses of transgenic *Drosophila melanogaster* ab3A neurons expressing PsimOR14 (*ab3A:PsimOR14*) and *W*¹¹¹⁸ *D. melanogaster*. The bars show the average Δ spikes/s values from five independent replicates $\pm$ SEM. **C.** Characteristic SSR traces of *ab3A:PsimOR14* and *W*¹¹¹⁸ flies for 1 μ g dose of neocembrene and geranylgeraniol. **D.** Dose response curve of *ab3A:PsimOR14* SSR responses to neocembrene. The graph shows average Δ spikes/s values $\pm$ SEM based on nine replicates (8 in case of 100 ng and 4 in case of 500 ng stimulations). The curve fit and ED50 value were calculated using log(agonist) vs. response non-linear algorithm with least square fit method and the constraint of minimal response > 0. The raw data for all graphs is provided in [Supplementary Tables S1–S6](#).

Δ spikes/s for geranylgeraniol. Characteristic responses for both lines to the two compounds are depicted in [Fig. 2C](#). In the next step, we tested the dose-response behavior of PsimOR14 flies to neocembrene, and recorded an exponentially increasing neuronal response over the range of 0.01–10 ng to $ED_{50} = \sim 24$ ng and a lack of saturation at the dose of 500 ng ([Fig. 2D](#), [Supplementary Tables S5](#), [S6](#)).

PsimOR14 is narrowly tuned to neocembrene, the main trail-following pheromone component

To further address the specificity of PsimOR14 tuning, we tested three additional panels containing 56 frequently occurring insect semiochemicals from various chemical classes. As shown in [Fig. 3A](#), none of these compounds, including multiple terpenoids (mono-, sesqui-, di-), generated a strong response, suggesting a narrow tuning of PsimOR14 to neocembrene. The narrow tuning of PsimOR14 is evident also from the tuning curve depicted in [Fig. 3B](#), with the receptor lifetime sparseness value 0.88 ([Supplementary Tables S7–S9](#)). These results confirm that PsimOR14 is a pheromone receptor adaptively tuned to detect the main trail-following pheromone component neocembrene.

Identification of *P. simplex* olfactory sensillum responding to neocembrene

Our next goal was to identify the antennal olfactory sensillum responsible for neocembrene detection by *P. simplex* workers using a combination of SSR measurements with scanning electron microscopy (SEM) and high-resolution SEM (HR-SEM) imaging. Since no previous study reported SSR responses of termite olfactory sensilla to pheromones or environmental cues, we decided to search for the neocembrene-detecting sensillum on the last flagellomere, known to harbor by far the most sensilla in termite workers ([Castillo et al., 2021](#)).

In SSR experiments with the termite-relevant compounds from Panel 1, we obtained strong response to both neocembrene and geranylgeraniol from a short multiporous grooved sensillum situated in the apical part of the last antennal segment ([Fig. 4A,B](#)). The response of this sensillum to neocembrene was confirmed on workers originating from two different colonies. Detailed view on spontaneous firing pattern of the neocembrene-responding sensillum revealed three different spike amplitudes ([Fig. 4C](#)), suggesting the potential presence of as many as three olfactory sensory neurons (a–c). Comparison with spike amplitudes upon neocembrene and geranylgeraniol stimulations then indicated that the responses are generated by the neuron labeled as b.

The SSR response spectrum of neocembrene sensillum to Panel 1 was markedly similar to that of ab3A neuron of PsimOR14-expressing *Drosophila* ([Fig. 5A](#)). None of the compounds elicited higher average responses than 10 Δ spikes/s, except for neocembrene and geranylgeraniol; their average Δ spikes/s were even slightly higher than those of heterologously expressed PsimOR14, reaching ~ 65 and ~ 29 , respectively ([Fig. 5A, B](#)). Likewise, the dose-response experiment with neocembrene indicated a higher sensitivity threshold and lower $ED_{50} = 0.016$ ng ([Fig. 5C](#), [Supplementary Tables S10](#), [S11](#)).

PsimOR14 gene and protein structure, protein modeling, ligand docking, MM/PBSA, molecular dynamics (MD) simulations

Mapping the *PsimOR14* transcript sequence on *P. simplex* draft genome revealed that the gene consists of six exons and is situated on the same locus and in close vicinity of *PsimOR15*, with which it shares the exon-intron boundaries, suggesting a recent diversification of the two genes via duplication, as also supported by their high sequence similarity ([Fig. 6A](#), [Fig. 1A](#)). Transcript ([Fig. 6B](#)) and protein ([Fig. 6C](#)) structures of PsimOR14 showed the presence of seven transmembrane domains (S1–S7) with the largest extracellular loop between S3 and S4 and the longest intracellular loop between S4 and S5.

Fig. 3.

SSR responses of transgenic *D. melanogaster* ab3 sensillum expressing PsimOR14 to the complete set of 67 compounds (Panels 1–4).

A. Heatmap showing the average responses as Δ spikes/s from 3–6 independent replicates. **B.** Tuning curve of PsimOR14 for the 67 compounds contained in panels 1–4. The raw data is provided in **Supplementary Tables S7** [↗](#)–**S9** [↗](#).

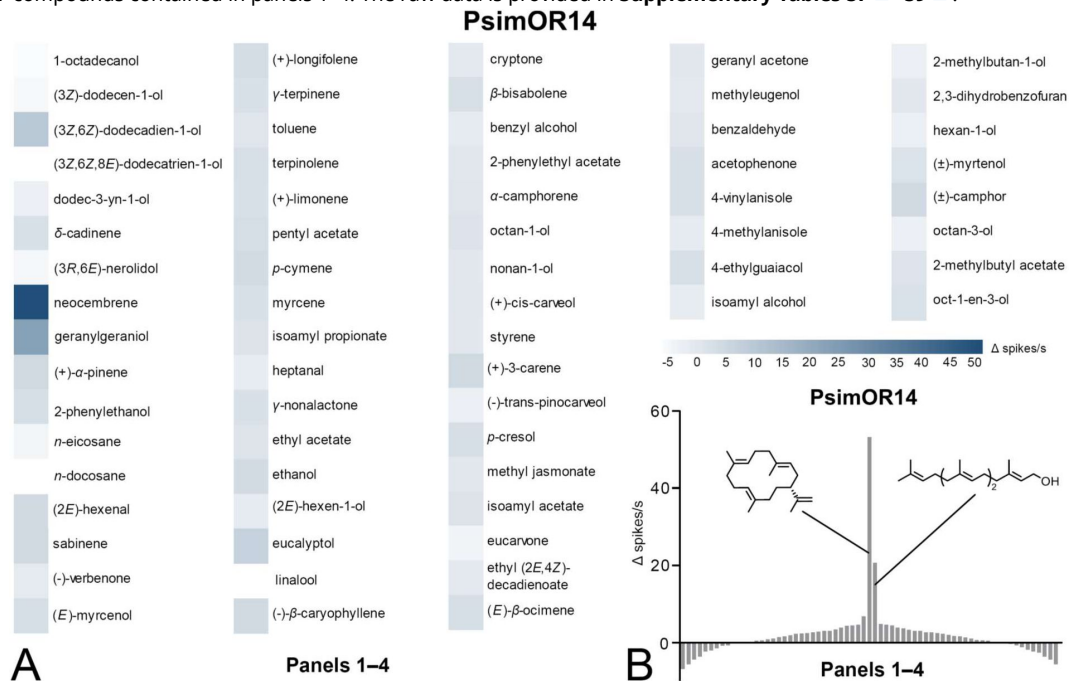
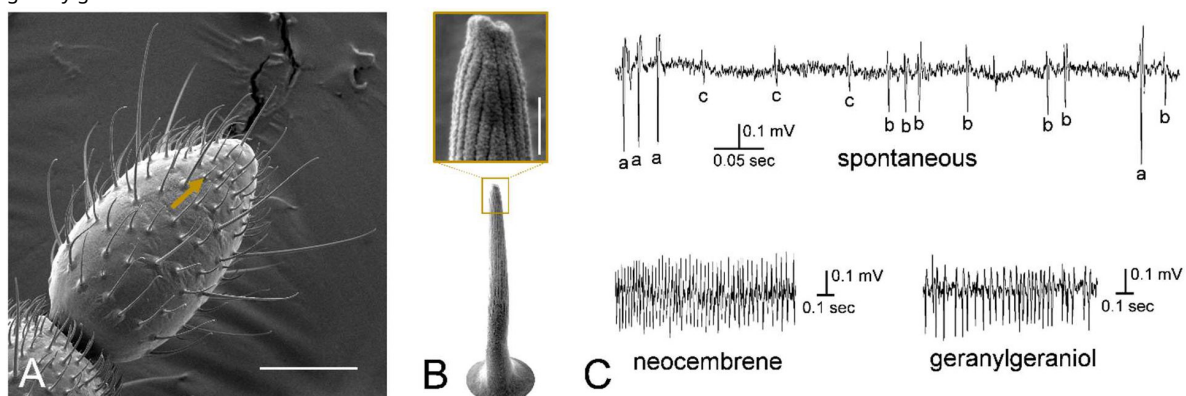


Fig. 4.

Neocembrene-responding sensillum in *P. simplex* workers.

A. SEM photograph of the last flagellomere of *P. simplex* worker. Arrow shows a small multiporous grooved sensillum responding to neocembrene and geranylgeraniol. Scale bar represents 50 μ m. **B.** HR-SEM view on the neocembrene-responding sensillum. Scale bar in the inset represents 500 nm. **C.** Detailed view on SSR traces recorded from the neocembrene-responding sensillum during spontaneous firing, and upon stimulation with neocembrene and geranylgeraniol.



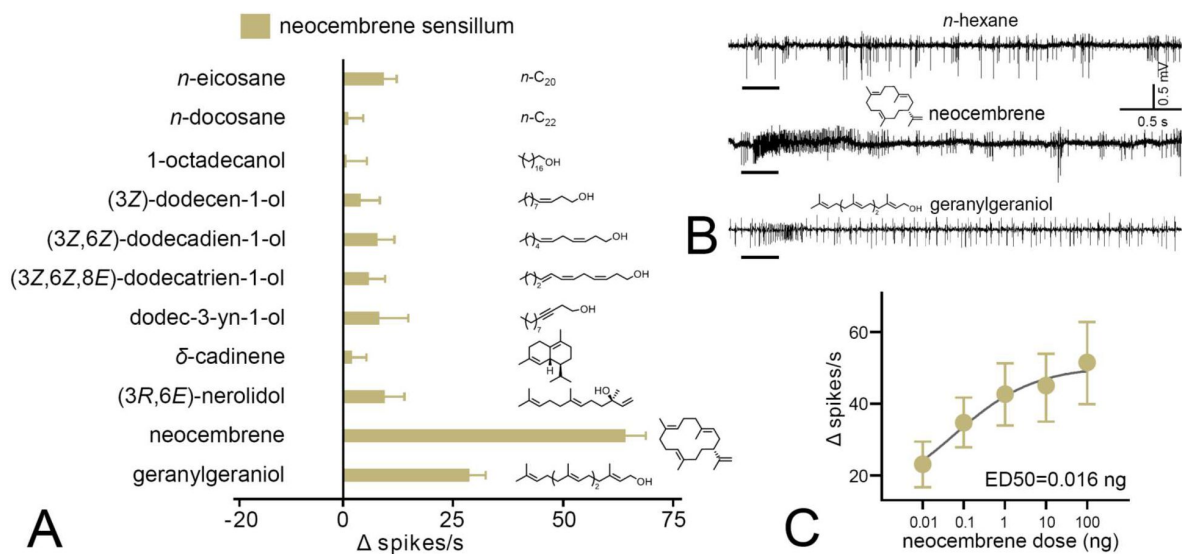


Fig. 5.

SSR responses of the neocembrene-responsive sensillum on the last flagellomere of *P. simplex* worker.

A. SSR responses to panel 1. The bars show the average Δ spikes/s values from 8–17 replicates ± SEM. The raw data is provided in [Supplementary Table S10](#). **B.** Characteristic SSR traces of the neocembrene-detecting sensillum for neocembrene and geranylgeraniol. **C.** Dose response curve of the SSR responses to neocembrene by the neocembrene-responsive sensillum. The graph shows average Δ spikes/s values ± SEM based on 9–11 replicates. The curve fit and ED50 value were calculated using log(agonist) vs. response non-linear algorithm with least square fit method and the constraint of minimal response > 0. The raw data is provided in [Supplementary Table S11](#).

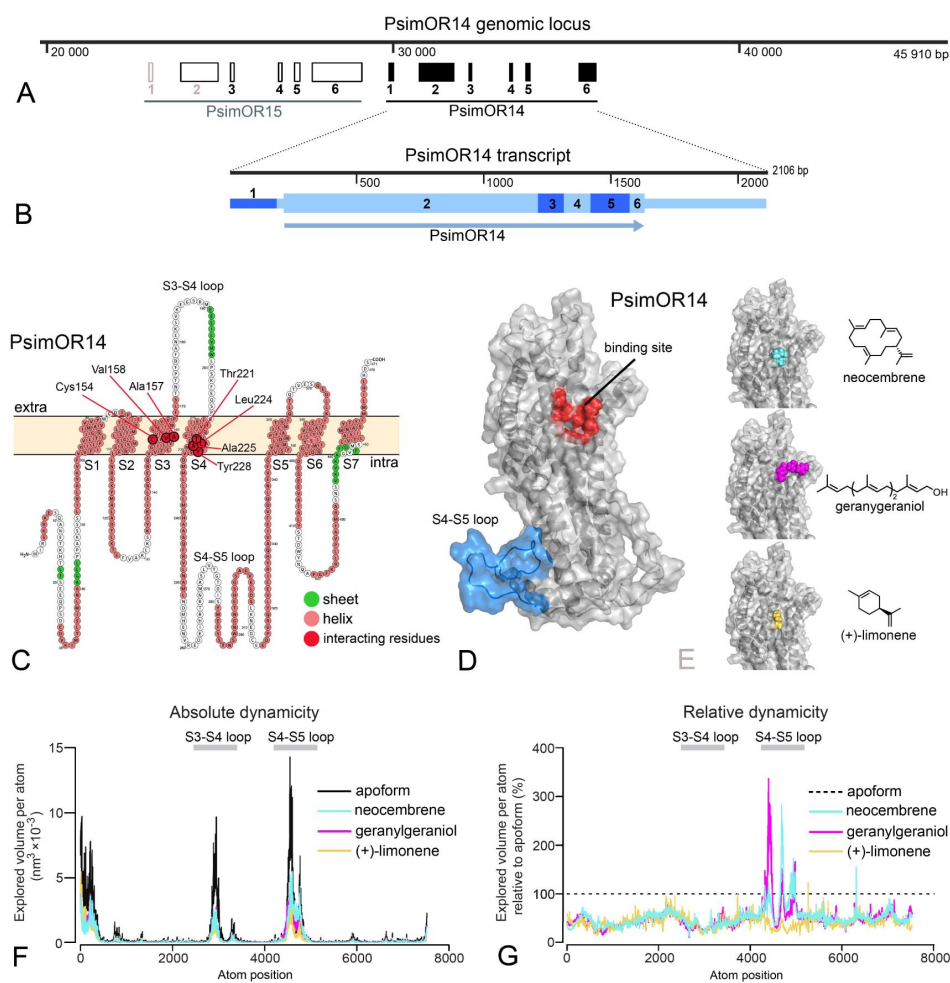


Fig. 6.

PsimOR14 gene, transcript and protein structures, docking and MD simulations.

A. Genomic locus containing *PsimOR14* and *PsimOR15*. *PsimOR14* gene consists of 1 non-coding and 5 protein coding exons. **B.** *PsimOR14* transcript with 6 exons, showing the protein-coding (higher boxes) and untranslated regions (lower boxes), and ORF (arrow). **C.** Transmembrane architecture of *PsimOR14*. In red are shown seven residues interacting with neocembrene. Light blue ellipse shows the intracellular loop the most impacted by ligand binding. **D.** Modelled apoform of *PsimOR14*. Red region denotes the binding site identified via docking, light blue region represents the intracellular S4-S5 loop. **E.** Holoforms of *PsimOR14* with three docked ligands. **F.** Absolute *PsimOR14* dynamics expressed as average volumes explored by atoms per simulation step in *PsimOR14* apoform and upon binding the three studied ligands. **G.** Relative *PsimOR14* dynamics expressed as average explored atom volumes upon ligand binding relative to the volumes in *PsimOR14* apoform. Nucleotide and protein sequences of *PsimOR14* are provided in **Supplementary Table S12** and as NCBI entry under accession OR921181.

Fig. 6D shows the initial PsimOR14 model obtained using AlphaFold 2. Three terpenoid ligands, i.e., the best agonists neocembrene and geranylgeraniol, and a weak agonist (+)-limonene, were selected for docking into the identified binding site. Dockings are visualized in **Fig. 6E**, the final scores of the best-ranked poses are shown in **Table 1**. The predicted docking scores indicated neocembrene as the best ligand, followed by geranylgeraniol and (+)-limonene, in line with the ranking of their biological effect in SSR assays (**Table 1**). (+)-Limonene ranked as the worst agonist also according to the binding free energy calculated in MM/PBSA analysis, while the best energy score was obtained for geranylgeraniol followed by neocembrene (**Table 1**). Both docking and MM/PBSA analysis suggested that primarily Van der Waals interactions facilitate the binding, only in case of geranylgeraniol non-negligible contribution of electrostatic interactions has been recorded. Per-residue decomposition results (**Fig. 6C**, **Supplementary Tables S13**) showed that all three ligands bind two hydrophobic patches made out of residues from S3 and S4 (Cys154, Ala157, Val158; Thr221, Leu224, Ala225, Tyr228). Neocembrene and (+)-limonene only bind these patches, while geranylgeraniol also interacts with additional residues (**Table S13**).

MD simulations of per atom explored volumes in PsimOR14 apoform and upon ligand binding delimited three regions with high dynamicity, i.e. the N-terminal region, the extracellular S3-S4 loop and especially the intracellular S4-S5 loop (**Fig. 6F**). Binding of each of the three ligands reduced the overall protein dynamicity; this protein stabilization did not differ dramatically among the three ligands. By contrast, binding of geranylgeraniol and neocembrene led to a conspicuous dynamicity increase in a portion of the S4-S5 loop, compared to both the apoform and (+)-limonene binding (**Fig. 6G**).

Caste-biased PsimOR14 expression and antennal sensitivity to neocembrene

We next decided to compare the expression pattern of PsimOR14 between *P. simplex* workers and soldiers, along with the sensitivity of the two castes to its preferred ligand, neocembrene. Both DESeq2 and EdgeR differential expression analyses of RNAseq read counts from heads (including antennae) of workers and soldiers revealed that PsimOR14 is significantly more expressed in workers, being among three the most upregulated ORs in workers (**Fig. 7A**, **Supplementary Table S14**). Subsequent EAG measurements were in line with this observation and indicated significantly stronger responses to neocembrene in workers ($p=0.012$) (**Fig. 7B**, **Supplementary Table S15**).

Discussion

Identification of PsimOR14 as the pheromone receptor narrowly tuned to neocembrene in *P. simplex* represents the first OR deorphanisation in termites and confirms that the trail-following communication is mediated by odorant receptors. This assumption, validated here for the monocyclic diterpene neocembrene, is indirectly supported by previous experiments in two termite species having a C₁₂ fatty alcohol as TFP; ORCo silencing in these species impaired the ability to follow the foraging trail (Gao et al., 2020). Because trail-following in termites has the shared evolutionary origin with courtship communication and both neocembrene and C₁₂ alcohols also occur as sex-pairing pheromone components (Bagnères & Hanus, 2015; Bordereau & Pasteels, 2011; Sillam-Dussès, 2010), it is likely that the two communication modalities share identical or closely related ORs. Future OR deorphanizations should test this hypothesis and provide an insight on how the functional diversification of trail-following and sex attraction is imprinted into the OR evolution.

Neurophysiological characteristics of *D. melanogaster* ab3A neuron expressing PsimOR14 showed expected patterns of spontaneous firing rates of units of Hz (6.87 ± 4.73 , mean $\pm$ SD) and maximum firing rates of 90 spikes/s at the highest used neocembrene doses (1 μ g), though not reaching the

Table 1.

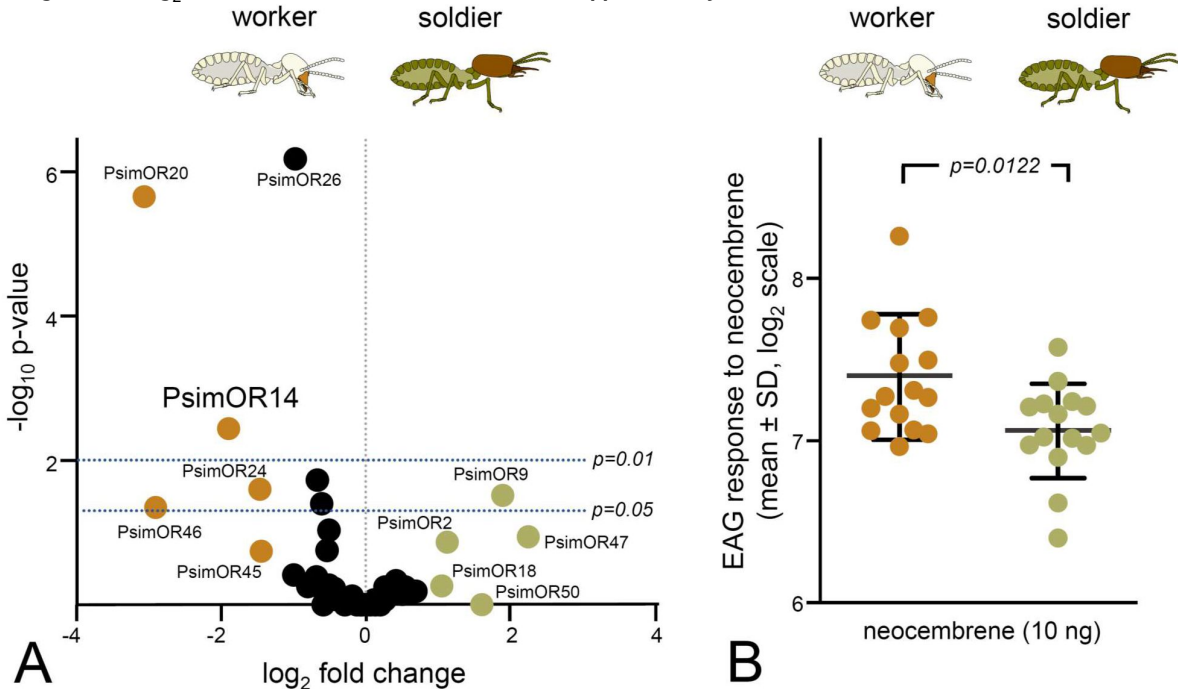
Docking scores and energy values inferred from the docking experiment and from MM/PBSA simulations for binding interactions of neocembrene, geranylgeraniol and (+)-limonene with PsimOR14.

ligand	docking experiment			MM/PBSA E (kcal/mol) ± SD			
	docking score	VDWAALS	electrostatic (kcal/mol)	ΔTOTAL	ΔVDWAALS	ΔEEL	ΔGSOLV
neocembrene	-8.658	-19.777	-0.223	-28.72 ± 1.46	-26.92 ± 1.24	-0.29 ± 0.56	-1.51 ± 0.10
geranylgeraniol	-8.331	-18.786	-11.137	-36.98 ± 1.22	-35.47 ± 0.96	-0.77 ± 0.56	-0.73 ± 0.27
(+)-limonene	-7.638	-16.134	-0.561	-20.02 ± 2.63	-20.57 ± 2.30	-0.35 ± 0.47	0.89 ± 2.21

Fig. 7.

Caste comparison of *PsimOR14* expression and EAG responses between *P. simplex* workers and soldiers.

A. Volcano plot representing edgeR differential gene expression analysis of all 50 *P. simplex* ORs in RNAseq data from soldier and workers heads (including antennae) sequenced in three independent biological replicates per caste. Colored dots mark ORs reaching absolute value of $\log_2$ fold change ≥ 1 , horizontal lines represent p-value thresholds of 0.05 and 0.01. Numeric values of the edgeR and DESeq2 differential expression analysis are provided in [Supplementary Table S14](#). Based on SRA archives accessible under SRX18952230–32 and SRX18952237–39. B. EAG responses of whole antenna preparations of workers and soldiers to neocembrene at a dose of 10 ng (mean $\pm$ SD shown on $\log_2$ scale). Inter-caste differences were compared using t-test on $\log_2$ -transformed data. Raw data is shown in [Supplementary Table S15](#).



reported maxima for ab3A responses with genuine and exogenous receptors, which may be well over 100 spikes/s (e.g., [Chahda et al., 2019](#)). This confirms that co-expression of termite ORs with the *Drosophila* ORCo using the Empty Neuron system is a suitable technique for termite OR characterizations. PsimOR14 has a narrow tuning to neocembrene with receptor lifetime sparseness equal to 0.88. ORs detecting pheromone components (pheromone receptors) and other volatiles with high biological importance, such as key food or host attractants, are often expressed in specialized sensilla and are highly selective, in contrast to the broad tuning of ORs sensing the general environmental stimuli (e.g., [Carey et al., 2010](#); [Fleischer & Krieger, 2018](#)). Such a high selectivity is the best known for sex pheromone receptors, e.g. in moths (reviewed in [Zhang & Löfstedt, 2015](#)) or *Drosophila* (reviewed in [Haverkamp, et al. 2018](#)), mostly tuned to a single compound, even though in some cases the respective ORs can be adaptively shaped to selectively detect more pheromone compounds ([Díaz-Morales et al., 2024](#); [Mariette et al., 2024](#)). Thus, the narrow tuning of PsimOR14 is in line with expectations from a pheromone receptor, especially when considering the ancient origin of trail-following from sex-pairing behavior and the great importance of TFP for concerted foraging in these blind insects. Nevertheless, due to the low coverage of insect diversity by OR deorphanization studies and their strong bias towards derived Holometabola, it remains difficult to make any general considerations about the OR selectivity relative to the receptor function across Insecta. This has been recently demonstrated by a comprehensive functional characterization of ORs in migratory locust unveiling an unexpected design of the olfactory information decoding, consisting of a large set of narrowly tuned ORs for environmental cues ([Chang et al., 2023](#)).

With average Δ spike/s over 20, the linear diterpene alcohol geranylgeraniol was the only other agonist with non-negligible response. None of the remaining compounds elicited a Δ spike of more than 7, despite the presence of multiple other compounds derived from terpenoid scaffolds in the tested panels. It is difficult to attribute any adaptive significance to this observation since there is no record of geranylgeraniol in communication context in termites. Therefore, the observed significant biological activity, high docking score and high ranking in free energy comparison, may be due to non-adaptive binding affinity of PsimOR14 to the non-native ligand (see also below). Interestingly, another related terpene alcohol, the monoterpene linalool, had the biggest negative Δ spike score (-6.8) and the absolute spikes/s ranged from 0 to 5 (1.5 ± 1.9 , mean $\pm$ SD), suggesting a possible inverse agonist function of linalool.

The neocembrene-sensing sensilla identified using SSR in *P. simplex* workers provided a response pattern to Panel 1 very similar to that of *PsimOR14*-expressing ab3 *Drosophila* sensillum, e.g., strong response to neocembrene followed by geranylgeraniol, supporting independently the results obtained from transgenic *Drosophila*. No other important responses to termite pheromone components from the Panel 1, which also included the minor *P. simplex* TFP component (3Z,6Z,8E)-dodecatrien-1-ol, were recorded, in spite of the likely presence of two additional olfactory neurons housed in the neocembrene-detecting sensillum. The function of these additional neurons remains elusive and examples from other insects offer multiple scenarios. A neuron expressing a highly selective pheromone receptor may co-exist with functionally independent neurons detecting environmental cues (e.g., [Tateishi et al. 2020](#)), or the co-habitation of several neurons may generate more complex interactions such as lateral inhibition of the neighboring neurons ([Su et al., 2012](#); [Zhang et al., 2019b](#)). However, it must be noted, that our SSR analyses focused a specific sensillum defined by its topology on the last flagellomere and we did not perform a comprehensive mapping of olfactory sensilla on the entire worker antenna. Therefore, also our image on the distribution of neocembrene-detecting sensilla and the pattern of neurons housed in these sensilla is incomplete.

The basic PsimOR14 protein architecture with seven transmembrane domains is typical for insect ORs. Likewise, the docking experiments identified a binding site defined by a binding pocket deep in the transmembrane region, homologous to that in previously studied insect ORs, and confirmed the nature of the ligand binding to mainly rely on hydrophobic interactions (e.g., [del Mármol et al.,](#)

2021 [\[1\]](#); Pettersson & Cattaneo, 2023 [\[2\]](#); Yuvaraj et al., 2021 [\[3\]](#)). In case of PsimOR14, mainly the residues from S3 and S4 interacted with the studied ligands; these transmembrane domains were shown to participate in ligand binding also in other studied insect ORs (e.g., del Mármol et al., 2021 [\[4\]](#); Yuvaraj et al., 2021 [\[3\]](#); Wang et al., 2024 [\[5\]](#); Zhao et al. 2024 [\[6\]](#)).

The calculated binding scores for the three tested ligands correlated with observed biological effects of the ligands. The MM/PBSA data also in part corroborated the SSR measurements by estimating (+)-limonene as the worst agonist; by contrast, geranylgeraniol and not neocembrene ranked as the best ligand. This may potentially be explained by the additional residues interacting with geranylgeraniol compared to neocembrene and (+)-limonene, and the relatively higher contribution of electrostatic interactions in geranylgeraniol binding. Moreover, in systems where ligand binding induces allosteric effects, as is the case of insect ORs, the simple binding affinity does not provide a complete picture of receptor signal transduction function.

In contrast to the general protein stabilization upon binding of the three tested ligands, some parts of the intracellular S4-S5 loop showed a significant increase in the dynamics upon binding of the biologically active ligands geranylgeraniol and neocembrene, compared to apoform and the weak agonist (+)-limonene binding. What is the mechanistic role of this allosteric transmission of the binding effect to the intracellular loop remains elusive. However, the impact of the S4-S5 loop on OR function has previously been reported for the MhOR5 from *Machilis hrabei* (del Mármol et al., 2021 [\[4\]](#)); replacement of the loop with a short linker increased the receptor response to native ligand. Interestingly, in some insect ORs this loop has been reduced to a short sequence of a few residues (e.g., Wang et al., 2024 [\[5\]](#)).

PsimOR14 does not belong to the most expressed *P. simplex* ORs (Fig. 1B [\[7\]](#)). However, interestingly, it is among those having the most caste-biased expression with significantly higher transcript abundance in antennae of workers compared to soldiers. Accordingly, also the electrophysiological responses of workers to neocembrene were significantly stronger in workers.

Termite soldiers are known to lay pheromone trails, to detect TFPs and to participate in foraging and field exploration in a number of termite species, though the behavioral patterns of soldiers and workers during these activities differ (Kaib, 1990 [\[8\]](#); Traniello, 1981 [\[9\]](#); Traniello & Busher, 1985 [\[10\]](#)). This has also been demonstrated in a close relative to our model, the congeneric species *Prorhinotermes inopinatus* (Rupf & Roisin, 2008 [\[11\]](#)). Yet, differences in sensitivity of termite workers and soldiers have not previously been addressed at the electrophysiological level. Caste-biased *PsimOR14* expression and neocembrene sensitivity may represent olfactory information filtering adaptive to the different tasks of the two castes, as documented, e.g., in ants (Caminer et al., 2023 [\[12\]](#)).

Genus *Prorhinotermes* is the most basally situated termite taxon known to have a terpenoid component as a part of its pheromone repertoire (Sillam-Dussès et al., 2009 [\[13\]](#)). The acquisition of terpenoid pheromone components is undoubtedly due to the evolution of terpene biosynthesis in basal Neoisoptera, which has led to the fascinating diversity of defensive terpenoids produced by soldiers of Neoisoptera in their frontal gland (Gössinger, 2019 [\[14\]](#)). In line with general observations on pheromone evolution (Steiger et al., 2010 [\[15\]](#)), some of these defensive terpenoids from the frontal gland gained via exaptation a novel function of alarm pheromones (Dolejšová et al., 2014 [\[16\]](#); Roisin et al., 1990 [\[17\]](#); Šobotník et al., 2008 [\[18\]](#), 2010 [\[19\]](#)). Likewise, the cyclic diterpenes probably became part of trail-following pheromones and sex-pairing pheromones by co-opting the terpenoid biosynthesis in exocrine glands other than the defensive frontal gland of soldiers. Beside the occurrence of neocembrene in the sub-basal *Prorhinotermes*, it only occurs as a pheromone component in much later diverging representatives of the subfamily Nasutitermitinae (Bordereau & Pasteels, 2011 [\[20\]](#)). Once reliable genome or antennal transcriptome of

Nasutitermitinae become available, it would be of interest to search in these species for OR orthologs of *PsimOR14* and test whether they have retained the function of neocembrene detection.

Insect ORs are frequently organized in tandem arrays (e.g., Bohbot et al., 2007 [↗](#); McKenzie and Kronauer, 2018 [↗](#); Robertson and Wanner, 2006 [↗](#)). Likewise, the genomic locus of *PsimOR14* contains a likely tandem copy paralog of *PsimOR15* gene. The two genes share the gene architecture, and their transcripts are equally represented in worker antennal transcriptome, though *PsimOR15* does not display the caste-biased expression. Thanks to the close similarity of both genes, *PsimOR15* is a candidate for potential pheromone receptor function.

Future research in termites should also aim at finding the ORs or other chemosensory proteins involved in the detection of CHCs. In spite of the similarities in CHC roles in termites and ants, ranging from nestmate recognition to fertility signaling, termite OR sequences do not show any conspicuous expansions analogous to 9-exon subfamily in ants. Therefore, finding the chemosensory principle of CHC detection in termites would bring another piece of knowledge about the fascinating convergent evolution in these two unrelated major groups of eusocial insects. Yet another appealing target for OR deorphanization is the search for pheromone receptors of queen pheromones, the central signals ensuring the reproductive dominance of queens in the colonies of social insects. In the honeybee and ants, narrowly tuned ORs responding to main queen pheromone components were already described (Wanner et al., 2007 [↗](#); Pask et al., 2017 [↗](#)). In termites, queen pheromones were so far only identified in two species (Dolejšová et al., 2022 [↗](#); Matsuura et al. 2010 [↗](#)). They are volatile and were shown to act as airborne signals via olfaction (Dolejšová et al., 2022 [↗](#)), it is thus reasonable to expect a selective OR responsible for their detection.

Materials and methods

Termites

Multiple laboratory colonies of *P. simplex*, originating from previous field collections in Cuba and Florida, are held in the Institute of Organic Chemistry and Biochemistry, Czech Academy of Sciences. Colonies are reared in glass vivaria at 27°C and 80% relative humidity in clusters of spruce wood slices.

The data reported here were collected from three mature Cuban colonies. The first one was used for antennal transcriptome sequencing and assembly followed by phylogenetic analysis, as described in Johnny et al. (2023) [↗](#). The second one was used for RNA extraction for OR cloning, SSR, SEM and HR-SEM. The third one was used for caste-specific head transcriptomes (head + antennae) of workers and soldiers, for caste-specific EAG recordings, and for SSR confirmation of the neocembrene-detecting sensillum. For all experiments with workers, 4th or 5th stage workers were selected as the most abundant developmental stages, recognized according to body size and head width.

RNA extraction, OR cloning and construct generation

Total RNA was extracted from 20 pairs of dissected *P. simplex* antennae using PureLink RNA Mini Kit (Invitrogen, Carlsbad, CA, USA) following the manufacturer's protocol and quantified using NanoDrop spectrophotometer (Thermo, Delaware, USA). From the total RNA, 2 µg was used to synthesize the cDNA using SuperScript IV Reverse Transcriptase (Invitrogen, Carlsbad, CA, USA) according to manufacturer's instructions. The efficiency of cDNA synthesis was evaluated by amplification of ORCo. The list of primers is provided in **Supplementary Table S16** [↗](#). The full-length open reading frame (ORF) of each selected *PsimORs* was PCR-amplified from the cDNA

using the DreamTaq Green PCR Master Mix (Invitrogen, USA) and gene-specific primers (**Supplementary Table S16** [↗](#)). Amplification products were purified by QIAquick Gel Extraction Kit (Qiagen, Germany), cloned into pCR8/GW/TOPO vector using the TOPO TA Cloning Kit (Invitrogen, USA) and transformed into OneShot TOP10 competent cells (Invitrogen, USA). Positive colonies were selected based on colony-PCRs using primers GW1 and GW2, recombinant plasmids were isolated using the QIAprep 2.0 Spin Miniprep Columns (Qiagen, Germany) and sequences were verified by Sanger sequencing (Eurofins Genomics, Germany).

The expression vector constructs were prepared using the Gateway LR recombination cloning technology (Invitrogen, USA) based on recombination of the phage-like attachment sites attL/R in pCR8/GW/TOPO with the bacteria-like attachment site attB in pUASg.attB vector (obtained from *Drosophila* Genomics Resource Center, Bloomington, USA). The resulting constructs pUASg.attB-PsimOR were purified using the QIAprep 2.0 Spin Miniprep Columns (Qiagen, Germany) and insert sequences were verified by Sanger sequencing at Eurofins Genomics (Germany). All primers used for Sanger sequencing and colony-PCR are listed in **Supplementary Table S16** [↗](#).

Fly lines

D. melanogaster lines used in the Empty Neuron system were kindly provided by Dr. Thomas O. Auer (from Richard Benton Lab, University of Lausanne, Switzerland). The wild type *W¹¹¹⁸* line, used as a control, was kindly provided by Prof. Michal Žurovec (Biology Centre, Czechia). All *Drosophila melanogaster* lines were reared in an incubator which was set at 24±2°C with relative humidity of 50±5%. Flies were fed with in-house prepared diet based on standard cornmeal food. The fly lines used are listed in **Supplementary Table S17** [↗](#).

Transgenic expression of termite ORs in *D. melanogaster* ab3A neuron

Selected PsimORs were expressed in the *Drosophila melanogaster* Empty Neuron system for functional screening. Transgenic *D. melanogaster* UAS-PsimOR lines were generated by BestGene Inc. (Chino Hills, CA, USA) by injecting pUASg.attB-PsimOR vectors into fly embryos expressing the integrase PhiC31 and carrying an attP landing site resulting in flies with genotype *w⁻; +; UAS-PsimOR (w⁺)/+*.

The recent CRISPR-Cas9-engineered empty neuron line *Or22ab^{-Gal4}* (Chahda et al., 2019 [↗](#)) was used as Δ halo genetic background for the expression of UAS-PsimOR in Dmel ab3 sensilla. The fly crossing scheme was adapted from (Gonzalez et al., 2016 [↗](#)) with a modification at the F3 crossing. Final homozygote lines with UAS-PsimOR and *Or22ab^{-Gal4}* were generated and used for the electrophysiological recordings. The full description of the crossing scheme is provided in **Supplementary Fig. S2** [↗](#).

Chemicals

For SSR measurements, we used a total of 67 chemicals organized into four panels. The initial screening Panel 1 contained 11 compounds biologically relevant to termites, i.e. components of termite pheromones and cuticular hydrocarbons known from Neoisoptera and structurally related compounds. This panel was used for initial SSR screening of PsimOR9, 14, 30 and 31 in transgenic *D. melanogaster* and for SSR experiments with *P. simplex* workers. For detailed analysis of PsimOR14, three additional panels were used, consisting of 56 frequently occurring insect semiochemicals (e.g., terpenoids, fatty acid esters, fatty alcohols and aldehydes, etc.). Panel 1 compounds were diluted in *n*-hexane to 100 ng/μL, panel 2–4 compounds were diluted in paraffin oil to 10⁻³ v/v. List of all compounds tested and their origin are listed in **Supplementary Table S18** [↗](#).

Organic synthesis

For the purpose of SSR experiments, we synthesized (Z)-dodec-3-en-1-ol, (3Z,6Z)-dodeca-3,6-dien-1-ol, (3Z,6Z,8E)-dodecatrien-1-ol, and dodec-3-yn-1-ol, and included these compounds into Panel 1. The *de novo* organic synthesis of these fatty alcohols is described in the Supplementary Information file.

Electrophysiology

SSR recordings on *Drosophila* ab3 sensillum were performed as described previously (Benton & Dahanukar, 2023 [DOI](#); Olsson & Hansson, 2013 [DOI](#)). We used 2–4 days old flies for one recording per each to avoid neuronal adaptations from multiple stimulations. To expose more ab3 sensilla, the fly preparation was done with arista down (Keesey et al., 2022 [DOI](#)).

In termites, the olfactory sensilla situated on the last antennal flagellomere of workers were targeted for SSR, since their number increases towards the distal end of termite antennae, the last segment being significantly more populated by olfactory sensilla than any other segment (Castillo et al., 2021 [DOI](#)). The grounding electrode was carefully inserted into the clypeus and the antenna was fixed on a microscope slide using a glass electrode. To avoid the antennal movement, the microscope slide was covered with double-sided tape and the three distal antennal segments were attached to the slide.

The sensilla were observed under the Nikon FN1 eclipse microscope at 60× magnification. For all electrophysiological measurements, the recording electrode was brought into contact with the base of the sensillum using a Kleindiek Nanotechnik MM3A micromanipulator connected to a cubic micromanipulator device. Using Syntech stimulus delivery system (CS55 model, Syntech, Germany), the odorant stimulus was administered as a 0.3 s pulse by inserting the tip of the glass Pasteur pipette through an opening into the delivery tube (situated 4 cm from the tube outlet) carrying a purified air stream (0.4 L/min). The tube outlet was placed approximately 1 cm from the antenna. The experiments were conducted at 25–26 °C.

From each diluted odorant (100 ng/μL), 10 μL were pipetted on 1 cm diameter filter paper disk placed in a glass Pasteur pipette in the screening experiment, while the doses ranging from 0.01 to 500 ng of neocembrene per filter paper were used in the dose-response experiments.

The signal was amplified and digitally converted using Syntech IDAC-4. The neuronal cells were sorted based on their amplitude and the spikes were counted using the AutoSpike v3.9 software (Syntech Ockenfels, Germany). Δ spike was calculated by subtracting the number of spikes during 1 second post-stimulation from the number of spikes generated 1 second before the stimulation. In dose response experiments and measurements on termite antennae, the counting periods were 0.5s. Δ spike values were corrected by subtracting the response generated by the solvent and converted into Δ spike/s (Benton & Dahanukar, 2023 [DOI](#); Olsson & Hansson, 2013 [DOI](#)). The receptor lifetime sparseness value was calculated according to Chang et al. (2023) [DOI](#).

EAG experiment addressing the caste-specificity of antennal responses to neocembrene was performed with 15 workers and 15 soldiers; each individual was only used for one stimulation series consisting of air–hexane–neocembrene (10 ng)–hexane–air. Brain and antennal tip were placed between two Ag/AgCl electrodes containing Ringer’s solution and connected to a high impedance (10^{14} Ω) amplifier (Syntech, Buchenbach, Germany). The antennal preparation was placed into a stream of cleaned air (500 mL/min), into which the stimuli were injected from Pasteur pipettes containing a 1.5 cm² filter paper impregnated with 10 μL of the tested solution. Odor injections were controlled by a foot switch-operated Syntech stimulus controller and maximal negative deflection was measured using Syntech EAGPro software. Pasteur pipettes containing odorant stimuli were changed after three stimulations. Air responses were used for

data normalization, the responses $\log_2$ -transformed to reduce heteroscedasticity and comply with assumptions for parametric testing (Bartlett test for equal variances, and Shapiro-Wilk normality test), and then compared between workers and soldiers using Student's t-test.

Scanning electron microscopy

For SEM, 10 workers with intact antennae were cold-anesthetized and decapitated with micro-scissors. Heads were desiccated in increasing ethanol concentrations (60, 80, 90, and 96%, each for 2 h) followed by 12 h in acetone. Heads were then attached to aluminium holders for microscopy using adhesive tape, and differentially oriented to allow axial, dorsal, ventral, and lateral views. The samples were gold-coated for regular SEM (4 nm gold layer) and high-resolution SEM (HR-SEM, 2 nm) using sputter coater Bal-Tec SCD 050. Last antennal segments were inspected and photographed under scanning electron microscope JEOL 6380 LV (SEM). Surface of particular sensilla were studied using high resolution field emission scanning electron microscope JSM-IT800 (HR-SEM) and Olympus Soft Imaging Solution software. Working distance for all samples was 4.0–4.1 mm and accelerating voltage 2.0 kV.

Bioinformatics

For phylogenetic reconstruction of termite ORs, we used 182 OR protein sequences originating from five termite species, i.e. *Neotermes cubanus*, *Prorhinotermes simplex*, *Inquilinitermes inquilinus* (Johnny et al., 2023 [DOI](#)), *Zootermopsis nevadensis* (Terrapon et al., 2014 [DOI](#)) and *Cryptotermes secundus* (Harrison et al., 2018 [DOI](#)), and the bristletail *Lepisma saccharina* (Thoma et al., 2019 [DOI](#)) as a basal insect outgroup. For all species also the ORCo sequence was included. Sequences were aligned by means of the MUSCLE algorithm and used for reconstructing the phylogenetic tree with the IQ-TREE maximum likelihood algorithm (Nguyen et al., 2015 [DOI](#)) using the JTT+F+R8 substitution model and 10,000 ultrafast bootstrap replicates.

The gene structures of *PsimORs* were characterized by local alignment of full-length transcript sequences from Johnny et al. (2023) [DOI](#) to our in-house genome assembly using BLAST (for details on genome assembly see Koubová et al., 2021 [DOI](#)) and confirmed with genomic mapping of the RNAseq data from *P. simplex* antennae available under accession SRX17749141 in NCBI SRA archives using STAR aligner v2.7.10b (Dobin et al., 2013 [DOI](#)). The mapping results were further used for abundance estimations of all ORs and ORCo in antennal transcriptome reported in Johnny et al. (2023) [DOI](#). Read counts were obtained with featureCounts tool from the Subread package (<https://subread.sourceforge.net/> [DOI](#)) and normalized according to the FPKM (Fragments Per Kilobase Million) method.

Differential OR expression analysis in *P. simplex* soldier and worker heads (including antennae) was performed using the RNAseq data from our previous sequencing project and available as SRA archives under accessions SRX18952230–32 and SRX18952237–39. The data was obtained from sequencing of three biological replicates of each caste. Read counts obtained using STAR mapping and featureCounts estimations were statistically evaluated using the DESeq2 Bioconductor package in R and edgeR.

PsimOR14 secondary structure was predicted using online tools Jpred 4.0.0 (www.compbio.dundee.ac.uk/jpred [DOI](#)) and TMHMM2.0 (<https://services.healthtech.dtu.dk/services/TMHMM-2.0> [DOI](#)), schematic model was generated using Protter (<https://wlab.ethz.ch/protter> [DOI](#)).

Protein modeling

PsimOR14 structure was modelled in its monomeric membrane-free form using AlphaFold2 (Jumper et al., 2021 [DOI](#); Mirdita et al., 2022 [DOI](#)). The best model was refined by MD relaxation, employing GROMACS 2021.3 and CHARMM36m (Abraham et al., 2015 [DOI](#); Huang et al., 2017 [DOI](#)). After solvation and neutralization by Na^+ ions in TIP3P CHARMM water in a 1.5 nm padded box,

temperature and pressure equilibration followed. Six different simulations with differing starting velocities were produced, followed by 150 ns periodic simulated annealing independently for each replica (0.5 ns at 300 K, then 0.5 ns at 320 K, repeating). The lowest potential energy structure was chosen. Neocembrene, geranylgeraniol and (+)-limonene structures were sourced from PubChem (13th Feb 2023) and parametrized using CgenFF 4.6, employing CHARMM-GUI for the conversion (Jo et al., 2008 [↗](#); Kim et al., 2023 [↗](#); Vanommeslaeghe et al., 2010 [↗](#)). The binding site was predicted based on DEET binding region of *MhOR5* from *Machilis hrabei* (7LIG) (del Marmol et al., 2021 [↗](#)). Using DockThor webserver, ligands were docked with all bonds treated as rotatable, centered around the expected binding site with maximized box size (40 units). The best binder was then selected for each complex (Santos et al., 2020 [↗](#)).

MM/PBSA simulations

Complex topologies were built in GROMACS. Simulations included 9 replicas for liganded (14 μ s each) and 6 for unliganded PsimOR14 (6.9 μ s), with different starting velocities. Convergence was assessed by ligand backbone RMSD distributions. Replicas were concatenated, split into 1.5 ns frames, and analyzed with MM/PBSA (ff19SB+GAFF2, linearized PB with die=2, SASA for apolar contributions, optimized CHARMM radii) using gmx_MMPBSA v1.6.3 and AmberTools 20 (Case et al., 2023 [↗](#); Guedes et al., 2021 [↗](#); Tian et al., 2020 [↗](#); Wang et al., 2004 [↗](#)). For each liganded PsimOR14 (neocembrene, geranylgeraniol and limonene) 9 replica trajectories were separately analyzed, along with apoform PsimOR14 trajectories (6 replicas). All trajectories were PBC-corrected, cleaned, and fitted to the first frame of the PsimOR14 trajectory.

Atom positions were marked in 3D space per frame, and the convex hull algorithm approximated the volume explored by each atom. This compared dynamic behavior between trajectories, with unliganded PsimOR14 as the baseline. The average total volume explored by atoms was compared.

Supplementary information

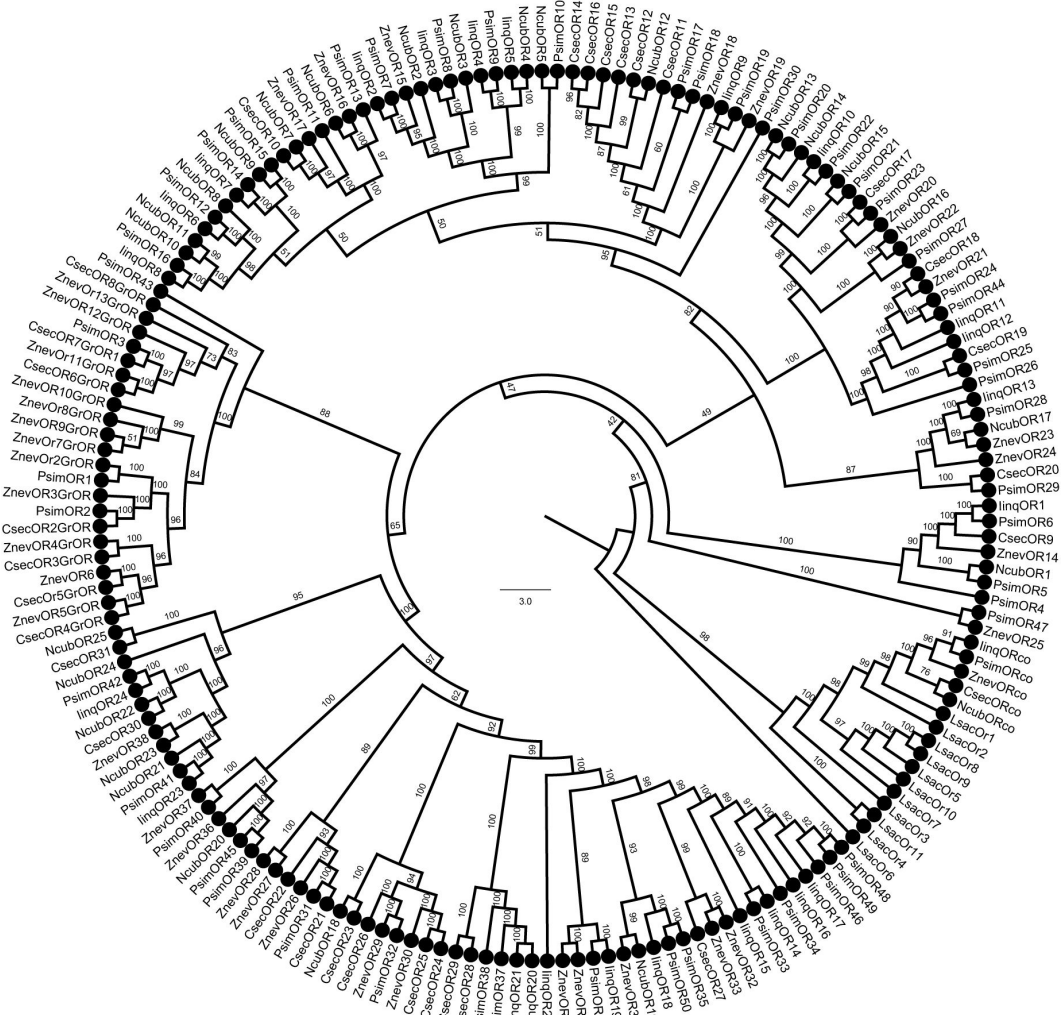


Fig. S1.
Full version of the phylogenetic tree of termite ORs shown in Fig. 1 of the main text.

Protein sequences of termite ORs can be found under the same labeling in [Johnny et al. \(2023\)](#), *Lepisma saccharina* sequences used as basal insect outgroup are listed in [Thoma et al. \(2019\)](#). The topology and branching supports were inferred using the IQ-TREE maximum likelihood algorithm with the JTT+F+R8 model and supported by 10,000 iterations of ultrafast bootstrap approximation.

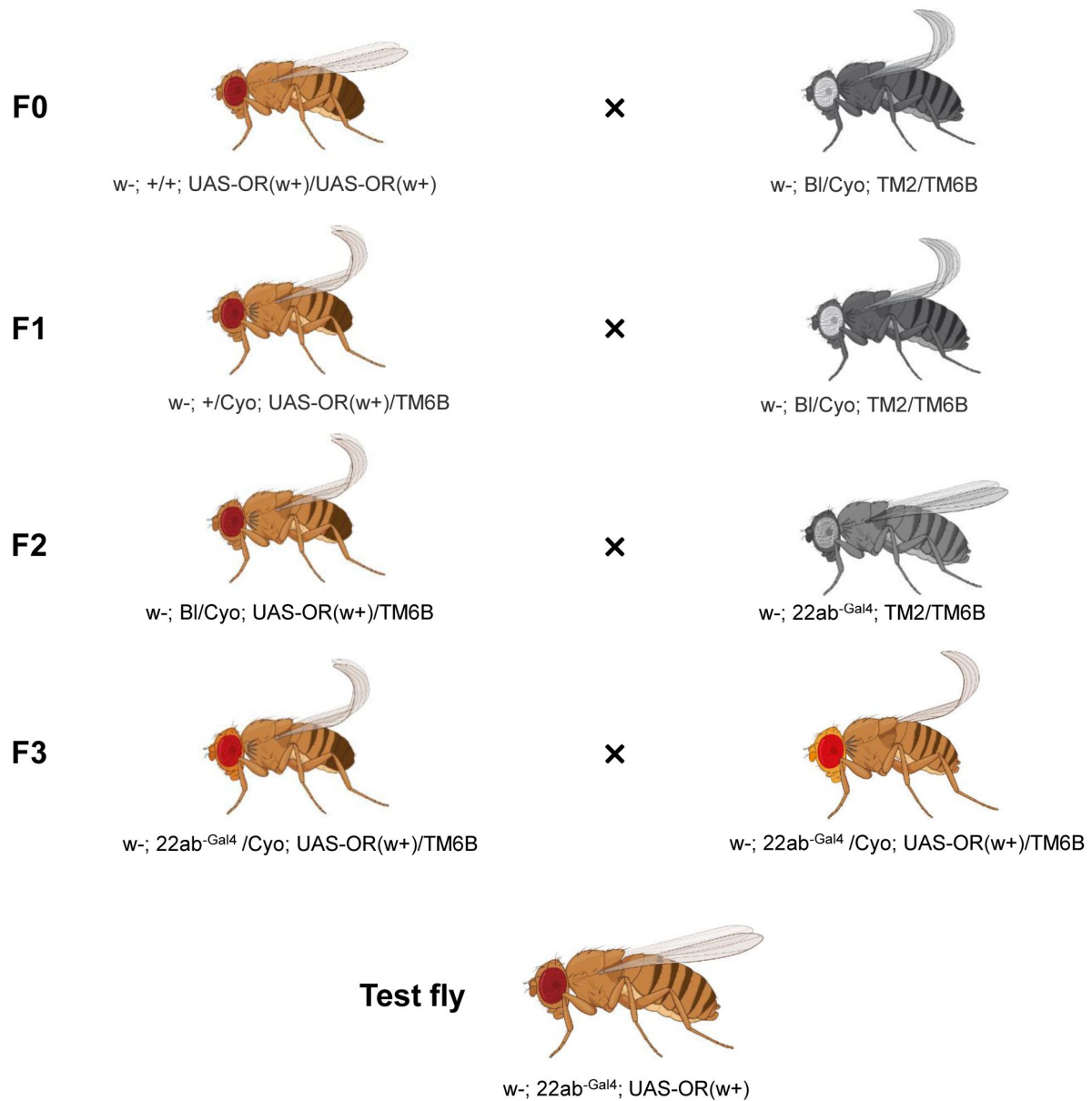


Fig. S2.

Crossing scheme of termite ORs heterologous expression using *Drosophila melanogaster* empty neurons in ab3 sensilla.

Supplementary methods

Organic synthesis

Unless noted otherwise, all reactions were carried out under argon in oven-dried glassware. Solvents were distilled from drying agents as indicated and transferred under nitrogen: THF (Na/benzophenone), toluene (Na/benzophenone). All starting materials were used as purchased (Sigma Aldrich, Combi-Blocks), unless otherwise indicated. Chromatography was performed using Fluka silica gel 60 (0.040 - 0.063 mm). For TLC analysis, F254 – coated aluminum sheets were used. The spots were detected both in UV and by the solution of $\text{Ce}(\text{SO}_4)_2 \cdot 4\text{H}_2\text{O}$ (1%) and $\text{H}_3\text{P}(\text{Mo}_3\text{O}_{10})_4$ (2%) in 10% sulfuric acid. ^1H - and ^{13}C NMR spectra were recorded at 400 MHz and 100 MHz, respectively, with a Bruker 400 MHz instrument at 25 °C (the solvents are indicated in parentheses). Chemical shifts are reported in ppm relative to TMS. The residual solvent signals in the ^1H and ^{13}C NMR spectra were used as an internal reference (CDCl_3 : $\delta = 7.26$ for ^1H and $\delta = 77.23$ for ^{13}C). The compounds were analyzed using the gas chromatograph TRACE 1310 (ThermoFisher Scientific, Waltham, MA, USA), equipped with a nonpolar Zebtron ZB-5MS column (30 m $\times$ 0.25 mm $\times$ 0.25 μm film; Phenomenex, Torrance, CA, USA) connected to a ThermoFisher Scientific ISQ LT mass-selective detector (70eV ionization voltage, source temperature 200 °C, transferline heated to 260 °C). The column temperature was held at 50 °C for 1 min, gradually increased to 320 °C at 8 °C/min and then held at 320 °C for 20 min. Helium was used as carrier gas at a flow 1.2 mL/min. Split/splitless port was heated to 200 °C, and samples were injected in splitless mode with a purge time of 1 min.

2-(But-3-yn-1-yloxy)tetra-2H-hydropyran (Allegretti & Ferreira, 2011 [↗](#)) (1)

To a solution of 3-butyne-1-ol (3.0 g; 42.79 mmol) in DCM (100 mL) at 0 °C was added 3,4-dihydro-2H-pyran (3.78 g; 44.93 mmol) followed by addition of *p*-toluenesulfonic acid monohydrate (0.080 g; 0.427 mmol). After stirring at 0 °C for 5 min, the reaction was warmed to room temperature and stirred for 90 min. The reaction was quenched with saturated solution of NaHCO_3 (20 mL), and the layers were separated. The aqueous layer was extracted with DCM (50 mL), and the combined organic layers were dried over MgSO_4 and concentrated in vacuo. The residue was purified by a column chromatography (silica gel; cyclohexane/EtOAc 4:1) to give **4** as colorless oil (0.53 g, 64%). ^1H NMR (401 MHz, CDCl_3) δ 4.67 (dd, $J = 4.2, 2.9$ Hz, 1H), 3.97 – 3.81 (m, 2H), 3.65 – 3.49 (m, 2H), 2.52 (td, $J = 7.0, 2.7$ Hz, 2H), 2.00 (t, $J = 2.7$ Hz, 1H), 1.93 – 1.78 (m, 1H), 1.78 – 1.69 (m, 1H), 1.69 – 1.46 (m, 4H). EI-MS (70 eV): m/z (%): 153 (1.4), 125 (2), 99 (9), 85 (100), 79 (9), 67 (21), 53 (33), 41 (22).

2-(Dodec-3-yn-1-yloxy)tetrahydro-2H-pyran (2)

To a solution of acetylene **1** (1.0 g; 6.48 mmol) in THF (10 mL) at -40 °C under argon atmosphere was added *n*-BuLi (2.6 mL, 2.5 M in hexanes, 6.5 mmol), followed with addition of HMPA (1 mL) and the resulting solution was stirred for 40 min at -40 °C. Then a solution of octyl bromide (1.25 g; 6.48 mmol) in THF (2.00 mL) was added rapidly. The resulting mixture was stirred at -40 °C for 60 min, and then warmed to room temperature and stirred for 2 h. The reaction mixture was quenched by addition of water (10 mL) and was diluted with EtOAc (20 mL). The layers were separated, and the aqueous layer was extracted with EtOAc. The combined organic layers were washed with water and brine and concentrated under reduced pressure. The residue was purified by a column chromatography (silica gel; cyclohexane/EtOAc 9:1) to give **2** as colorless oil (1.12 g, 86%). ^1H NMR (401 MHz, CDCl_3) δ 4.67 (dd, $J = 4.2, 2.9$ Hz, 1H), 3.91 (ddd, $J = 11.2, 8.1, 3.3$ Hz, 1H), 3.82 (dt, $J = 9.7, 7.2$ Hz, 1H), 3.61 – 3.47 (m, 2H), 2.48 (tt, $J = 7.2, 2.4$ Hz, 2H), 2.16 (tt, $J = 7.1, 2.4$ Hz, 2H), 1.94 – 1.79 (m, 1H), 1.74 (tdd, $J = 9.2, 3.9, 2.9$ Hz, 1H), 1.67 – 1.43 (m, 6H), 1.43 – 1.24 (m, 10H),

Replicate	Compound	Pre-stimulation spikes	Post-stimulation spikes	Generated spikes	Δ Spikes/s
1	hexane	8	5	-3	
1	(3Z,6Z)-dodecadien-1-ol	6	9	3	6
1	(3Z)-dodecen-1-ol	11	10	-1	2
1	(3Z,6Z,8E)-dodecatrien-1-ol	5	12	7	10
1	<i>n</i> -eicosane	4	9	5	8
1	1-octadecanol	10	5	-5	-2
1	<i>n</i> -docosane	10	11	1	4
1	dodec-3-yn-1-ol	1	8	7	10
1	neocembrene	9	11	2	5
1	δ-cadinene	3	0	-3	0
1	(3R,6E)-nerolidol	0	0	0	3
2	hexane	6	3	-3	
2	(3Z,6Z)-dodecadien-1-ol	9	7	-2	1
2	(3Z)-dodecen-1-ol	10	8	-2	1
2	(3Z,6Z,8E)-dodecatrien-1-ol	4	13	9	12
2	<i>n</i> -eicosane	12	2	-10	-7
2	1-octadecanol	8	6	-2	1
2	<i>n</i> -docosane	11	12	1	4
2	dodec-3-yn-1-ol	9	10	1	4
2	neocembrene	8	13	5	8
2	δ-cadinene	12	15	3	6
2	(3R,6E)-nerolidol	8	14	6	9
2	geranylgeraniol	12	10	-2	1
3	hexane	6	5	-1	
3	(3Z,6Z)-dodecadien-1-ol	10	17	7	8
3	(3Z)-dodecen-1-ol	9	14	5	6
3	(3Z,6Z,8E)-dodecatrien-1-ol	6	19	13	14
3	<i>n</i> -eicosane	4	15	11	12
3	1-octadecanol	13	11	-2	-1
3	<i>n</i> -docosane	7	12	5	6
3	dodec-3-yn-1-ol	8	11	3	4
3	neocembrene	11	18	7	8
3	δ-cadinene	10	18	8	9
3	(3R,6E)-nerolidol	14	11	-3	-2
3	geranylgeraniol	7	12	5	6
4	hexane	11	10	-1	
4	(3Z,6Z)-dodecadien-1-ol	12	17	5	6
4	(3Z)-dodecen-1-ol	4	15	11	12
4	(3Z,6Z,8E)-dodecatrien-1-ol	3	8	5	6
4	<i>n</i> -eicosane	3	3	0	1
4	1-octadecanol	2	2	0	1
4	<i>n</i> -docosane	5	10	5	6
4	dodec-3-yn-1-ol	4	12	8	9
4	neocembrene	2	9	7	8
4	δ-cadinene	6	5	-1	0
4	(3R,6E)-nerolidol	1	4	3	4
4	geranylgeraniol	3	5	2	3
5	hexane	10	8	-2	
5	(3Z,6Z)-dodecadien-1-ol	7	8	1	3
5	(3Z)-dodecen-1-ol	9	9	0	2
5	(3Z,6Z,8E)-dodecatrien-1-ol	5	15	10	12
5	<i>n</i> -eicosane	6	12	6	8
5	1-octadecanol	9	16	7	9
5	<i>n</i> -docosane	8	10	2	4
5	dodec-3-yn-1-ol	6	9	3	5
5	neocembrene	1	5	4	6
5	δ-cadinene	5	0	-5	-3
5	(3R,6E)-nerolidol	4	2	-2	0
5	geranylgeraniol	3	1	-2	0

Table S1.

SSR responses to Panel 1 for PsimOR9.

Related to [Fig. 2](#).

Replicate	Compound	Pre-stimulation spikes	Post-stimulation spikes	Generated spikes	Δ Spikes/s
1	hexane	7	3	-4	
1	1-octadecanol	10	6	-4	0
1	(3Z,6Z)-dodecadien-1-ol	6	7	1	5
1	(3Z)-dodecen-1-ol	16	8	-8	-4
1	dodec-3-yn-1-ol	8	18	10	14
1	(3Z,6Z,8E)-dodecatrien-1-ol	13	12	-1	3
1	(3R,6E)-nerolidol	15	8	-7	-3
1	geranylgeraniol	4	22	18	22
1	<i>n</i> -docosane	13	10	-3	1
1	<i>n</i> -eicosane	10	8	-2	2
1	neocembrene	14	62	48	52
1	δ-cadinene	13	22	9	13
2	hexane	8	9	1	
2	1-octadecanol	5	3	-2	-3
2	(3Z,6Z)-dodecadien-1-ol	7	3	-4	-5
2	(3Z)-dodecen-1-ol	13	3	-10	-11
2	dodec-3-yn-1-ol	9	3	-6	-7
2	(3Z,6Z,8E)-dodecatrien-1-ol	13	3	-10	-11
2	(3R,6E)-nerolidol	6	5	-1	-2
2	geranylgeraniol	4	38	34	33
2	<i>n</i> -docosane	8	2	-6	-7
2	<i>n</i> -eicosane	7	2	-5	-6
2	neocembrene	4	55	51	50
2	δ-cadinene	4	5	1	0
3	hexane	17	15	-2	
3	1-octadecanol	11	6	-5	-3
3	(3Z,6Z)-dodecadien-1-ol	10	13	3	5
3	(3Z)-dodecen-1-ol	17	7	-10	-8
3	dodec-3-yn-1-ol	10	7	-3	-1
3	(3Z,6Z,8E)-dodecatrien-1-ol	18	5	-13	-11
3	(3R,6E)-nerolidol	17	14	-3	-1
3	geranylgeraniol	15	41	26	28
3	<i>n</i> -docosane	20	12	-8	-6
3	<i>n</i> -eicosane	13	4	-9	-7
3	neocembrene	18	90	72	74
3	δ-cadinene	12	6	-6	-4
4	hexane	6	7	1	
4	1-octadecanol	15	3	-12	-13
4	(3Z,6Z)-dodecadien-1-ol	7	7	0	-1
4	(3Z)-dodecen-1-ol	14	5	-9	-10
4	dodec-3-yn-1-ol	16	9	-7	-8
4	(3Z,6Z,8E)-dodecatrien-1-ol	12	6	-6	-7
4	(3R,6E)-nerolidol	12	6	-6	-7
4	geranylgeraniol	2	28	26	25
4	<i>n</i> -docosane	16	4	-12	-13
4	<i>n</i> -eicosane	7	1	-6	-7
4	neocembrene	12	57	45	44
4	δ-cadinene	5	7	2	1
5	hexane	8	6	-2	
5	1-octadecanol	8	3	-5	-3
5	(3Z,6Z)-dodecadien-1-ol	6	3	-3	-1
5	(3Z)-dodecen-1-ol	2	5	3	5
5	dodec-3-yn-1-ol	8	6	4	6
5	(3Z,6Z,8E)-dodecatrien-1-ol	10	4	-4	-2
5	(3R,6E)-nerolidol	10	8	-2	0
5	geranylgeraniol	6	20	14	16
5	<i>n</i> -docosane	6	7	1	3
5	<i>n</i> -eicosane	6	4	-2	0
5	neocembrene	5	49	44	46
5	δ-cadinene	10	7	-3	-1

Table S2.

SSR responses to Panel 1 for PsimOR14.

Related to [Fig. 2](#).

Replicate	Compound	Pre-stimulation spikes	Post-stimulation spikes	Generated spikes	Δ Spikes/s
1	hexane	1	0	-1	
1	(3Z,6Z)-dodecadien-1-ol	1	5	4	5
1	(3Z)-dodecen-1-ol	7	3	-4	-3
1	(3Z,6Z,8E)-dodecatrien-1-ol	4	6	2	3
1	<i>n</i> -eicosane	4	8	4	5
1	1-octadecanol	5	6	1	2
1	<i>n</i> -docosane	5	6	1	2
1	dodec-3-yn-1-ol	4	2	-2	-1
1	neocembrene	5	3	-2	-1
1	δ -cadinene	5	3	-2	-1
1	(3R,6E)-nerolidol	2	3	1	2
1	geranylgeraniol	4	2	-2	-1
2	hexane	4	4	0	
2	(3Z,6Z)-dodecadien-1-ol	5	6	1	1
2	(3Z)-dodecen-1-ol	5	8	3	3
2	(3Z,6Z,8E)-dodecatrien-1-ol	7	7	0	0
2	<i>n</i> -eicosane	7	7	0	0
2	1-octadecanol	9	6	-3	-3
2	<i>n</i> -docosane	5	11	6	6
2	dodec-3-yn-1-ol	9	7	-2	-2
2	neocembrene	9	11	2	2
2	δ -cadinene	5	7	2	2
2	(3R,6E)-nerolidol	11	8	-3	-3
2	geranylgeraniol	10	7	-3	-3
3	hexane	8	5	-3	
3	(3Z,6Z)-dodecadien-1-ol	7	5	-2	1
3	(3Z)-dodecen-1-ol	6	5	-1	2
3	(3Z,6Z,8E)-dodecatrien-1-ol	9	5	-4	-1
3	<i>n</i> -eicosane	2	10	8	11
3	1-octadecanol	7	7	0	3
3	<i>n</i> -docosane	8	8	0	3
3	dodec-3-yn-1-ol	3	7	4	7
3	neocembrene	7	11	4	7
3	δ -cadinene	8	7	-1	2
3	(3R,6E)-nerolidol	7	3	-4	-1
3	geranylgeraniol	5	10	5	8

Table S3.

SSR responses to Panel 1 for PsimOR30.

Related to [Fig. 2](#).

Replicate	Compound	Pre-stimulation spikes	Post-stimulation spikes	Generated spikes	Δ Spikes/s
1	hexane	5	6	1	
1	(3Z,6Z)-dodecadien-1-ol	5	24	19	18
1	(3Z)-dodecen-1-ol	5	25	20	19
1	(3Z,6Z,8E)-dodecatrien-1-ol	3	27	24	23
1	<i>n</i> -eicosane	3	25	22	21
1	1-octadecanol	5	25	20	19
1	<i>n</i> -docosane	6	27	21	20
1	dodec-3-yn-1-ol	4	28	24	23
1	neocembrene	6	19	13	12
1	δ-cadinene	3	19	16	15
1	(3R,6E)-nerolidol	5	16	11	10
1	geranylgeraniol	3	10	7	6
2	hexane	3	2	-1	
2	(3Z,6Z)-dodecadien-1-ol	5	5	0	1
2	(3Z)-dodecen-1-ol	4	9	5	6
2	(3Z,6Z,8E)-dodecatrien-1-ol	2	7	5	6
2	<i>n</i> -eicosane	5	10	5	6
2	1-octadecanol	6	6	0	1
2	<i>n</i> -docosane	5	4	-1	0
2	dodec-3-yn-1-ol	7	6	-1	0
2	neocembrene	2	5	3	4
2	δ-cadinene	2	10	8	9
2	(3R,6E)-nerolidol	2	2	0	1
3	hexane	4	3	-1	
3	(3Z,6Z)-dodecadien-1-ol	9	7	-2	-1
3	(3Z)-dodecen-1-ol	3	14	11	12
3	(3Z,6Z,8E)-dodecatrien-1-ol	8	14	6	7
3	<i>n</i> -eicosane	9	21	12	13
3	1-octadecanol	6	21	15	16
3	<i>n</i> -docosane	6	27	21	22
3	dodec-3-yn-1-ol	10	18	8	9
3	neocembrene	9	17	8	9
3	δ-cadinene	7	20	13	14
3	(3R,6E)-nerolidol	6	14	8	9
3	geranylgeraniol	10	21	11	12
4	hexane	13	15	2	
4	(3Z,6Z)-dodecadien-1-ol	20	17	-3	-5
4	(3Z)-dodecen-1-ol	15	19	4	2
4	(3Z,6Z,8E)-dodecatrien-1-ol	14	21	7	5
4	<i>n</i> -eicosane	18	22	4	2
4	1-octadecanol	19	22	3	1
4	<i>n</i> -docosane	16	20	4	2
4	dodec-3-yn-1-ol	11	23	12	10
4	neocembrene	8	22	14	12
4	δ-cadinene	10	24	14	12
4	(3R,6E)-nerolidol	16	21	5	3
4	geranylgeraniol	12	21	9	7
5	hexane	11	10	-1	
5	(3Z,6Z)-dodecadien-1-ol	17	18	1	2
5	(3Z)-dodecen-1-ol	5	16	11	12
5	(3Z,6Z,8E)-dodecatrien-1-ol	8	12	4	5
5	<i>n</i> -eicosane	17	14	-3	-2
5	1-octadecanol	7	13	6	7
5	<i>n</i> -docosane	11	16	5	6
5	dodec-3-yn-1-ol	10	15	5	6
5	neocembrene	0	15	15	16
6	hexane	24	25	1	
6	(3Z,6Z)-dodecadien-1-ol	21	46	25	24
6	(3Z)-dodecen-1-ol	18	44	26	25
6	(3Z,6Z,8E)-dodecatrien-1-ol	14	44	30	29
6	<i>n</i> -eicosane	17	43	26	25
6	1-octadecanol	10	44	34	33
6	<i>n</i> -docosane	12	38	26	25
6	dodec-3-yn-1-ol	17	47	30	29
6	neocembrene	11	32	21	20
6	δ-cadinene	19	27	8	7
6	(3R,6E)-nerolidol	7	26	19	18
6	geranylgeraniol	9	19	10	9

Table S4.

SSR responses to Panel 1 for PsimOR31.

Related to [Fig. 2](#).

Table S5.

SSR responses to Panel 1 for PsimOR14 vs. W¹¹¹⁸.

Related to Fig. 2.

Fly line	Replicate	Compound	Pre-stimulation spikes	Post-stimulation spikes	Generated spikes	Δ Spikes/s
PsimOR14	1	hexane	7	3	-4	
PsimOR14	1	(3Z,6Z)-dodecadien-1-ol	6	7	1	5
PsimOR14	1	(3Z,6Z,8E)-dodecatrien-1-ol	16	8	-8	-4
PsimOR14	1	(3Z,6Z,8E)-dodecatrien-1-ol	13	12	-1	3
PsimOR14	1	n-eicosane	10	8	-2	2
PsimOR14	1	1-octadecanol	10	6	-4	0
PsimOR14	1	n-docosane	13	10	-3	1
PsimOR14	1	dodec-3-yn-1-ol	8	18	10	14
PsimOR14	1	neocembrene	14	62	48	52
PsimOR14	1	δ-cadinene	13	22	9	13
PsimOR14	1	(3R,6E)-nerolidol	15	8	-7	-3
PsimOR14	1	geranylgeraniol	4	22	18	22
PsimOR14	2	hexane	8	9	1	
PsimOR14	2	(3Z,6Z)-dodecadien-1-ol	7	3	-4	-5
PsimOR14	2	(3Z,6Z,8E)-dodecatrien-1-ol	13	3	-10	-11
PsimOR14	2	n-eicosane	7	2	-5	-6
PsimOR14	2	1-octadecanol	5	3	-2	-3
PsimOR14	2	n-docosane	8	2	-6	-7
PsimOR14	2	dodec-3-yn-1-ol	9	3	-6	-7
PsimOR14	2	neocembrene	4	55	51	50
PsimOR14	2	δ-cadinene	4	5	1	0
PsimOR14	2	(3R,6E)-nerolidol	6	5	-1	-2
PsimOR14	2	geranylgeraniol	4	38	34	33
PsimOR14	3	hexane	17	15	-2	
PsimOR14	3	(3Z,6Z)-dodecadien-1-ol	10	13	3	5
PsimOR14	3	(3Z,6Z,8E)-dodecatrien-1-ol	17	7	-10	-8
PsimOR14	3	(3Z,6Z,8E)-dodecatrien-1-ol	18	5	-13	-11
PsimOR14	3	n-eicosane	13	4	-9	-7
PsimOR14	3	1-octadecanol	11	6	-5	-3
PsimOR14	3	n-docosane	20	12	-8	-6
PsimOR14	3	dodec-3-yn-1-ol	10	7	-3	-1
PsimOR14	3	neocembrene	18	90	72	74
PsimOR14	3	δ-cadinene	12	6	-6	-4
PsimOR14	3	(3R,6E)-nerolidol	17	14	-3	-1
PsimOR14	3	geranylgeraniol	15	41	26	28
PsimOR14	4	hexane	6	7	1	
PsimOR14	4	neocembrene	12	57	45	44
PsimOR14	4	geranylgeraniol	2	28	26	25
PsimOR14	4	(3Z,6Z)-dodecadien-1-ol	7	7	0	-1
PsimOR14	4	(3Z,6Z,8E)-dodecatrien-1-ol	14	5	-9	-10
PsimOR14	4	(3Z,6Z,8E)-dodecatrien-1-ol	12	6	-6	-7
PsimOR14	4	n-eicosane	7	1	-6	-7
PsimOR14	4	1-octadecanol	15	3	-12	-13
PsimOR14	4	n-docosane	16	4	-12	-13
PsimOR14	4	dodec-3-yn-1-ol	16	9	-7	-8
PsimOR14	4	δ-cadinene	5	7	2	1
PsimOR14	4	(3R,6E)-nerolidol	12	6	-6	-7
PsimOR14	5	hexane	8	6	-2	
PsimOR14	5	neocembrene	5	49	44	46
PsimOR14	5	geranylgeraniol	6	20	14	16
PsimOR14	5	(3Z,6Z)-dodecadien-1-ol	6	3	-3	-1
PsimOR14	5	(3Z,6Z,8E)-dodecatrien-1-ol	2	5	3	5
PsimOR14	5	(3Z,6Z,8E)-dodecatrien-1-ol	8	4	-4	-2
PsimOR14	5	n-eicosane	6	4	-2	0
PsimOR14	5	1-octadecanol	8	3	-5	-3
PsimOR14	5	n-docosane	6	7	1	3
PsimOR14	5	dodec-3-yn-1-ol	2	6	4	6
PsimOR14	5	δ-cadinene	10	7	-3	-1
PsimOR14	5	(3R,6E)-nerolidol	10	8	-2	0
W1118	1	hexane	3	6	3	
W1118	1	(3Z,6Z)-dodecadien-1-ol	1	3	2	-1
W1118	1	(3Z,6Z,8E)-dodecatrien-1-ol	2	4	2	-1
W1118	1	(3Z,6Z,8E)-dodecatrien-1-ol	4	1	-3	-6
W1118	1	n-eicosane	1	19	18	15
W1118	1	1-octadecanol	4	2	-2	-5
W1118	1	n-docosane	2	6	4	1
W1118	1	dodec-3-yn-1-ol	2	5	3	0
W1118	1	neocembrene	2	2	0	-3
W1118	1	δ-cadinene	1	4	3	0
W1118	1	(3R,6E)-nerolidol	3	1	-2	-5
W1118	1	geranylgeraniol	3	4	1	-2
W1118	2	hexane	1	4	3	
W1118	2	(3Z,6Z)-dodecadien-1-ol	2	4	2	-1
W1118	2	(3Z,6Z,8E)-dodecatrien-1-ol	5	3	-2	-5
W1118	2	(3Z,6Z,8E)-dodecatrien-1-ol	5	3	-2	-5
W1118	2	n-eicosane	3	3	0	-3
W1118	2	1-octadecanol	3	2	-1	-4
W1118	2	n-docosane	0	5	5	2
W1118	2	dodec-3-yn-1-ol	3	3	0	-3
W1118	2	neocembrene	4	4	0	-3
W1118	2	δ-cadinene	6	0	-6	-9
W1118	2	(3R,6E)-nerolidol	1	3	2	-1
W1118	2	geranylgeraniol	3	4	1	-2
W1118	3	hexane	5	1	-4	
W1118	3	(3Z,6Z)-dodecadien-1-ol	5	5	0	4
W1118	3	(3Z,6Z,8E)-dodecatrien-1-ol	5	2	-3	1
W1118	3	(3Z,6Z,8E)-dodecatrien-1-ol	2	4	2	6
W1118	3	n-eicosane	3	5	2	6
W1118	3	1-octadecanol	3	5	2	6
W1118	3	n-docosane	4	5	1	5
W1118	3	dodec-3-yn-1-ol	6	5	-1	3
W1118	3	neocembrene	1	5	4	8
W1118	3	δ-cadinene	4	4	0	4
W1118	3	(3R,6E)-nerolidol	6	3	-3	1
W1118	3	geranylgeraniol	4	2	-2	2
W1118	4	hexane	4	4	0	
W1118	4	(3Z,6Z)-dodecadien-1-ol	1	6	5	5
W1118	4	(3Z,6Z,8E)-dodecatrien-1-ol	3	3	0	0
W1118	4	(3Z,6Z,8E)-dodecatrien-1-ol	6	4	-2	-2
W1118	4	n-eicosane	4	7	3	3
W1118	4	1-octadecanol	5	7	2	2
W1118	4	n-docosane	4	9	5	5
W1118	4	dodec-3-yn-1-ol	0	4	4	4
W1118	4	neocembrene	2	0	-2	-2
W1118	4	δ-cadinene	5	2	-3	-3
W1118	4	(3R,6E)-nerolidol	4	1	-3	-3
W1118	4	geranylgeraniol	2	0	-2	-2
W1118	5	hexane	3	8	5	0
W1118	5	(3Z,6Z)-dodecadien-1-ol	6	11	5	0
W1118	5	(3Z,6Z,8E)-dodecatrien-1-ol	9	7	-2	-7
W1118	5	(3Z,6Z,8E)-dodecatrien-1-ol	6	9	3	-2
W1118	5	n-eicosane	7	14	7	2
W1118	5	1-octadecanol	4	13	9	4
W1118	5	n-docosane	8	7	-1	-6
W1118	5	dodec-3-yn-1-ol	7	8	1	-4
W1118	5	neocembrene	6	8	2	-3
W1118	5	δ-cadinene	5	8	3	-2
W1118	5	(3R,6E)-nerolidol	6	6	0	-5
W1118	5	geranylgeraniol	6	6	0	-5

Table S6.

SSR dose response data for neocembrene and PsimOR14 fly line.

Related to Fig. 2.

	replicate	1	2	3	4	5	6	7	8	9
Dose	0.01 ng	22	10	10	6	-4	10	18	16	24
	0.1 ng	4	26	26	8	0	0	4	38	12
	1 ng	14	32	14	10	12	2	22	28	10
	10 ng	10	30	32	26	26	14	20	32	6
	100 ng	30		24	26	42	38	46	26	10
	500 ng	44		36	22	34				

Panel	Replicate	Compound	Pre-stimulation spikes	Post-stimulation spikes	Generated spikes	Δ Spikes/s
2	1	paraffin oil	6	2	-4	
2	1	myrcene	10	15	5	9
2	1	eucalyptol	2	2	0	4
2	1	(+)-3-carene	2	5	3	7
2	1	ethanol	2	6	4	8
2	1	(2E)-hexenal	3	5	2	6
2	1	(+)-α-pinene	3	2	-1	3
2	1	p-cymene	3	2	-1	3
2	1	(+)-limonene	4	3	-1	3
2	1	(+)-longifolene	3	2	-1	3
2	1	(-)-β-caryophyllene	1	2	1	5
2	1	(2E)-hexen-1-ol	1	0	-1	3
2	1	isoamyl propionate	0	0	0	4
2	1	sabinene	2	1	-1	3
2	1	γ-terpinene	0	2	2	6
2	1	terpinolene	2	0	-2	2
2	1	toluene	1	5	4	8
2	1	pentyl acetate	1	3	2	6
2	1	ethyl acetate	2	1	-1	3
2	1	heptanal	0	0	0	4
2	1	γ-nonolactone	2	0	-2	2
2	2	paraffin oil	0	1	1	
2	2	myrcene	6	11	5	4
2	2	eucalyptol	0	13	13	12
2	2	(+)-3-carene	6	10	4	3
2	2	ethanol	2	12	10	9
2	2	(2E)-hexenal	6	11	5	4
2	2	(+)-α-pinene	0	10	10	9
2	2	p-cymene	8	12	4	3
2	2	(+)-limonene	2	11	9	8
2	2	(+)-longifolene	1	10	9	8
2	2	(-)-β-caryophyllene	5	7	2	1
2	2	(2E)-hexen-1-ol	8	12	4	3
2	2	isoamyl propionate	8	0	-1	
2	2	sabinene	6	16	10	9
2	2	γ-terpinene	9	7	-2	-3
2	2	terpinolene	5	10	5	4
2	2	toluene	8	8	0	-1
2	2	pentyl acetate	7	9	2	1
2	2	ethyl acetate	8	9	1	0
2	2	heptanal	10	9	-1	-2
2	2	γ-nonolactone	6	12	6	5
2	3	paraffin oil	5	6	1	
2	3	myrcene	2	6	4	3
2	3	eucalyptol	2	10	8	7
2	3	(+)-3-carene	0	14	14	13
2	3	ethanol	2	6	4	3
2	3	(2E)-hexenal	2	12	10	9
2	3	(+)-α-pinene	2	9	7	6
2	3	p-cymene	3	4	1	0
2	3	(+)-limonene	1	1	0	-1
2	3	(+)-longifolene	0	10	10	9
2	3	(-)-β-caryophyllene	0	10	10	9
2	3	(2E)-hexen-1-ol	4	1	-3	-4
2	3	(2E)-hexen-1-ol	3	1	-2	-3
2	3	sabinene	1	2	1	0
2	3	γ-terpinene	9	5	-4	-5
2	3	terpinolene	1	9	8	7
2	3	toluene	2	0	-2	-3
2	3	pentyl acetate	1	9	8	7
2	3	ethyl acetate	4	0	-4	-5
2	3	heptanal	1	4	3	2
2	3	γ-nonolactone	2	2	0	-1
2	4	paraffin oil	4	5	1	0
2	4	myrcene	3	1	-2	-3
2	4	eucalyptol	0	7	7	6
2	4	(+)-3-carene	2	5	3	2
2	4	ethanol	1	4	3	2
2	4	(2E)-hexenal	0	3	3	2
2	4	(+)-α-pinene	3	12	9	8
2	4	p-cymene	1	11	10	9
2	4	(+)-limonene	0	3	3	2
2	4	(+)-longifolene	0	8	8	7
2	4	(-)-β-caryophyllene	0	2	2	1
2	4	(2E)-hexen-1-ol	0	5	5	4
2	4	isoamyl propionate	0	5	5	4
2	4	sabinene	2	2	0	-1
2	4	γ-terpinene	3	9	6	5
2	4	terpinolene	7	4	-3	-4
2	4	toluene	2	2	0	-1
2	4	pentyl acetate	3	2	-1	-2
2	4	ethyl acetate	1	0	-1	-1
2	4	heptanal	3	1	-2	-3
2	4	paraffin oil	6	7	1	0
2	5	myrcene	11	11	0	-1
2	5	eucalyptol	7	13	6	5
2	5	(+)-3-carene	12	11	-1	-2
2	5	ethanol	11	9	-2	-3
2	5	(2E)-hexenal	11	13	2	1
2	5	(+)-α-pinene	12	11	-1	-2
2	5	p-cymene	11	16	5	4
2	5	(+)-limonene	9	9	0	-1
2	5	(+)-longifolene	17	9	-8	-9
2	5	(-)-β-caryophyllene	6	13	7	6
2	5	(2E)-hexen-1-ol	9	3	-6	-7
2	5	isoamyl propionate	0	0	0	-1
2	5	sabinene	6	19	13	12
2	5	γ-terpinene	3	13	10	9
2	5	terpinolene	9	15	6	5
2	5	toluene	10	13	3	2
2	5	pentyl acetate	6	11	5	4
2	5	ethyl acetate	8	19	11	10
2	5	heptanal	14	7	-7	-8
2	5	γ-nonolactone	10	14	4	3

Table S7.

SSR responses to Panel 2 for PsimOR14.

Related to [Fig. 3](#).

Panel	Replicate	Compound	Pre-stimulation spikes	Post-stimulation spikes	Generated spikes	Δ Spikes/s
3	1	paraffin oil	4	5	1	
3	1	ethyl (2E,4Z)-decadienoate	4	6	2	1
3	1	p-cresol	2	9	7	6
3	1	β -bisabolene	7	2	-5	-6
3	1	(E)-myrcenol	1	6	5	4
3	1	nonan-1-ol	1	8	7	6
3	1	isoamyl acetate	5	11	6	5
3	1	2-phenylethyl acetate	3	9	6	5
3	1	octan-1-ol	6	12	6	5
3	1	benzyl alcohol	12	11	-1	-2
3	1	methyl jasmonate	8	13	5	4
3	1	(E)- β -ocimene	6	14	8	7
3	1	eucarvone	10	4	-6	-7
3	1	α -camphorene	2	14	12	11
3	1	(-)-trans-pinocarveol	6	4	-2	-3
3	1	cryptone	8	8	0	-1
3	1	(+)-cis-carveol	12	17	5	4
3	2	paraffin oil	9	5	-4	
3	2	ethyl (2E,4Z)-decadienoate	19	11	-8	-4
3	2	p-cresol	15	15	0	4
3	2	β -bisabolene	8	14	6	10
3	2	(E)-myrcenol	17	16	-1	3
3	2	nonan-1-ol	17	14	-3	1
3	2	isoamyl acetate	20	16	-4	0
3	2	2-phenylethyl acetate	13	7	-6	-2
3	2	octan-1-ol	10	3	-7	-3
3	2	benzyl alcohol	3	2	-1	3
3	2	methyl jasmonate	0	1	1	5
3	2	(E)- β -ocimene	0	2	2	6
3	2	eucarvone	0	1	1	5
3	2	α -camphorene	0	1	1	5
3	2	(-)-trans-pinocarveol	0	1	1	5
3	2	cryptone	0	0	0	4
3	2	(+)-cis-carveol	0	0	0	4
3	3	paraffin oil	0	1	1	
3	3	ethyl (2E,4Z)-decadienoate	4	8	4	3
3	3	p-cresol	2	6	4	3
3	3	β -bisabolene	12	3	-9	-10
3	3	(E)-myrcenol	6	10	4	3
3	3	nonan-1-ol	5	7	2	1
3	3	isoamyl acetate	4	9	5	4
3	3	2-phenylethyl acetate	6	9	3	2
3	3	octan-1-ol	0	14	14	13
3	3	benzyl alcohol	8	7	-1	-2
3	3	methyl jasmonate	9	11	2	1
3	3	(E)- β -ocimene	2	4	2	1
3	3	eucarvone	10	1	-9	-10
3	3	α -camphorene	7	7	0	-1
3	3	(-)-trans-pinocarveol	5	0	-5	-6
3	3	cryptone	5	8	3	2
3	3	(+)-cis-carveol	5	6	1	0
3	4	paraffin oil	2	7	5	
3	4	ethyl (2E,4Z)-decadienoate	1	0	-1	-6
3	4	p-cresol	6	10	4	-1
3	4	β -bisabolene	3	9	6	1
3	4	(E)-myrcenol	2	9	7	2
3	4	nonan-1-ol	10	4	-6	-11
3	4	isoamyl acetate	1	6	5	0
3	4	2-phenylethyl acetate	0	0	0	-5
3	4	octan-1-ol	2	0	-2	-7
3	4	benzyl alcohol	1	5	4	-1
3	4	methyl jasmonate	10	3	-7	-12
3	4	(E)- β -ocimene	0	5	5	0
3	4	eucarvone	3	1	-2	-7
3	4	α -camphorene	1	0	-1	-6
3	4	(-)-trans-pinocarveol	9	5	-4	-9
3	4	cryptone	9	6	-3	-8
3	4	(+)-cis-carveol	1	0	-1	-6
3	5	paraffin oil	4	1	-3	
3	5	ethyl (2E,4Z)-decadienoate	1	4	3	6
3	5	p-cresol	0	0	0	3
3	5	β -bisabolene	0	15	15	18
3	5	(E)-myrcenol	5	5	0	3
3	5	nonan-1-ol	0	1	1	4
3	5	isoamyl acetate	4	1	-3	0
3	5	2-phenylethyl acetate	3	3	0	3
3	5	octan-1-ol	9	6	-3	0
3	5	benzyl alcohol	5	0	-5	-2
3	5	methyl jasmonate	0	1	1	4
3	5	(E)- β -ocimene	2	0	-2	1
3	5	eucarvone	9	7	-2	1
3	5	α -camphorene	11	4	-7	-4
3	5	(-)-trans-pinocarveol	0	0	0	3
3	5	cryptone	3	2	-1	2
3	5	(+)-cis-carveol	4	1	-3	0

Table S8.

SSR responses to Panel 3 for PsimOR14.

Related to [Fig. 3](#).

Panel	Replicate	Compound	Pre-stimulation spikes	Post-stimulation spikes	Generated spikes	Δ Spikes/s
4	1	paraffin oil	6	1	-5	
4	1	(-)-verbenone	14	4	-10	-5
4	1	oct-1-en-3-ol	4	2	-2	3
4	1	acetophenone	12	20	8	13
4	1	4-vinylanisole	12	14	2	7
4	1	4-ethylguaiacol	6	2	-4	1
4	1	(±)-camphor	10	6	-4	1
4	1	octan-1-ol	16	6	-10	-5
4	1	methyl Eugenol	18	2	-16	-11
4	1	styrene	16	14	-2	3
4	1	benzaldehyde	20	6	-14	-9
4	1	linalool	21	0	-21	-16
4	1	4-methylanisole	20	6	-14	-9
4	1	2-methylbutan-1-ol	4	0	-4	1
4	1	isoamyl alcohol	12	0	-12	-7
4	1	2-methylbutyl acetate	18	14	-4	1
4	1	(±)-myrtenol	7	2	-5	0
4	2	paraffin oil	7	4	-3	
4	2	(-)-verbenone	2	12	10	13
4	2	oct-1-en-3-ol	7	8	1	4
4	2	acetophenone	4	10	6	9
4	2	4-vinylanisole	8	5	-3	0
4	2	4-ethylguaiacol	4	10	6	7
4	2	(±)-camphor	4	11	7	10
4	2	octan-1-ol	13	8	-5	-2
4	2	methyl Eugenol	5	14	9	12
4	2	styrene	10	6	-4	-1
4	2	benzaldehyde	7	11	4	7
4	2	linalool	5	5	0	3
4	2	4-methylanisole	10	4	-6	-3
4	2	2-methylbutan-1-ol	8	6	-2	1
4	2	isoamyl alcohol	11	12	1	4
4	2	2-methylbutyl acetate	10	10	0	3
4	2	(±)-myrtenol	7	9	2	5
4	2	2-phenylethanol	4	8	4	7
4	2	hexan-1-ol	9	4	-5	-2
4	2	2,3-dihydrobenzofuran	9	4	-5	-2
4	2	geranyl acetone	8	6	-2	1
4	3	paraffin oil	4	2	-2	
4	3	(-)-verbenone	12	8	-4	-2
4	3	oct-1-en-3-ol	7	5	-2	0
4	3	acetophenone	2	10	8	10
4	3	4-vinylanisole	8	12	4	6
4	3	4-ethylguaiacol	1	11	10	12
4	3	(±)-camphor	1	10	9	11
4	3	octan-1-ol	9	6	-3	-1
4	3	methyl Eugenol	9	6	-3	-1
4	3	styrene	3	5	2	4
4	3	benzaldehyde	11	6	-5	-3
4	3	linalool	10	1	-9	-7
4	3	4-methylanisole	3	9	6	8
4	3	2-methylbutan-1-ol	7	2	-5	-2
4	3	isoamyl alcohol	10	9	-1	1
4	3	2-methylbutyl acetate	10	6	-4	-2
4	3	(±)-myrtenol	5	3	-2	0
4	3	2-phenylethanol	2	8	6	8
4	3	hexan-1-ol	7	4	-3	-1
4	3	2,3-dihydrobenzofuran	6	9	3	5
4	3	geranyl acetone	6	9	3	5
4	4	paraffin oil	10	6	-4	
4	4	(-)-verbenone	11	5	-6	-2
4	4	oct-1-en-3-ol	7	8	1	5
4	4	acetophenone	13	3	-10	-6
4	4	4-vinylanisole	8	8	0	4
4	4	4-ethylguaiacol	7	6	-1	3
4	4	(±)-camphor	1	1	0	4
4	4	octan-1-ol	7	7	0	4
4	4	methyl Eugenol	14	7	-7	-3
4	4	styrene	11	6	-5	-1
4	4	benzaldehyde	5	14	9	13
4	4	linalool	8	0	-8	-4
4	4	4-methylanisole	7	5	-2	2
4	4	2-methylbutan-1-ol	5	0	-5	-1
4	4	isoamyl alcohol	9	3	-6	-2
4	4	2-methylbutyl acetate	6	9	3	7
4	4	(±)-myrtenol	3	6	3	7
4	4	2-phenylethanol	6	5	-1	3
4	4	hexan-1-ol	9	8	-1	3
4	4	2,3-dihydrobenzofuran	6	4	-2	3
4	4	geranyl acetone	10	7	-3	1
4	5	paraffin oil	8	7	-1	
4	5	(-)-verbenone	6	3	-3	-2
4	5	oct-1-en-3-ol	3	1	-2	-1
4	5	acetophenone	6	2	-4	-3
4	5	4-vinylanisole	5	3	-2	-1
4	5	4-ethylguaiacol	9	5	-4	-3
4	5	(±)-camphor	8	5	-3	-2
4	5	octan-1-ol	6	4	-2	-1
4	5	methyl Eugenol	7	8	1	2
4	5	styrene	7	6	-1	0
4	5	benzaldehyde	10	5	-5	-4
4	5	linalool	9	2	-7	-6
4	5	4-methylanisole	10	9	-1	0
4	5	2-methylbutan-1-ol	10	4	-6	-5
4	5	isoamyl alcohol	10	7	-3	-2
4	5	2-methylbutyl acetate	10	8	-2	-1
4	5	(±)-myrtenol	6	5	-1	0
4	5	2-phenylethanol	8	7	-1	0
4	5	hexan-1-ol	9	5	-4	-3
4	5	2,3-dihydrobenzofuran	7	6	-1	0
4	5	geranyl acetone	9	6	-3	-2
4	6	paraffin oil	9	10	1	
4	6	(-)-verbenone	7	4	-3	-4
4	6	oct-1-en-3-ol	6	8	2	1
4	6	acetophenone	12	6	-6	-7
4	6	4-vinylanisole	7	6	-1	-2
4	6	4-ethylguaiacol	8	5	-3	-4
4	6	(±)-camphor	11	14	3	2
4	6	octan-1-ol	13	5	-8	-9
4	6	methyl Eugenol	5	6	1	0
4	6	styrene	9	6	-3	-4
4	6	benzaldehyde	6	7	1	0
4	6	linalool	11	1	-10	-11
4	6	4-methylanisole	9	8	-1	-2
4	6	2-methylbutan-1-ol	10	4	-6	-7
4	6	isoamyl alcohol	7	7	0	-1
4	6	2-methylbutyl acetate	6	7	1	0
4	6	(±)-myrtenol	7	6	-1	-2
4	6	2-phenylethanol	10	8	-2	-3
4	6	hexan-1-ol	11	6	-5	-6
4	6	2,3-dihydrobenzofuran	12	9	-3	-4
4	6	geranyl acetone	12	9	-3	-4

Table S9.

SSR responses to Panel 4 for PsimOR14.

Related to [Fig. 3](#).

Table S10.

SSR responses to Panel 1 by neocembrene sensillum in *P. simplex* workers.Related to Fig. 5 [↗](#).

Replicate	Compound	Pre-stimulation spikes	Post-stimulation spikes	Generated spikes	Δ Spikes/s
1	hexane	8	8	0	
1	(3Z,6Z)-dodecadien-1-ol	5	8	3	6
1	(3Z)-dodecen-1-ol	12	21	9	18
1	(3Z,6Z,8E)-dodecatrien-1-ol	10	13	3	6
1	n-eicosane	6	8	2	4
1	1-octadecanol	5	8	3	6
1	n-docosane	3	5	2	4
1	(3R,6E)-nerolidol	4	46	12	24
1	neocembrene	11	34	23	46
1	δ-cadinene	9	12	3	6
1	dodec-3-yn-1-ol	5	6	1	2
1	geranylgeraniol	4	12	8	16
2	hexane	18	18	0	
2	(3Z,6Z)-dodecadien-1-ol	9	11	2	4
2	(3Z)-dodecen-1-ol	2	5	3	6
2	(3Z,6Z,8E)-dodecatrien-1-ol	8	6	-2	-4
2	neocembrene	2	41	38	76
2	geranylgeraniol	2	11	9	18
3	hexane	6	6	0	
3	(3Z,6Z)-dodecadien-1-ol	10	12	2	4
3	(3Z)-dodecen-1-ol	15	9	-6	-12
3	(3Z,6Z,8E)-dodecatrien-1-ol	9	11	2	4
3	neocembrene	10	46	36	72
3	geranylgeraniol	6	16	10	20
3	δ-cadinene	12	13	1	2
4	hexane	5	6	1	2
4	(3Z,6Z)-dodecadien-1-ol	10	12	2	4
4	(3Z)-dodecen-1-ol	2	2	0	0
4	(3Z,6Z,8E)-dodecatrien-1-ol	9	2	-8	-16
4	neocembrene	7	45	38	74
4	geranylgeraniol	3	9	6	10
5	hexane	6	10	4	8
5	(3Z,6Z)-dodecadien-1-ol	7	13	6	12
5	(3Z)-dodecen-1-ol	6	15	9	18
5	(3Z,6Z,8E)-dodecatrien-1-ol	9	11	2	4
5	neocembrene	5	51	46	84
6	hexane	12	4	-8	-16
6	(3Z,6Z)-dodecadien-1-ol	7	14	7	14
6	(3Z)-dodecen-1-ol	11	7	-4	-8
6	(3Z,6Z,8E)-dodecatrien-1-ol	10	18	8	16
6	neocembrene	3	48	45	90
6	geranylgeraniol	9	14	5	10
7	hexane	36	33	-3	-6
7	(3Z,6Z)-dodecadien-1-ol	40	53	13	26
7	(3Z)-dodecen-1-ol	47	49	2	4
7	(3Z,6Z,8E)-dodecatrien-1-ol	55	64	9	18
7	n-eicosane	54	54	0	0
7	1-octadecanol	54	49	-5	-10
7	n-docosane	56	49	-7	-14
7	(3R,6E)-nerolidol	57	50	-7	-14
7	δ-cadinene	52	46	-6	-12
7	dodec-3-yn-1-ol	48	54	6	12
7	neocembrene	40	74	34	68
7	geranylgeraniol	17	40	23	46
8	hexane	34	32	-2	-4
8	neocembrene	18	46	28	56
8	geranylgeraniol	20	39	19	38
9	hexane	65	70	5	10
9	(3Z,6Z)-dodecadien-1-ol	28	33	5	10
9	(3Z)-dodecen-1-ol	37	36	-1	-2
9	(3Z,6Z,8E)-dodecatrien-1-ol	29	27	-2	-4
9	n-eicosane	36	42	6	12
9	1-octadecanol	35	34	-1	-2
9	n-docosane	34	35	1	2
9	(3R,6E)-nerolidol	19	30	11	22
9	neocembrene	22	54	32	64
9	δ-cadinene	19	15	-4	-8
9	dodec-3-yn-1-ol	35	31	-4	-8
9	geranylgeraniol	20	44	24	48

Replicate	Compound	Pre-stimulation spikes	Post-stimulation spikes	Generated spikes	Δ Spikes/s
10	hexane	17	22	5	10
10	neocembrene	20	39	19	38
10	geranylgeraniol	17	41	24	48
11	hexane	6	11	5	10
11	neocembrene	19	46	27	54
11	geranylgeraniol	24	34	10	20
12	hexane	28	26	-2	-4
12	neocembrene	31	56	25	50
13	hexane	41	37	-4	-8
13	neocembrene	17	38	21	42
13	geranylgeraniol	8	28	20	40
13	(3Z,6Z)-dodecadien-1-ol	15	13	-2	-4
13	(3Z)-dodecen-1-ol	18	20	2	4
13	(3Z,6Z,8E)-dodecatrien-1-ol	12	13	1	2
13	n-eicosane	15	18	3	6
13	1-octadecanol	7	11	4	8
13	n-docosane	7	11	4	8
13	(3R,6E)-nerolidol	27	19	-8	-16
13	δ-cadinene	27	23	-4	-8
13	dodec-3-yn-1-ol	18	30	12	24
14	hexane	25	21	-4	-8
14	neocembrene	18	68	50	100
14	geranylgeraniol	22	72	50	100
14	(3Z,6Z)-dodecadien-1-ol	28	15	-13	-26
14	(3Z)-dodecen-1-ol	32	26	-6	-12
14	(3Z,6Z,8E)-dodecatrien-1-ol	20	24	4	8
14	n-eicosane	21	17	-4	-8
14	1-octadecanol	30	23	-7	-14
14	n-docosane	27	24	-3	-6
14	(3R,6E)-nerolidol	7	9	2	4
14	δ-cadinene	13	11	-2	-4
14	dodec-3-yn-1-ol	8	8	0	0
15	hexane	9	8	-1	-2
15	(3Z,6Z)-dodecadien-1-ol	24	21	-3	-6
15	(3Z)-dodecen-1-ol	24	29	5	10
15	(3Z,6Z,8E)-dodecatrien-1-ol	13	14	1	2
15	n-eicosane	8	19	11	22
15	1-octadecanol	23	19	-4	-8
15	n-docosane	30	26	-4	-8
15	(3R,6E)-nerolidol	9	16	7	14
15	neocembrene	12	40	28	56
15	δ-cadinene	9	9	0	0
15	dodec-3-yn-1-ol	16	8	-8	-16
15	geranylgeraniol	10	24	14	28
16	hexane	13	7	-6	-12
16	neocembrene	18	38	20	40
16	geranylgeraniol	9	26	17	34
16	(3Z,6Z)-dodecadien-1-ol	15	12	-3	-6
16	(3Z)-dodecen-1-ol	12	25	13	26
16	(3Z,6Z,8E)-dodecatrien-1-ol	11	13	2	4
16	n-eicosane	15	18	3	6
16	1-octadecanol	7	13	6	12
16	n-docosane	9	10	1	2
16	(3R,6E)-nerolidol	7	13	6	12
16	δ-cadinene	12	13	1	2
16	dodec-3-yn-1-ol	8	18	10	20
17	hexane	16	14	-2	-4
17	neocembrene	7	32	25	50
17	geranylgeraniol	8	14	6	12
17	(3Z,6Z)-dodecadien-1-ol	10	7	-3	-6
17	(3Z)-dodecen-1-ol	18	8	-10	-20
17	(3Z,6Z,8E)-dodecatrien-1-ol	17	9	-8	-16
17	n-eicosane	7	8	1	2
17	1-octadecanol	17	9	-8	-16
17	n-docosane	14	10	-4	-8
17	(3R,6E)-nerolidol	8	8	0	0
17	δ-cadinene	9	14	5	10
17	dodec-3-yn-1-ol	9	10	1	2

Table S11.

SSR dose response data for neocembrene and *P. simplex* neocembrene sensillum.Related to Fig. 5 [↗](#).

	replicate	1	2	3	4	5	6	7	8	9	10	11
Dose	0.01 ng	62	44		18		14	8	30	10	4	18
	0.1 ng	76	24	20	64	90	22	42	36	12	10	42
	1 ng	90	46	54	86	106	14	30	38	20	12	36
	10 ng	84	44	50	88	120	22	14		16	28	46
	100 ng	104	54	34	112	124	36	8		36	38	40

PsimOR14

ATGATTCGATCAAAGAGAAAGGAGAGCCAAGCAAACGAAACAAAACACACATTAACAAGTGAAGAGCAACCTTCTGATTG
TGACGTGAAGATCATGACACTCAGTATTATGCTGAATGCGGCTGGCCTCCTACCTCCAGCCAAATCGTCATCATTGATCA
GACTGGCCTACAAAGTATTTGTAGTATTTATTCACATACTTTTCGCTCTTAACGCTGATAGGACAGATAATGGCAGTAGTG
GTTTACTGGGGAGACATTCTCTAATTGCAACCACAATAAGCTTGATGACTAGTCTGATTGGATCGATGAGTTCATCCAT
AAATTTTCTTCTAAACAGAAAGAAGTACATGCGTCTTGCGGACACGTTGAAAACAGAATTTGTTGCCAAATTGAAATCAA
AATATATCAAAATTATTTTAAATGCTGAACGTCAGGTTGTATTCTGTGGGATACTCGTATGTATTGTAGCTGTATGTATT
GGATTTATTTGGATAGTCGTGCCATTTTTAAGTACCAACACCCCATTTGACTTTGCAAATGAAAAAGTGTCAAAGAAGG
AAGCCGCATGGAGGAATTAATTCTTGTGATGTGGCTCCCTTCTAAATTTGAACAGTCCCCTCAATTTGAAATAATAGTTT
TTTTACAAATCTTTGTCGTAACGTTTGCATTAGCAATGATCTATTGATGATGTTACTATCTCTGATGAGCCAC
GCTGCTGCACAGTTCAGGGTTTTGAATGCCATGCTGAATGACATGCACGAAAATGTTTCGTGAAGACGAGATTACAGAAC
AAGAAACATGGCTTCATTGGTCACTGGCACTGACATCTCGTACATGGAGTTCTCTTCTACCAATTCTTGGAATGGAAACA
CAGAGCATTCTGGAAGCGCTGGTGTGAGTTGGACAGCCTAAAAATGAAGACTGTGAAGAAGATCCTGTCCGACAGTAC
CTCGTTGAGTGCATTAGATATCACCAGGCTGTAATTGAGTTTGTGACCAACTGAACGAGGTGTTGCGCGCAGTGAGCTT
CGTGAAGATGCTTGACTGGCCTTTTGCATTGTATGACAGGATTTGAGTTGACACAGACTGTAGAGAGCCAGGAGGATT
TACTTAAATTCATCTCCCTGTTTGTGCTGGGGTTGTATATCTAATAATCTCTTACATTTGGTTCGGACAGCAAGTAATTGAC
GAGAGCGAGGAAGTAGCAACAGCGTTGTACAGCACTGACTGGTACAACCAGGCACCAGGTTCAAACGTCTGCTGCCTGT
AGCCATCATGCGGGCTTCGAATTCTGTCAAAGTTAAAGCTGGGGTGTCTTTGACATGTCTTGCGTCACGTTAGCTTCGA
TTATGAATGCATCTTACACGTATTTTATGATGCTAATTCATCTACACGACTCCTAA

PsimOR14

MIRSKRKESQANETKHTLTSEEQPSDCDVKIMTSLIMLNAAGLLPPAKSSSLIRLAYKVFVFIHILFVLTLLIGQIMAVV
VYWGDIPLIATTISLMTSLIGSMSSSINFLNRKKYMLADTLKTEFVAKLKSKYIKIILNAERQVVFCGILVCIVAVCI
GFIWIVPFLSTNTPDFANEKSVKEGSRMEELILVMWLPSPKFEQSPQFEIIVFLQIFVVTALAMIYSVDMMLLSLSH
AAAQFRVLNAMLNDMHENVREDEIHRTRNMAVLVTGTDISYMEFSSNSWNGNTEHSGSAGVELDSLKNEDCEEDPVRQY
LVECIYHQAVIEFVDQLNEVFGAVSFVKMLDWPFAICMTGFQLTQTVESQEDLLKFISLFAGVVYLIISYIWFQQVID
ESEEVATALYSTDWYNQAPGFKRLLPVAIMRASNSVKVKGVFVDMSCVTLASIMNASYTYFMMLIHLHDSX

Table S12.

Nucleotide and protein sequences of *PsimOR14*.

Related to **Fig. 6** [↗](#).

Interaction energy (kcal/mol)	neocembrene	geranylgeraniol	(+)-limonene
	mean $\pm$ SD		
Cys154	-0.90 $\pm$ 0.58	-0.71 $\pm$ 0.35	-0.17 $\pm$ 0.27
Ala157	-0.68 $\pm$ 0.46	-1.02 $\pm$ 0.38	-0.45 $\pm$ 0.38
Val158	-1.16 $\pm$ 0.77	-1.72 $\pm$ 0.56	-0.23 $\pm$ 0.44
Gly161	x	-0.82 $\pm$ 0.38	x
Phe162	x	-1.08 $\pm$ 0.67	x
Ile165	x	-0.66 $\pm$ 0.49	x
Ile217	x	-1.03 $\pm$ 0.47	x
Val220	x	-0.72 $\pm$ 0.31	x
Thr221	-1.00 $\pm$ 0.53	-1.19 $\pm$ 0.47	-0.17 $\pm$ 0.35
Leu224	-1.09 $\pm$ 0.58	-1.41 $\pm$ 0.39	-0.78 $\pm$ 0.46
Ala225	-0.93 $\pm$ 0.6	-0.97 $\pm$ 0.45	-0.17 $\pm$ 0.29
Tyr228	-1.10 $\pm$ 0.72	-1.06 $\pm$ 0.49	-0.47 $\pm$ 0.4

Table S13.

MM/PBSA calculated interaction energies of ligands with PsimOR14 decomposed into per-residue contributions.

OR	DESeq2			edgeR	
	log ₂ fold change (worker vs soldier)	lfcSE	p-value	log ₂ fold change (worker vs soldier)	p-value
PsimOR1	-0.5905	0.3426	0.0848	-0.6679	0.0186
PsimOR10	-0.4006	0.9211	0.6636	-0.4375	0.5905
PsimOR11	0.3565	0.5005	0.4763	0.262	0.5589
PsimOR12	-0.2703	1.2658	0.8309	-0.4127	0.8431
PsimOR13	0.1677	0.6478	0.7957	0.0539	1
PsimOR14	-1.8565	0.8841	0.0357	-1.8989	0.0036
PsimOR15	-0.578	0.8735	0.5082	-0.6829	0.4084
PsimOR16	-0.4973	1.1753	0.6722	-0.5878	0.613
PsimOR17	-0.7328	3.7695	0.8459	-0.5892	1
PsimOR18	1.2163	1.6983	0.4739	1.0486	0.5486
PsimOR19	0.6517	1.0553	0.5369	0.5408	0.5589
PsimOR2	1.2943	0.7769	0.0957	1.1238	0.1342
PsimOR20	-2.9803	0.7045	0	-3.0603	0
PsimOR22	0.5913	1.1359	0.6027	0.5019	0.7071
PsimOR23	-0.0816	1.3315	0.9511	-0.1249	1
PsimOR24	-1.3934	0.8105	0.0856	-1.4649	0.025
PsimOR25	0.9132	1.4392	0.5257	0.6897	0.6457
PsimOR26	-0.879	0.1926	0	-0.9787	0
PsimOR27	0.2988	1.1656	0.7977	0.2	1
PsimOR28	0.3981	1.1226	0.7229	0.3443	0.722
PsimOR29	-0.4531	1.3799	0.7426	-0.45	0.6812
PsimOR3	-0.037	1.156	0.9744	-0.1663	0.8475
PsimOR30	0.7053	1.184	0.5514	0.5977	0.6758
PsimOR31	0.1924	0.5699	0.7357	0.0691	0.9348
PsimOR32	-0.2186	1.4043	0.8763	-0.288	1
PsimOR34	-0.4167	0.4244	0.3261	-0.5383	0.1738
PsimOR35	-1.0587	2.6395	0.6883	-0.8036	0.5553
PsimOR36	-0.3664	0.9102	0.6873	-0.5093	0.5249
PsimOR37	0.3495	1.2265	0.7756	0.2614	0.8477
PsimOR39	-0.0669	0.6154	0.9135	-0.1886	0.7502
PsimOR4	0.0586	0.7043	0.9336	-0.0669	1
PsimOR40	-0.5383	0.34	0.1134	-0.6076	0.0395
PsimOR41	-0.3886	0.2793	0.164	-0.5129	0.0916
PsimOR42	NA	NA	NA	0	1
PsimOR43	-0.5866	1.4600	0.6878	-0.5596	0.7796
PsimOR44	0.1981	1.6775	0.906	0.128	1
PsimOR45	-1.5455	1.331	0.2456	-1.4406	0.1817
PsimOR46	-3.0579	1.863	0.1007	-2.9	0.0445
PsimOR47	2.375	1.7303	0.1699	2.2464	0.1134
PsimOR48	0.0016	1.2033	0.9989	-0.0757	1
PsimOR49	-1.0692	1.6425	0.5151	-0.9962	0.3814
PsimOR5	0.4942	0.6163	0.4227	0.4152	0.4609
PsimOR50	0.229	4.0805	0.9553	1.6045	1
PsimOR6	-0.0086	0.5823	0.9882	-0.0515	1
PsimOR7	0.161	0.4781	0.7363	0.1071	0.8517
PsimOR8	0.4963	0.6856	0.4692	0.3467	0.6466
PsimOR9	2.0761	0.9969	0.0373	1.8932	0.0303

Table S14.

Differential expression of *P. simplex* ORs (DESeq2 and edge R analyses), based on heads (including antennae) of workers and soldiers (three replicates).

Related to **Fig. 7** [↗](#).

EAG response normalized to air stimulation		log ₂ value	
worker	soldier	worker	soldier
199.52	92.67	7.64	6.53
322.81	116.73	8.33	6.87
132.44	136.93	7.05	7.10
142.77	78.81	7.16	6.30
124.07	120.01	6.96	6.91
135.73	120.49	7.08	6.91
192.41	136.59	7.59	7.09
146.41	116.43	7.19	6.86
122.17	176.51	6.93	7.46
201.83	122.48	7.66	6.94
123.71	110.96	6.95	6.79
142.15	139.64	7.15	7.13
115.99	138.32	6.86	7.11
164.7	132.45	7.36	7.05
167.08	152.31	7.38	7.25

Table S15.

Differential sensitivity of *P. simplex* workers and soldiers to neocembrene inferred from EAG responses to the dose of 10 ng.

Related to **Fig. 7** [↗](#).

Primer Name	Sequence (5' to 3' direction)	Use
PsimOR14_F	ATGATTCGATCAAGAGAAAGG	Cloning
PsimOR14_R	TTAGGAGTCGTGTAGATGAAT	Cloning
PsimO31_F	ATGGAATACATAAAAAATGAAACATATTCTCA	Cloning
PsimO31_R	TCAACCTACGACATGTGAGTTATT	Cloning
PsimOR9_F	ATGGACAGCCTTTACGACCAATCTT	Cloning
PsimOR9_R	TCATTCACTGACTGAGGGATCCTT	Cloning
PsimO30_F	ATGGAGCACAGGAAATACAAAGTGACAA	Cloning
PsimO30_R	TTACGTTCCCTGATTTGTGTCGGTAT	Cloning
PsimOrco_F	ATGTACAAGTTCAGGTTACACG	cDNA check
PsimOrco_R	CTAGTTGAGCTGTACCAACAC	cDNA check
GW1	GTTGCAACAAATTGATGAGCAATGC	Sanger Sequencing & Colony PCR
GW2	GTTGCAACAAATTGATGAGCAATTA	Sanger Sequencing & Colony PCR
UAS1	TAGCGAGCGCCGGAGTATAAATAG	Sanger Sequencing
UAS2	ACTGATTTTCGACGGTTACCC	Sanger Sequencing
DmOr22a_F	TCTCCAGCATCGCCGAGTGT	Single-Wing PCR
DmOr22a_R	CGGCAGAGGTCCAGTCCGAT	Single-Wing PCR
PsimOR14_SW_F	GAGAGCCAAGCAACGAAAC	Single-Wing PCR
PsimOR14_SW_R	TTTAGAAGGGAGCCACATCAC	Single-Wing PCR
PsimO31_SW_F	GCTGGGTTAATCCCGATCAT	Single-Wing PCR
PsimO31_SW_R	GCATGGCACCAAAATAGTTCTTC	Single-Wing PCR
PsimOR9_SW_F	TGGGCGAACTGAGGATATG	Single-Wing PCR
PsimOR9_SW_R	CGAGCCGACATAGAAGAAGAG	Single-Wing PCR
PsimO30_SW_F	TGCCATCACCAGCAGATAAA	Single-Wing PCR
PsimO30_SW_R	CACCGACTGACTCAGCATATT	Single-Wing PCR

Table S16.

List of primers.

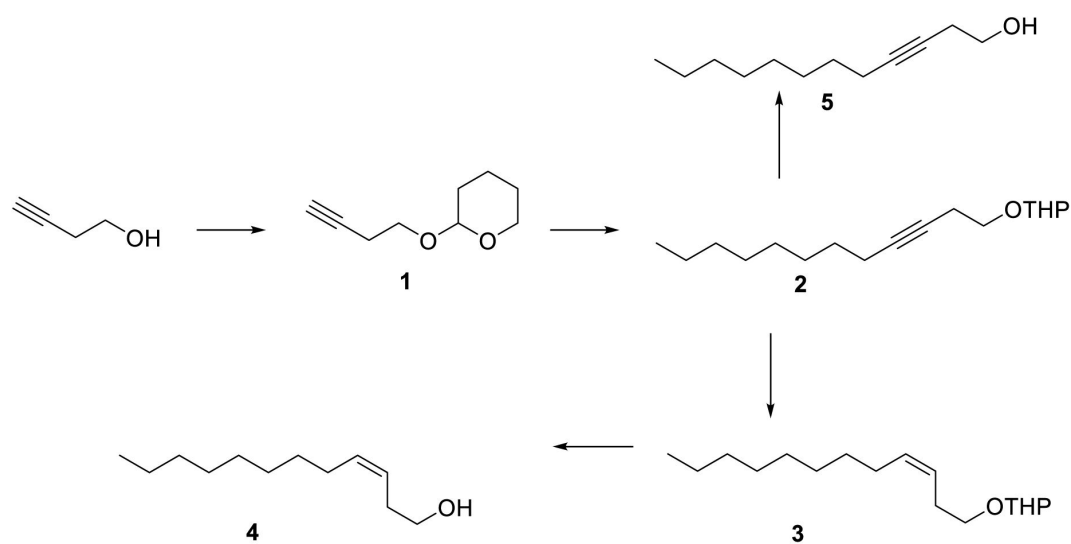
Fly line	Source
W¹¹¹⁸	Biology Centre CAS Czech Republic
w-; BI/Cyo; TM2/TM6B	MPI-Jena, Germany
w; Or22ab^{GAL4}	Benton Lab, Switzerland
w-; +/+; UAS-OR(w+)/UAS-OR(w+)	BestGene Inc, USA

Table S17.
Characteristics and origin of *D. melanogaster* lines used.

Panel	Compound	Provider	Purity
1	<i>n</i> -eicosane	Merck	99%
1	<i>n</i> -docosane	Merck	99%
1	1-octadecanol	Merck	99%
1	(3 <i>Z</i>)-dodecen-1-ol	IOCB	in house synthesis, see supplementary methods, GC purity 95%
1	(3 <i>Z</i> ,6 <i>Z</i>)-dodecadien-1-ol	IOCB	in house synthesis, see supplementary methods, GC purity 94%
1	(3 <i>Z</i> ,6 <i>Z</i> ,8 <i>E</i>)-dodecatrien-1-ol	IOCB	in house synthesis, see supplementary methods, GC purity 96%
1	dodec-3-yn-1-ol	IOCB	in house synthesis, see supplementary methods, GC purity 93%
1	δ-cadinene	Chemenu	95%
1	(3 <i>R</i> ,6 <i>E</i>)-nerolidol	IOCB	in house synthesis (Havličková et al. 2019), 96%
1	neocembrene	IOCB	purified from natural resource (Sillam-Dussès et al., 2005), GC purity 96%
1	geranylgeraniol	Merck	98%
2	myrcene	Sigma Aldrich	≥90%
2	eucalyptol	Sigma Aldrich	99%
2	(+)-3-carene	Sigma Aldrich	90%
2	(+)-limonene	Thermo-Fisher	97%
2	(+)-α-pinene	Sigma Aldrich	98%
2	p-cymene	Sigma Aldrich	99%
2	γ-terpinene	Sigma Aldrich	97%
2	(-)-β-caryophyllene	Sigma Aldrich	98%
2	(+)-longifolene	Phyto Lab	≥90%
2	terpinolene	Sigma Aldrich	≥85%
2	toluene	VWR Chemicals	99%
2	(2 <i>E</i>)-hexenal	Thermo-Fisher	98%
2	sabinene	Chemenu	≥95%
2	pentyl acetate	Sigma Aldrich	≥99%
2	ethyl acetate	VWR Chemicals	99%
2	γ-nonolactone	Sigma Aldrich	98%
2	heptanal	Thermo-Fisher	98%
2	isoamyl propionate	Sigma Aldrich	≥98%
2	ethanol	VWR Chemicals	≥99%
2	(2 <i>E</i>)-hexen-1-ol	Sigma Aldrich	96%
3	β-bisabolene	Thermo-Fisher	96%
3	(<i>E</i>)-myrcenol	FytoFarm	95%
3	(-)-trans-pinocarveol	Sigma Aldrich	≥96%
3	isoamyl acetate	Sigma Aldrich	≥99%
3	2-phenylethyl acetate	J&K Scientific Ltd.	≥98%
3	α-camphorene	Synergy Ltd	≥90%
3	(<i>E</i>)-β-ocimene	TRC Canada	98%
3	nonan-1-ol	Thermo-Fisher	95%
3	octan-1-ol	Honey well	99%
3	benzyl alcohol	Thermo-Fisher	99%
3	p-cresol	Sigma Aldrich	≥99%
3	cryptone	Chemenu	≥95%
3	(+)-cis-carveol	BOC Sciences	95%
3	methyl jasmonate	Sigma Aldrich	≥98%
3	eucarvone	MuseChem	≥95%
3	ethyl (2 <i>E</i> ,4 <i>Z</i>)-decadienoate	Sigma Aldrich	95%
4	(-)-verbenone	Merck	94%
4	acetophenone	Sigma Aldrich	99%
4	2-phenylethanol	Acros organics	99%
4	(±)-myrtenol	Sigma Aldrich	95%
4	geranyl acetone	Sigma Aldrich	≥97%
4	oct-1-en-3-ol	Thermo-Fisher	98%
4	4-vinylanisole	Sigma Aldrich	97%
4	4-ethylguaiaicol	Sigma Aldrich	≥98%
4	hexan-1-ol	Sigma Aldrich	99%
4	oct-1-en-3-ol	Sigma Aldrich	≥97%
4	styrene	Thermo-Fisher	99%
4	2,3-dihydrobenzofuran	Thermo-Fisher	99%
4	2-methylbutan-1-ol	J&K Scientific Ltd.	98%
4	2-methylbutyl acetate	Sigma Aldrich	99%
4	4-methylanisole	Sigma Aldrich	97%
4	linalool	Thermo-Fisher	97%
4	methyleugenol	Sigma Aldrich	98%
4	(±)-camphor	Sigma Aldrich	≥95%
4	benzaldehyde	Sigma Aldrich	≥99%
4	isoamyl alcohol	VWR Life Science	≥98%

Table S18.

Origin of chemicals.



Scheme 1.

0.95 – 0.86 (m, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 98.7, 81.4, 66.3, 62.2, 31.9, 30.6, 29.2, 29.2, 29.0, 28.9, 25.5, 20.3, 19.4, 18.8, 14.1. EI-MS (70 eV): m/z (%): 265 (0.002), 211 (0.4), 195 (0.7), 153 (2), 115 (4), 101 (7), 85 (100), 81 (8), 67 (17), 55 (13), 41 (12).

2-[[*(Z)*-Dodec-3-enyl]oxy]-tetrahydro-2H-pyran (3)

To a solution of borane-dimethylsulfide complex (0.44 g; 5.78 mmol) in THF (10 mL) at -10°C under argon atmosphere cyclohexane (0.95 g; 11.57 mmol) was added. After the addition was complete, the mixture was slowly warmed to room temperature over 3 hr, yielding a white slurry. The mixture was cooled to -20°C and a solution of substrate 2 (0.77 g; 2.89 mmol) in THF (3 mL) was added dropwise. The reaction mixture was slowly warmed to 0°C over 3 hr, and stirred at 0°C for a further 3 hr. Then acetic acid (1.40 g; 23.1 mmol) was then added dropwise, and the reaction mixture was stirred overnight. The reaction mixture was cooled to 0°C , and a solution of NaOH (1.9 g) in water (8 mL) was added over 10 min, followed by dropwise addition of hydrogen peroxide (0.7 mL, 30% solution in water). The stirred mixture was warmed to room temperature and poured into water. Crude product extracted with cyclohexane and then concentrated under reduced pressure. The residue was purified by column chromatography (silica gel; cyclohexane/EtOAc 9:1) to provide 3 as colorless oil (0.71 g, 91%). ^1H NMR (401 MHz, CDCl_3) δ 5.56 – 5.28 (m, 2H), 4.63 (dd, J = 4.4, 2.7 Hz, 1H), 3.97 – 3.85 (m, 1H), 3.75 (dt, J = 9.5, 7.2 Hz, 1H), 3.61 – 3.48 (m, 1H), 3.48 – 3.36 (m, 1H), 2.46 – 2.32 (m, 2H), 2.06 (td, J = 7.1, 1.3 Hz, 2H), 1.86 (ddt, J = 10.7, 7.8, 5.3 Hz, 1H), 1.80 – 1.70 (m, 1H), 1.69 – 1.50 (m, 4H), 1.38 – 1.28 (m, 10H), 0.94 – 0.87 (m, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 132.1, 125.4, 98.8, 67.1, 62.3, 31.9, 30.7, 29.7, 29.5, 29.4, 29.2, 28.90, 27.4, 22.7, 19.6, 14.1. EI-MS (70 eV): m/z (%): 268 (0.1), 166 (3), 115 (2), 101 (15), 85 (100), 67 (13), 55 (11), 41 (8).

(*Z*)-Dodec-3-en-1-ol (4)

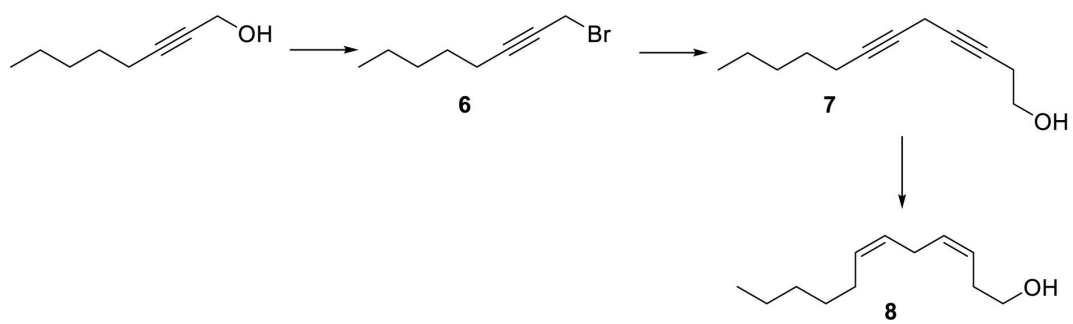
A mixture of *Z*-alkene 3 (0.71 g; 2.64 mmol), and *p*-toluenesulfonic acid monohydrate (0.025 g; 0.13 mmol) in methanol (2 mL) was stirred at room temperature for 3 hr. Then the reaction mixture was poured into an ice-cold solution of sodium bicarbonate and was extracted with diethyl ether (2×10 mL). The combined organic layers were washed with brine (5 mL), dried over MgSO_4 and evaporated. The residue was purified by column chromatography (silica gel; cyclohexane/EtOAc 9:1) to provide 4 as colorless oil (0.43 g, 87%). ^1H NMR (401 MHz, CDCl_3) δ 5.66 – 5.48 (m, 1H), 5.46 – 5.31 (m, 1H), 3.67 (t, J = 6.5 Hz, 2H), 2.40 – 2.32 (m, 2H), 2.13 – 1.98 (m, 2H), 1.43 – 1.21 (m, 12H), 0.97 – 0.86 (m, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 133.6, 124.9, 62.4, 31.9, 30.8, 29.7, 29.5, 29.3, 29.3, 27.4, 22.7, 14.1. EI-MS (70 eV): m/z (%): 184 (0.1), 166 (5), 138 (6), 124 (8), 110 (18), 96 (39), 81 (79), 68 (100).

Dodec-3-yn-1-ol (5)

A mixture of THP-protected alkyne 2 (0.30 g; 1.12 mmol), and *p*-toluenesulfonic acid monohydrate (0.04 g; 0.022 mmol) in methanol (5 mL) was stirred at room temperature for 3 hr. Then the reaction mixture was poured into an ice-cold solution of sodium bicarbonate and was extracted with diethyl ether (2×10 mL). The combined organic layers were washed with brine (5 mL), dried over MgSO_4 and evaporated. The residue was purified by column chromatography (silica gel; cyclohexane/EtOAc 4:1) to provide 5 as colorless oil (0.20 g, 98%). ^1H NMR (401 MHz, CDCl_3) δ 3.70 (t, J = 6.2 Hz, 2H), 2.46 (tt, J = 6.2, 2.4 Hz, 2H), 2.18 (tt, J = 7.2, 2.4 Hz, 2H), 1.57 – 1.46 (m, 2H), 1.39 – 1.22 (m, 8H), 0.94 – 0.86 (m, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 82.9, 76.2, 61.4, 31.8, 29.2, 29.1, 29.0, 28.9, 23.2, 22.7, 18.8, 14.1.

1-Bromooct-2-yne (6)

A mixture of 2-octynol (2.0 g; 15.87 mmol), triphenylphosphine (4.57 g; 17.46 mmol), and tetrabromomethane (6.30 g; 19.04 mmol) in DCM (40 mL) was stirred at 0°C for 2 hr and then it was diluted with cyclohexane and filtered. The filtrate was evaporated, and a column chromatography (silica gel; eluent cyclohexane) of the residue afforded 3.70 g (90%) of bromide 5



Scheme 2.

as a pale oil. ^1H NMR (401 MHz, CDCl_3) δ 3.95 (t, J = 2.4 Hz, 2H), 2.25 (ddt, J = 7.2, 4.7, 2.4 Hz, 2H), 1.59 – 1.46 (m, 2H), 1.44 – 1.27 (m, 4H), 0.96 – 0.88 (m, 3H). ^1H NMR data match published spectrum (Sigurjónsson & Haraldsson, 2024 [DOI](#)).

Dodeca-3,6-diyn-1-ol (Liu et al. 2018 [DOI](#)) (7)

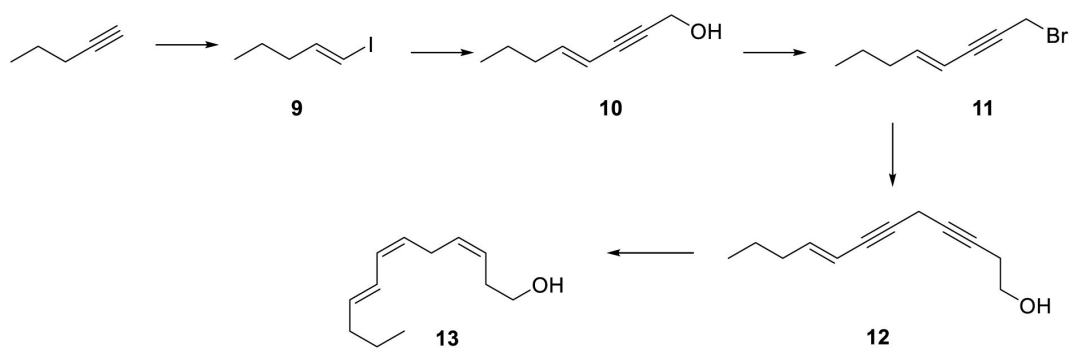
To a stirred solution of but-3-yn-1-ol (0.90 g; 12.76 mmol) in anhydrous DMF (30 mL) under an argon atmosphere at room temperature was added cesium carbonate (3.45 g; 10.63 mmol), sodium iodide (1.60 g; 10.63 mmol) and copper(I) iodide (2.02 g; 10.63 mmol). The reaction mixture was stirred for 30 min. Then a solution of 1-bromooct-2-yne (2.0 g; 10.63 mmol) in anhydrous DMF (10 mL) was added dropwise and stirring was continued for another 2 hr. The reaction mixture was quenched by addition of saturated aqueous NH_4Cl and extracted with Et_2O . The combined organic layers were washed with water, brine, and concentrated under reduced pressure. The residue was purified by a column chromatography (silica gel; cyclohexane/ EtOAc 4:1) to provide diynol 7 as colorless oil (0.90 g, 45%). ^1H NMR (401 MHz, CDCl_3) δ 3.73 (t, J = 6.2 Hz, 2H), 3.16 (t, J = 2.4 Hz, 2H), 2.47 (tt, J = 6.2, 2.4 Hz, 2H), 2.17 (tt, J = 7.2, 2.4 Hz, 2H), 1.56 – 1.46 (m, 2H), 1.43 – 1.22 (m, 4H), 0.98 – 0.85 (m, 3H). EI-MS (70 eV): m/z (%): 178 (0.2), 163 (0.7), 149 (2), 135 (4), 121 (14), 117 (20), 105 (37), 91 (100), 79 (41).

(3Z,6Z)-Dodeca-3,6-dien-1-ol (Liu et al. 2018 [DOI](#)) (8)

To a stirred suspension of $\text{Ni}(\text{OAc})_2 \cdot \text{H}_2\text{O}$ (0.81 g; 3.25 mmol) in EtOH (10 mL) under an argon atmosphere at room temperature was added NaBH_4 (0.12 g; 3.25 mmol). Then the flask was filled with hydrogen and balloon with hydrogen gas was attached to the reaction flask for the rest of the experiment. Ethylenediamine (0.78 g; 13.02 mmol) was added to the reaction mixture after 30 min and then the reaction mixture was stirred for another 30 min. Then a solution of diynol 7 (0.58 g; 3.25 mmol) in EtOH (2 mL) was added and the mixture was stirred for another 2 hr. The reaction mixture was diluted with water and EtOAc (5+5 mL) and the P-2 catalyst was filtered through a pad of Celite. The filtrate was concentrated under reduced pressure and partitioned between water and Et_2O . The organic layer was separated, the aqueous phase was extracted with Et_2O , and the combined organic layers were washed with brine and concentrated under reduced pressure. The residue was purified by a column chromatography (silica gel; cyclohexane/ EtOAc 4:1) to provide dienol 8 as colorless oil (0.29 g, 49%). ^1H NMR (401 MHz, CDCl_3) δ 5.65 – 5.48 (m, 1H), 5.47 – 5.27 (m, 2H), 3.68 (t, J = 6.5 Hz, 2H), 2.89 – 2.75 (m, 2H), 2.44 – 2.32 (m, 2H), 2.13 – 1.99 (m, 2H), 1.47 – 1.22 (m, 6H), 0.98 – 0.86 (m, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 131.6, 130.7, 127.4, 125.3, 62.3, 31.5, 30.8, 29.3, 27.3, 26.9, 25.8, 22.6, 14.1. EI-MS (70 eV): m/z (%): 182 (2), 164 (5), 138 (5), 135 (8), 121 (16), 107 (19), 93 (48), 79 (100), 67 (66).

(E)-1-iodopent-1-ene (Saran et al., 2018) (9)

Diisobutylaluminum hydride in hexanes (1 M, 50 mL, 50 mmol) was added dropwise to a solution of pent-1-yne (3.7 g; 54 mmol) in hexanes cooled to -40°C . The resulting mixture was stirred for 30 min, then slowly warmed to room temp over 3 hr, and was allowed to stir overnight. The mixture was warmed to $\sim 50^\circ\text{C}$ for 4 hr, then cooled to -40°C , and a solution of iodine (12.7 g; 50 mmol) in THF (20 mL) was added dropwise over 30 min. The resulting dark brown suspension was allowed to warm to room temperature with stirring overnight. The solution was cooled in an ice bath, and slowly quenched with dropwise addition of ice-cold diluted sulfuric acid (5 mL of 96% acid in 60 mL of water) with vigorous stirring. After stirring for 30 min, the layers were separated, and the organic layer was washed with dilute NaHSO_3 solution and brine and dried over MgSO_4 . The resulting solution was filtered through a short pad of silica gel, rinsing with pentane. Obtained solution was concentrated by rotary evaporation without heating (rotavap was set to 100 Torr pressure). This procedure gave 12.24 g of crude alkenyl iodide 9 that was used in the next step. EI-MS (70 eV): m/z (%): 155 (4), 141 (3), 127 (5), 73 (69), 55 (100).



Schema 3.

(E)-Oct-4-ene-2-yn-1-ol (Saran et al., 2018) (10)

To a mixture of bis(triphenylphosphine)palladium(II) dichloride (1.45 g; 2.0 mmol), copper(I) iodide (0.72 g; 3.8 mmol), and pyrrolidine (50 mL) under argon atmosphere the crude (E)-1-iodopent-1-ene (12.2 g) was added dropwise, followed by dropwise addition of propargyl alcohol (4.92 g; 87.8 mmol). The mixture was immersed into water bath until the exothermic reaction ended. After cooling the mixture to room temperature, the mixture was stirred for 3 hr. Then the reaction mixture was poured into an ice-cold mixture of saturated aqueous NH_4Cl and 2 M HCl (150+150 mL) and extracted several times with diethyl ether. The combined organic layers were washed with aqueous solution of citric acid (10% solution), water, brine, concentrated under reduced pressure. The residue was purified by a column chromatography (silica gel; eluent DCM) to provide alcohol **10** as colorless oil (2.22 g). ^1H NMR (401 MHz, CDCl_3) δ 6.19 (dt, J = 15.9, 7.1 Hz, 1H), 5.57 – 5.47 (m, 1H), 4.48 – 4.36 (m, 2H), 2.11 (qd, J = 7.2, 1.6 Hz, 2H), 1.71 – 1.51 (m, 1H), 1.45 (q, J = 7.4 Hz, 2H), 0.93 (t, J = 7.4 Hz, 3H). EI-MS (70 eV): m/z (%): 124 (74), 109 (14), 95 (44), 81 (100), 67 (49), 53 (33).

(E)-1-Bromooct-4-ene-2-yne (11)

To a stirred solution of alcohol **10** (2.22 g; 17.9 mmol) and triphenylphosphine (5.16 g; 19.7 mmol) in dry DCM (30 mL) under an argon atmosphere at 0 °C was added solution of tetrabromomethane (7.12 g; 21.5 mmol) in dry DCM (10 mL). The reaction mixture was stirred at 0 °C for 10 min and then an ice bath was removed, and the reaction mixture was allowed to stir for 3 hr. The reaction mixture was concentrated to half the volume and cyclohexane (30 mL) was added. The mixture was stirred for 30 min. Then the mixture was filtered through a pad of Celite, and the solvent was evaporated. A column chromatography (silica gel, eluent cyclohexane) of the residue gave 3.00 g (90%) of bromide **11** as colorless oil. ^1H NMR (401 MHz, CDCl_3) δ 6.22 (dt, J = 15.9, 7.1 Hz, 1H), 5.58 – 5.47 (m, 1H), 4.08 (d, J = 2.3 Hz, 2H), 2.12 (qd, J = 7.2, 1.6 Hz, 2H), 1.51 – 1.37 (m, 4H), 0.93 (t, J = 7.4 Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 146.7, 108.8, 85.8, 82.5, 35.2, 26.9, 21.8, 15.8, 13.6. EI-MS (70 eV): m/z (%): 188 (21), 186 (23), 107 (100), 91 (55), 79 (49), 65 (66).

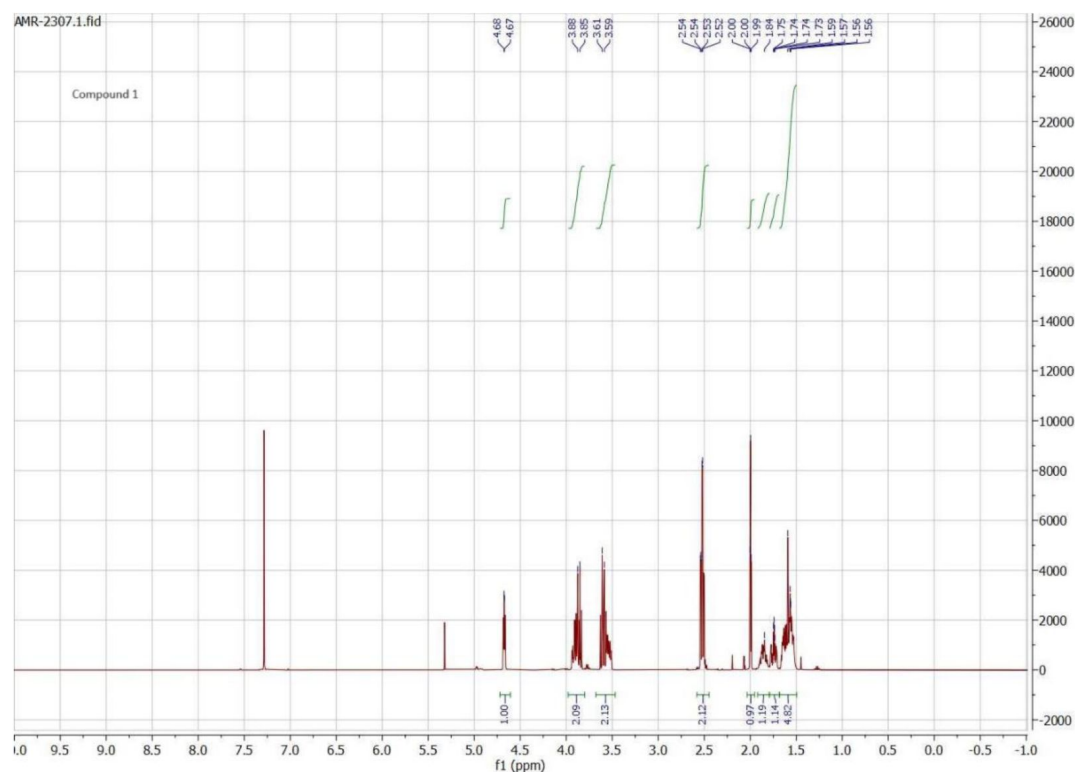
(E)-Dodeca-8-en-3,6-diyn-1-ol (12)

To a stirred solution of but-3-yn-1-ol (1.58 g; 22.58 mmol) in anhydrous DMF (30 mL) under an argon atmosphere at room temperature was added cesium carbonate (6.11 g; 18.81 mmol), sodium iodide (2.82 g; 18.81 mmol) and copper(I) iodide (3.57 g; 18.81 mmol). The reaction mixture was stirred for 30 min. Then a solution of bromide **11** (3.50 g; 18.81 mmol) in anhydrous DMF (10 mL) was added dropwise and stirring was continued for another 2 hr. The reaction mixture was quenched by addition of saturated aqueous NH_4Cl and extracted with Et_2O . The combined organic layers were washed with water, brine, and concentrated under reduced pressure. The residue was purified by a column chromatography (silica gel; cyclohexane/ EtOAc 3:1) to provide alcohol **12** as colorless oil (2.0 g, 60%). ^1H NMR (401 MHz, CDCl_3) δ 6.14 (dt, J = 15.9, 7.1 Hz, 1H), 5.52 – 5.42 (m, 1H), 3.73 (t, J = 6.2 Hz, 2H), 3.33 – 3.26 (m, 2H), 2.52 – 2.42 (m, 2H), 2.14 – 2.03 (m, 2H), 1.51 – 1.36 (m, 2H), 0.92 (t, J = 7.4 Hz, 3H). ^{13}C NMR (101 MHz, CDCl_3) δ 144.7, 109.3, 81.9, 79.4, 77.2, 76.3, 61.1, 35.1, 23.2, 21.9, 13.6, 10.4. HRMS (ESI): m/z calcd for $\text{C}_{12}\text{H}_{16}\text{ONa}^+$: 199.1093 $[\text{M}+\text{Na}]^+$; found: 199.1094. EI-MS (70 eV): m/z (%): 176 (4), 161 (3), 147 (14), 128 (36), 115 (67), 103 (45), 91 (100), 77 (62).

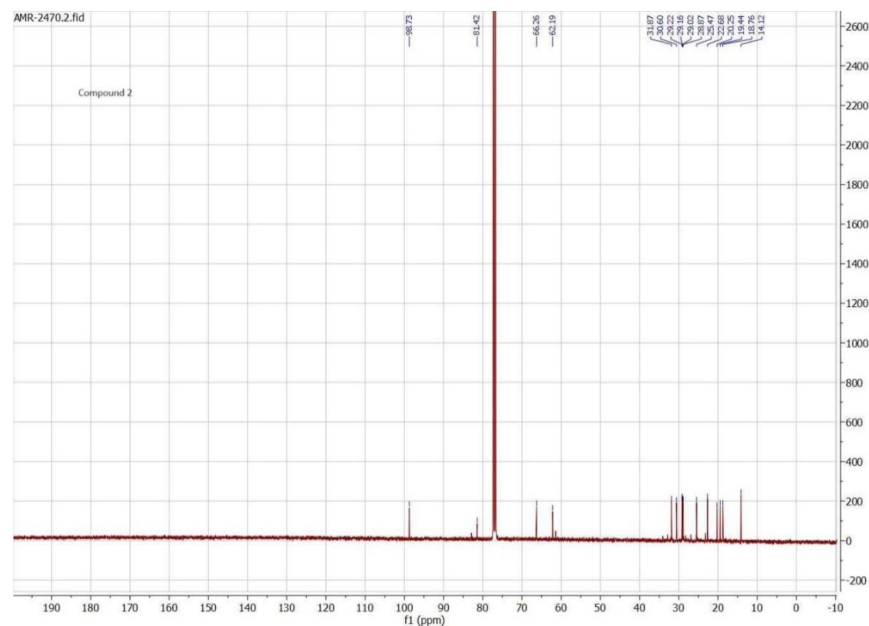
(3Z,6Z,8E)-Dodeca-3,6,8-trien-1-ol (13)

To a stirred suspension of $\text{Ni}(\text{OAc})_2 \cdot \text{H}_2\text{O}$ (1.41 g; 5.68 mmol) in EtOH (8 mL) under an argon atmosphere at room temperature was added a suspension of NaBH_4 (0.21 g; 5.68 mmol) in EtOH (5 mL). Then the flask was filled with hydrogen and balloon with hydrogen gas was attached to the reaction flask for the rest of the experiment. Ethylenediamine (1.36 g; 22.7 mmol) was added to the reaction mixture after 30 min and then the reaction mixture was stirred for another 30 min. Then a solution of diynol **12** (1.0 g; 5.68 mmol) in EtOH (2 mL) was added and the mixture was stirred for another 2 hr. The reaction mixture was diluted with water and EtOAc (5+5 mL) and the P-2

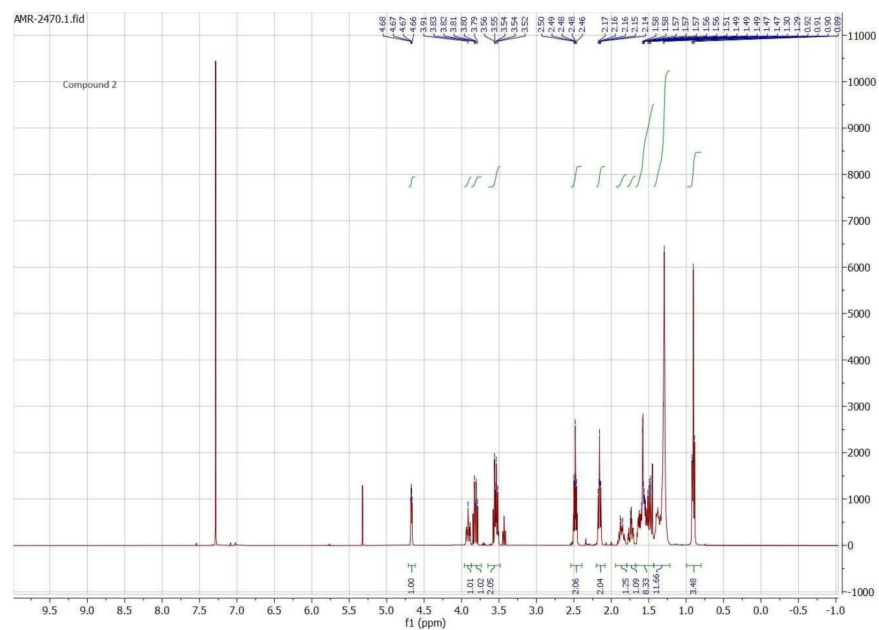
catalyst was filtered through a pad of Celite. The filtrate was concentrated under reduced pressure and partitioned between water and Et₂O. The organic layer was separated, the aqueous phase was extracted with Et₂O, and the combined organic layers were washed with brine and concentrated under reduced pressure. The residue was purified by a column chromatography (silica gel; cyclohexane/EtOAc 4:1) to provide trienol **13** as colorless oil (0.40 g, 40%). ¹H NMR (401 MHz, CDCl₃) δ 6.42 – 6.25 (m, 1H), 6.12 – 5.93 (m, 1H), 5.72 (dt, *J* = 14.6, 7.0 Hz, 1H), 5.65 – 5.53 (m, 1H), 5.50 – 5.38 (m, 1H), 5.28 (dt, *J* = 10.6, 7.5 Hz, 1H), 3.69 (t, *J* = 6.5 Hz, 2H), 2.45 – 2.35 (m, 2H), 2.12 (qd, *J* = 7.5, 1.4 Hz, 2H), 1.51 – 1.44 (m, 2H), 0.94 (t, *J* = 7.3 Hz, 3H). ¹³C NMR (101 MHz, CDCl₃) δ 135.5, 131.1, 129.1, 127.1, 125.7, 125.4, 62.3, 35.0, 30.9, 26.2, 22.5, 13.8. EI-MS (70 eV): *m/z* (%): 180 (13), 137 (8), 119 (20), 105 (41), 91 (100), 79 (75), 67 (42).



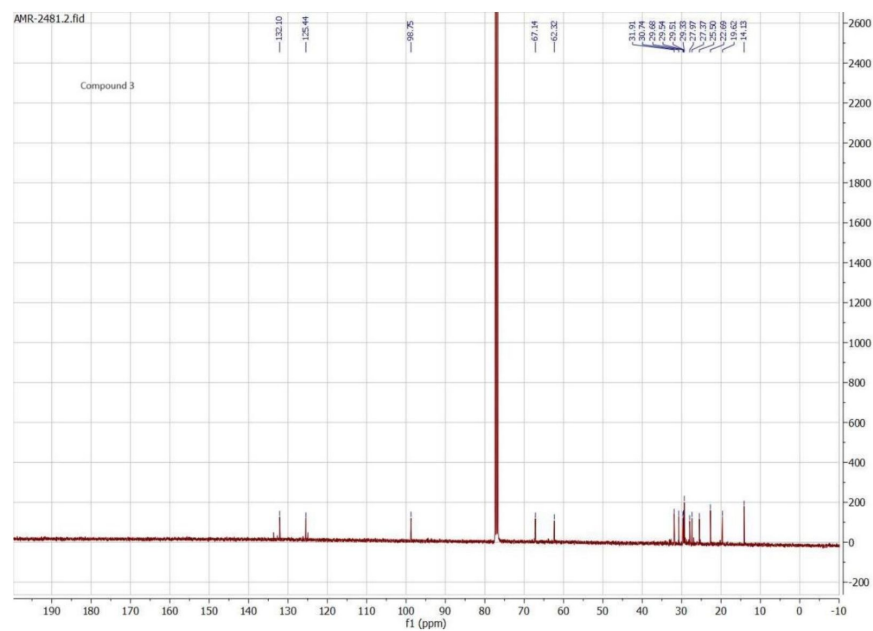
¹H NMR spectrum of the compound 1.



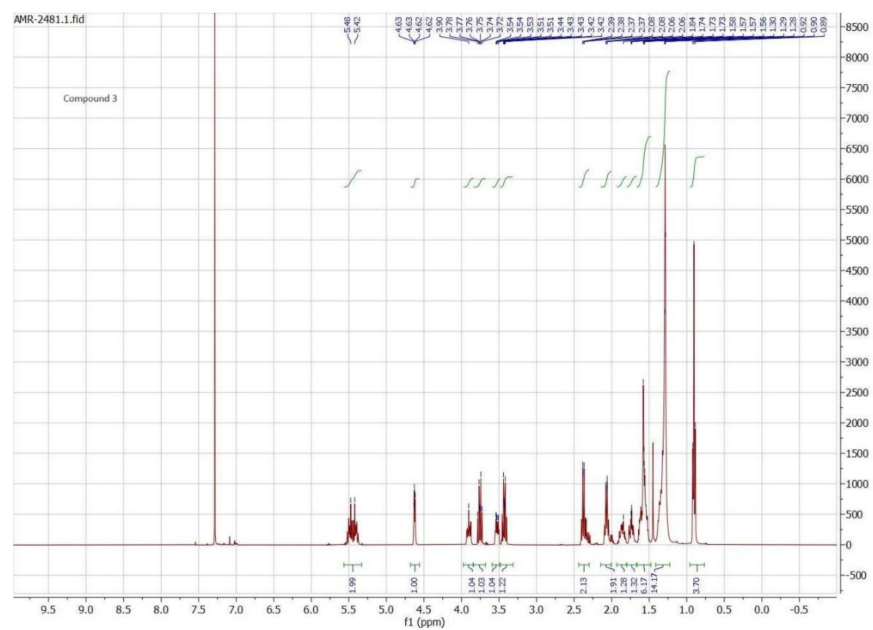
^{13}C NMR spectrum of the compound 2.



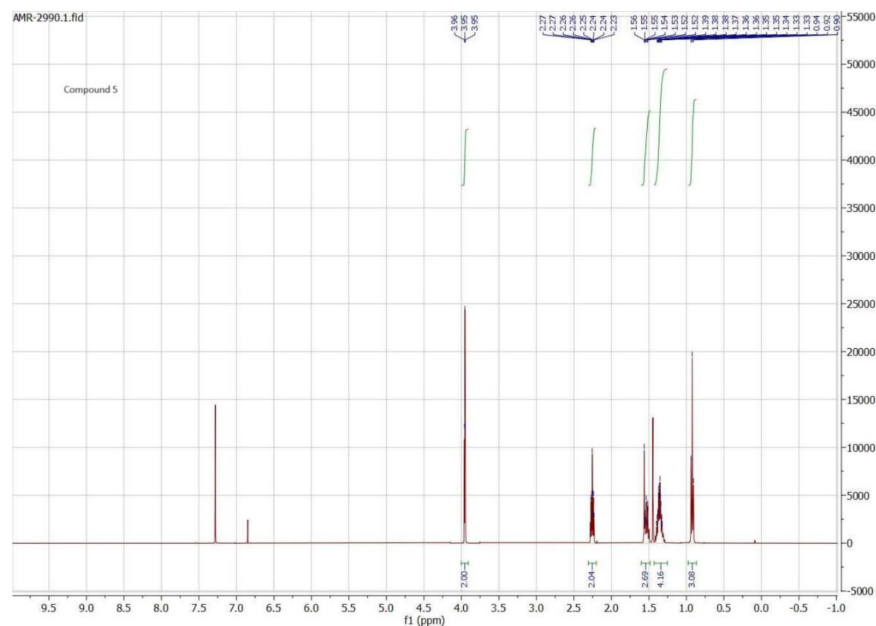
^1H NMR spectrum of the compound 2.



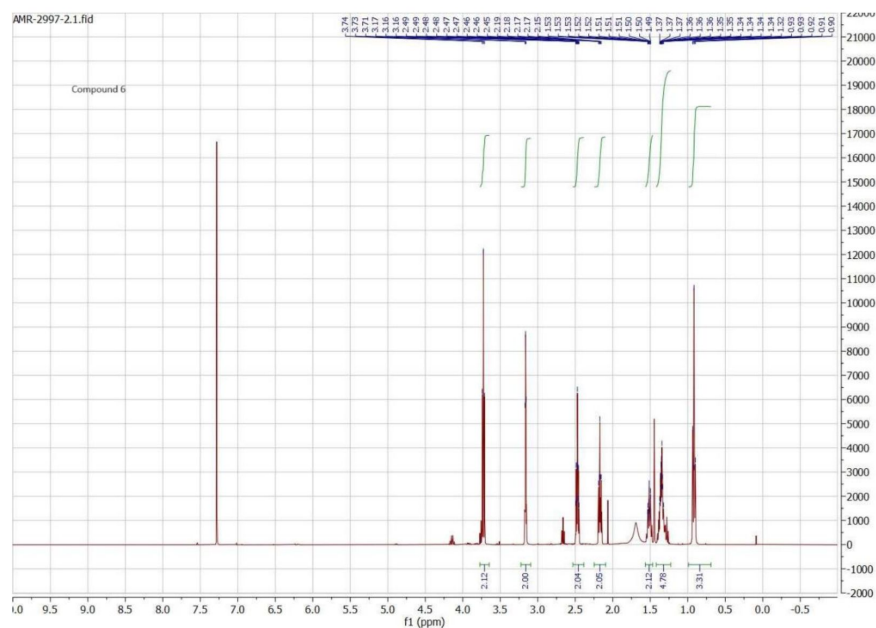
^{13}C NMR spectrum of the compound 3.



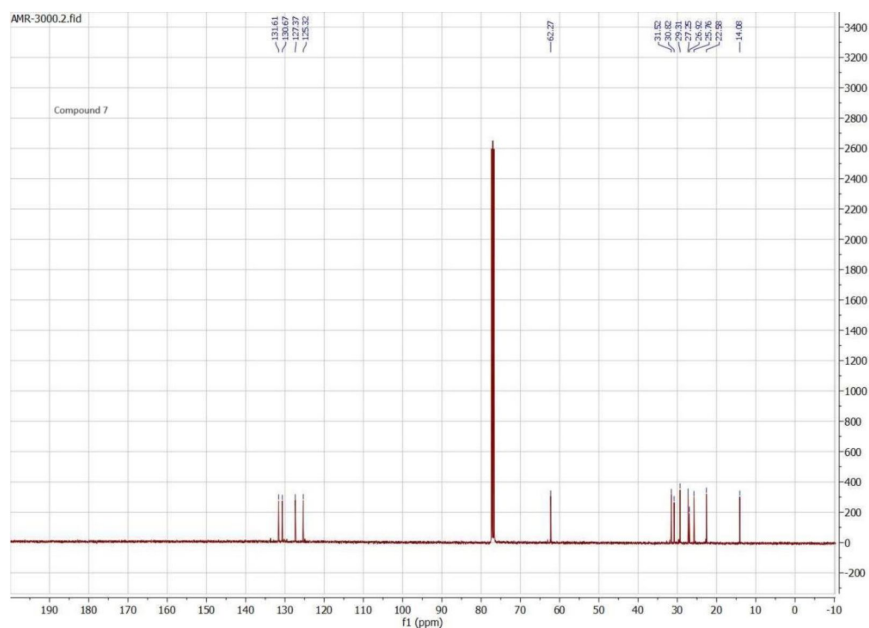
^1H NMR spectrum of the compound 3.



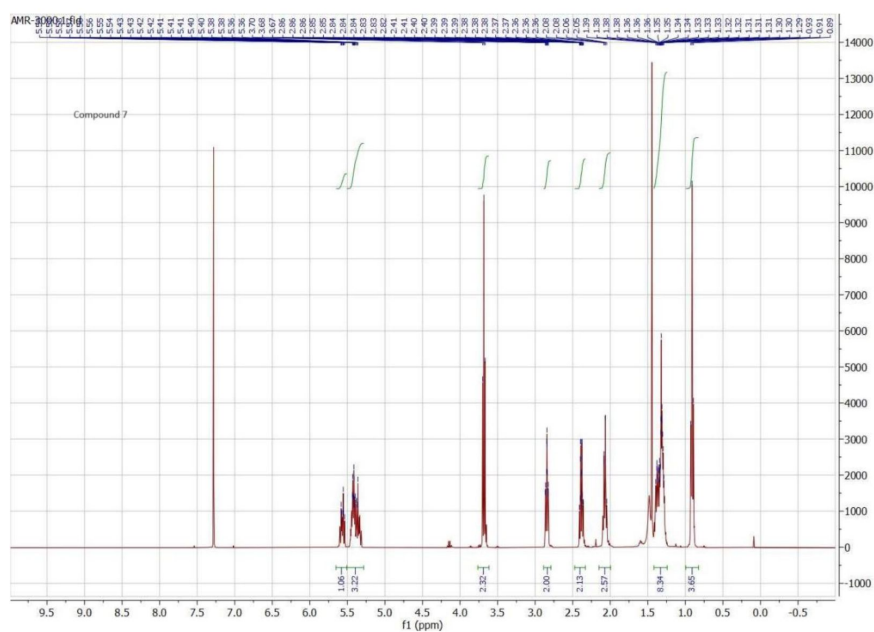
^1H NMR spectrum of the compound 5.



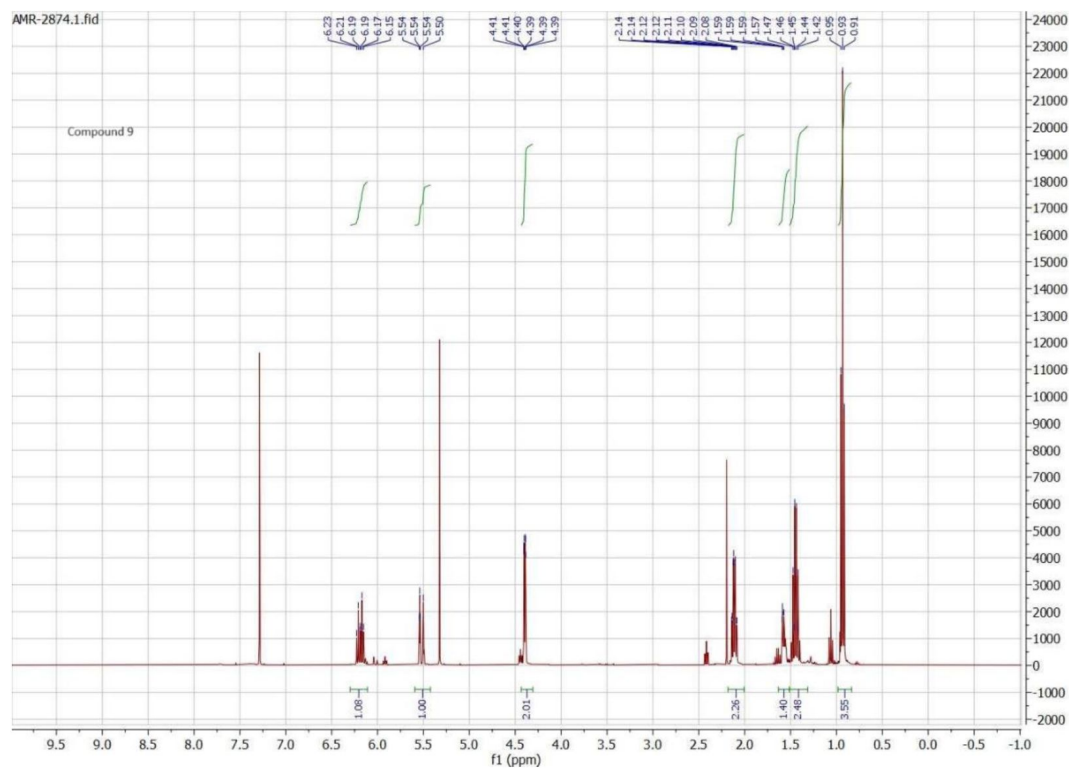
^1H NMR spectrum of the compound 6.



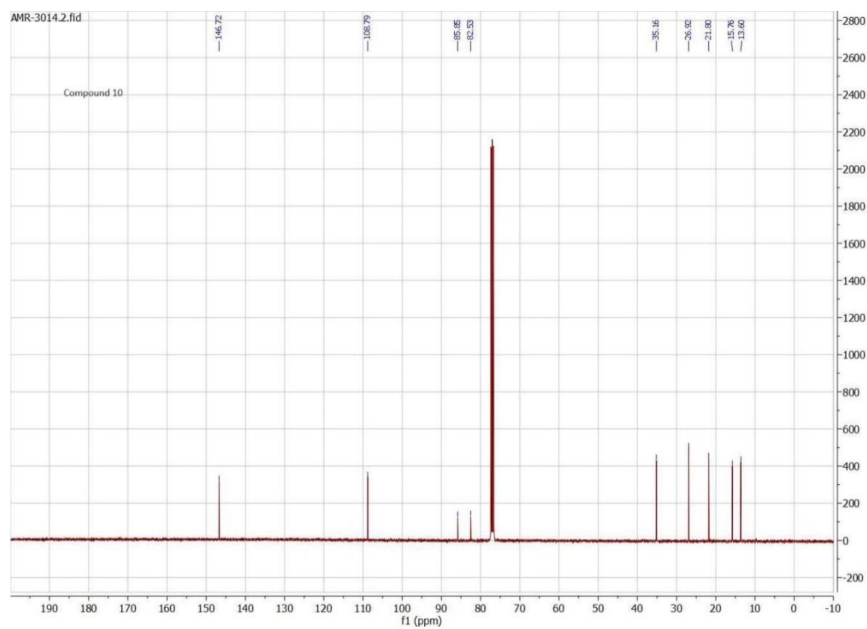
^{13}C NMR spectrum of the compound 7.



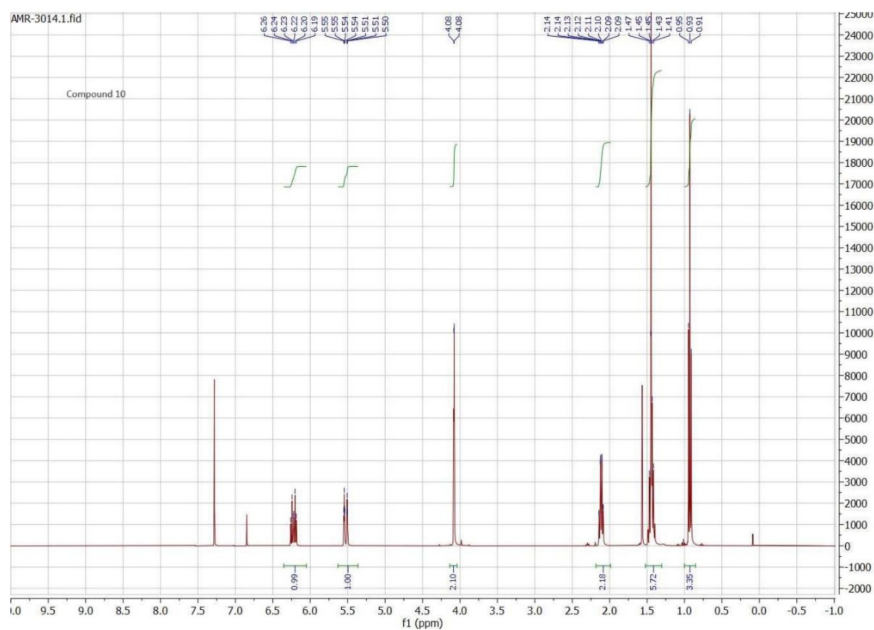
^1H NMR spectrum of the compound 7.



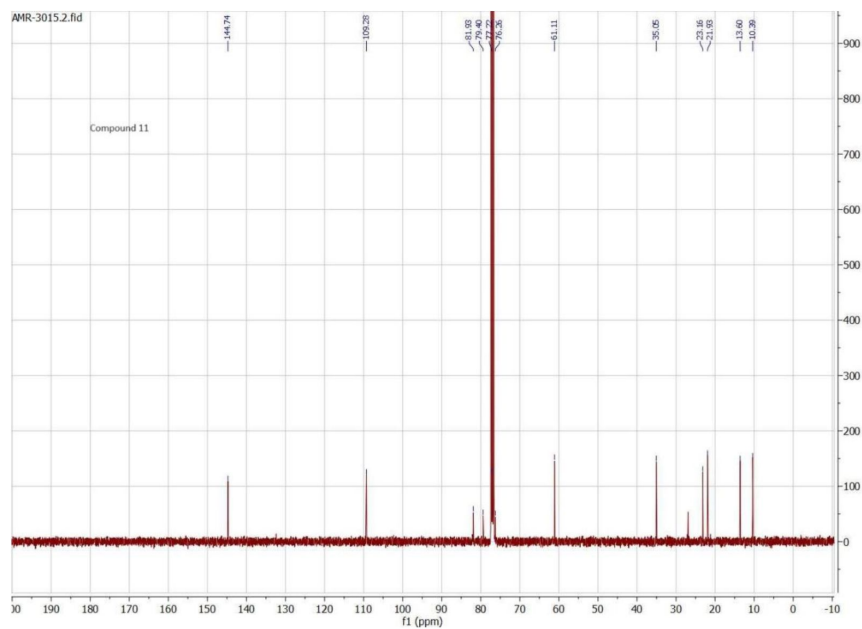
^1H NMR spectrum of the compound 9.



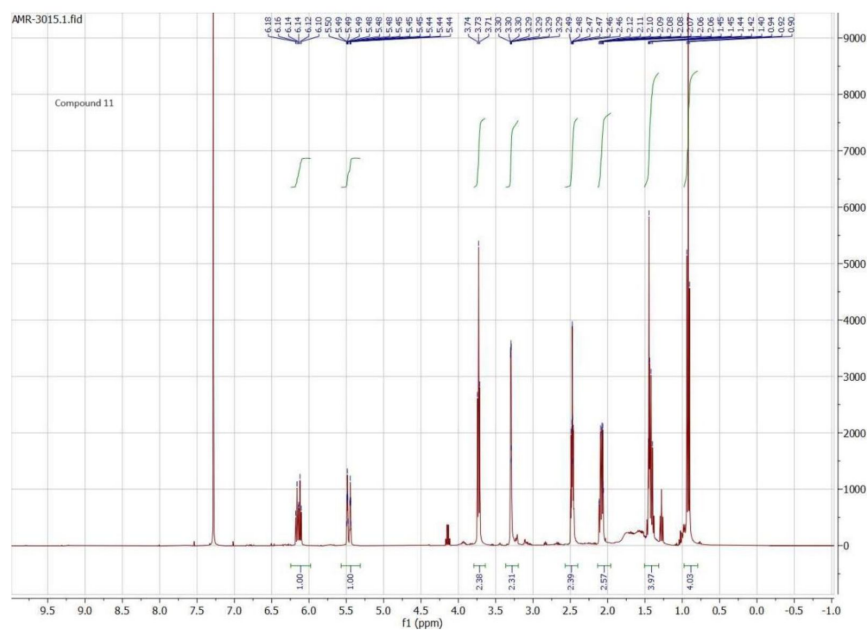
^{13}C NMR spectrum of the compound 10.



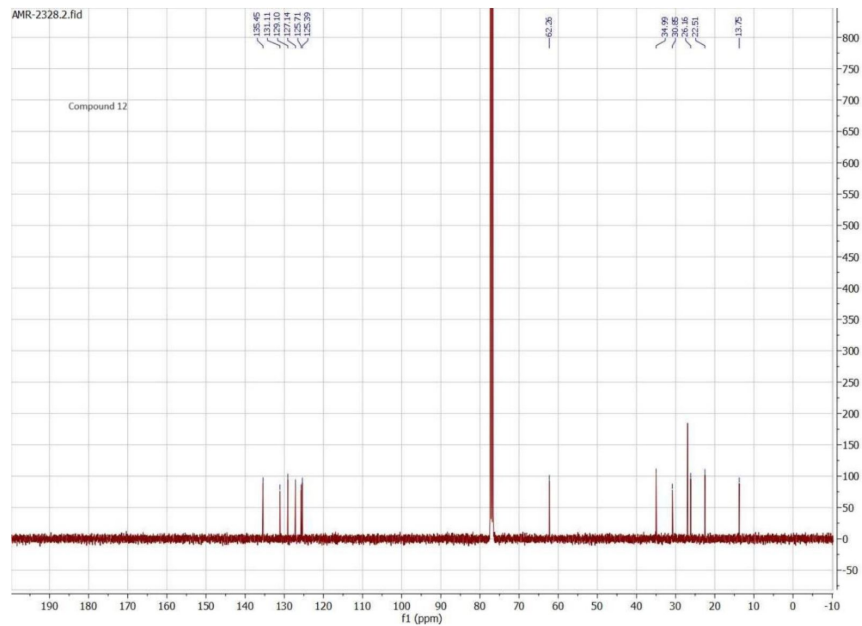
^1H NMR spectrum of the compound 10.



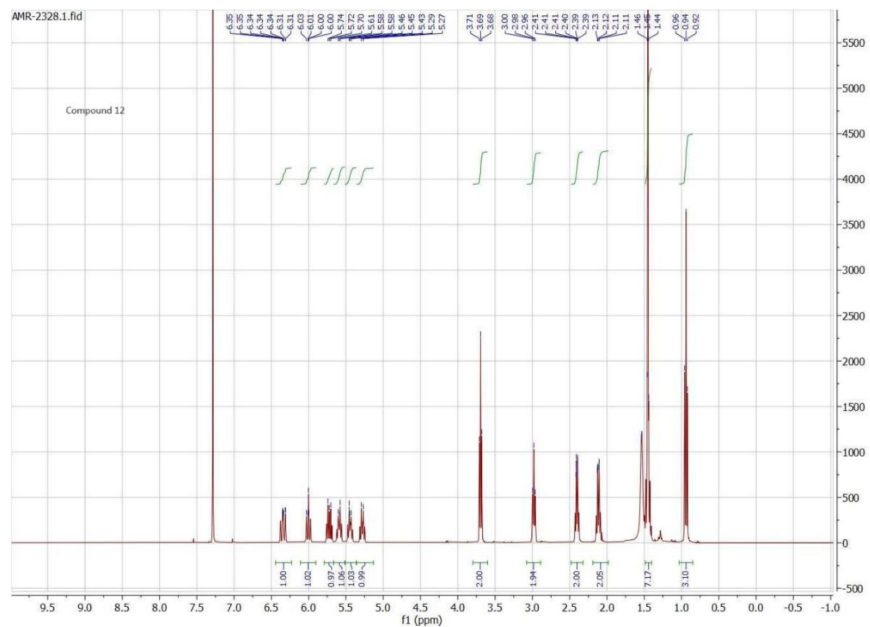
^{13}C NMR spectrum of the compound 11.



^1H NMR spectrum of the compound 11.



^{13}C NMR spectrum of the compound 12.



^1H NMR spectrum of the compound 12.

Data availability

Nucleotide and protein sequences of termite ORs used for phylogenetic reconstruction and functional characterizations were published in [Johny et al. \(2023\)](#) and the raw sequencing data was previously deposited in NCBI SRA archive as PRJNA885453 bioproject. Transcript abundances of *P. simplex* ORs were also inferred from the antennal transcriptome data, available at NCBI under SRX17749141. Sequences of PsimOR14 studied in detail in the present paper are listed in the [Supplementary Table S12](#) and deposited at NCBI as OR921181 entry. Origin of the *P. simplex*

draft genome assembly is reported in Koubová et al. (2021) [↗](#). Differential expression of *P. simplex* ORs in workers and soldiers was studied using caste-specific head transcriptomes available at NCBI as SRA archives under accessions SRX18952230-32 and SRX18952237-39.

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Additional information

Author contributions

SD – SSR, fly transgenesis, SEM; KK – EAG; JŠ, JV – docking, molecular dynamics; JJ – RNA extraction, cloning and vector constructions, fly transgenesis; JK – insect culture, SEM; JN – SEM, HR-SEM; DSD – advising; PK – chemical analysis; AM – organic synthesis; OL – bioinformatics; RH, EGW, OL, SD – conception, supervision, statistics, writing. All authors contributed to the manuscript writing and approved its final version.

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Additional files

Supplementary tables, figures and methods [↗](#)

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Author information

Souleymane Diallo

Institute of Organic Chemistry and Biochemistry of the Czech Academy of Sciences, Prague, Czech Republic, Czech University of Life Sciences, Prague, Czech Republic

Kateřina Kašparová

Institute of Organic Chemistry and Biochemistry of the Czech Academy of Sciences, Prague, Czech Republic, Faculty of Science, Charles University, Prague, Czech Republic

Josef Šulc

Institute of Organic Chemistry and Biochemistry of the Czech Academy of Sciences, Prague, Czech Republic

Jibin Johnny

Czech University of Life Sciences, Prague, Czech Republic

Jan Křivánek

Institute of Organic Chemistry and Biochemistry of the Czech Academy of Sciences, Prague, Czech Republic

Jana Nebesářová

Faculty of Science, Charles University, Prague, Czech Republic, Biology Centre of the Czech Academy of Sciences, České Budějovice, Czech Republic

David Sillam-Dussès

University Sorbonne Paris Nord, Villetaneuse, France

Pavčina Kyjaková

Institute of Organic Chemistry and Biochemistry of the Czech Academy of Sciences, Prague, Czech Republic

Jiří Vondrášek

Institute of Organic Chemistry and Biochemistry of the Czech Academy of Sciences, Prague, Czech Republic

Aleš Machara

Institute of Organic Chemistry and Biochemistry of the Czech Academy of Sciences, Prague, Czech Republic

Ondřej Lukšan

Institute of Organic Chemistry and Biochemistry of the Czech Academy of Sciences, Prague, Czech Republic

For correspondence: ondrej.luksan@uochb.cas.cz

Ewald Grosse-Wilde

Czech University of Life Sciences, Prague, Czech Republic

Robert Hanus

Institute of Organic Chemistry and Biochemistry of the Czech Academy of Sciences, Prague, Czech Republic

For correspondence: robert@uochb.cas.cz

Editors

Reviewing Editor

Hiromu Tanimoto

Tohoku University, Sendai, Japan

Senior Editor

Albert Cardona

University of Cambridge, Cambridge, United Kingdom

Reviewer #1 (Public review):

Summary:

In their comprehensive analysis Diallo et al. deorphanise the first olfactory receptor of a non-hymenopteran eusocial insect - a termite and identified the well established trail pheromone neocembrene as the receptor's best ligand. By using a large set of odorants the authors convincingly show that, as expected for a pheromone receptor, PsimOR14 is very narrowly tuned. While the authors first make use of an ectopic expression system, the empty neuron of *Drosophila melanogaster*, to characterise the receptor's responses, they next perform single sensillum recordings with different sensilla types on the termite antenna. By that they are able to identify a sensillum which houses three neurons, of which the B neuron exhibits the narrow responses described for PsimOR14. Hence the authors do not only identify the first pheromone receptor in a termite but can even localise its expression on the antenna. The authors in addition perform a structural analysis to explain the binding properties of the receptor and its major and minor ligands (as this is beyond my expertise, I cannot judge this part of the manuscript). Finally, they compare expression patterns of ORs in different castes and find that PsimOR14 is more strongly expressed in worker than in soldier termites, which corresponds well with stronger antennal responses in the worker caste.

Strengths:

The manuscript is well written and a pleasure to read.

Weaknesses:

Whenever it comes to the deorphanization of a receptor and its potential role in behaviour (in the case of the manuscript it would be trail following of the termite) one thinks immediately of knocking out the receptor to check whether it is necessary for the behaviour. However, I definitely do not want to ask for this (especially as the establishment of CRISPR Cas-9 in eusocial insects usually turns out to be a nightmare). I also do not know either, whether knock downs via RNAi have been established in termites, but maybe the authors could consider some speculation on this in the discussion.

Comments on revisions:

I appreciate how the authors have replied to my comments and I have the feeling that also the other reviewers' comments have been dealt with carefully. I therefore support the acceptance of this very nice and interesting manuscript.

<https://doi.org/10.7554/eLife.101814.2.sa3>

Reviewer #2 (Public review):

Summary:

In this manuscript, the authors performed the functional analysis of odorant receptors (ORs) of the termite *Prorhinotermes simplex* to identify the receptor of trail-following pheromone. The authors performed single-sensillum recording (SSR) using the transgenic *Drosophila* flies expressing a candidate of the pheromone receptor and revealed that PsimOR14 strongly responds to neocembrene, the major component of the pheromone. Also, the authors found that one sensillum type (S I) detects neocembrene and also performed SSR for S I in the wild termite workers. Furthermore, the authors revealed the gene, transcript, and protein structures of PsimOR14, predict the 3D model and ligand docking of PsimOR14, and demonstrated that PsimOR14 is higher expressed in workers than soldiers using RNA-seq for heads of workers and soldiers of *P. simplex* and that EAG response to neocembrene is higher in workers than soldiers. I considered that this study will contribute to further understanding of the molecular and evolutionary mechanisms of chemoreception system in termites.

Strength:

The manuscript is well written. As far as I know, this study is the first study that identified a pheromone receptor in termites. The authors not only present a methodology for analyzing the function of termite pheromone receptors but also provide important insights in terms of the evolution of ligand selectivity of termite pheromone receptors.

Weakness:

This revised manuscript appears to me to have no major weaknesses.

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Reviewer #3 (Public review):

Summary:

Chemical communication is essential for the organization of eusocial insect societies. It is used in various important contexts, such as foraging and recruiting colony members to food sources. While such pheromones have been chemically identified and their function demonstrated in bioassays, little is known about their perception. Excellent candidates are the odorant receptors that have been shown to be involved in pheromone perception in other insects including ants and bees but not termites. The authors investigated the function of the odorant receptor PsimOR14, which was one of four target odorant receptors based on gene sequences and phylogenetic analyses. They used the *Drosophila* empty neuron system to demonstrate that the receptor was narrowly tuned to the trail pheromone neocembrene. Similar responses to the odor panel and neocembrene in antennal recordings suggested that one specific antennal sensillum expresses PsimOR14. Additional protein modeling approaches characterized the properties of the ligand binding pocket in the receptor. Finally, PsimOR14 transcripts were found to be significantly higher in worker antennae compared to soldier antennae, which corresponds to the worker's higher sensitivity to neocembrene.

Strengths:

The study presents an excellent characterization of a trail pheromone receptor in a termite species. The integration of receptor phylogeny, receptor functional characterization, antennal sensilla responses, receptor structure modeling, and transcriptomic analysis is especially powerful. All parts build on each other and are well supported with a good sample size. (I cannot comment on protein modeling and docking due to a lack of expertise in this area)

Weaknesses:

None.

<https://doi.org/10.7554/eLife.101814.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:**Reviewer #1 (Public review):***Summary:*

*In their comprehensive analysis Diallo et al. deorphanise the first olfactory receptor of a nonhymenopteran eusocial insect - a termite and identified the well-established trail pheromone neocembrene as the receptor's best ligand. By using a large set of odorants the authors convincingly show that, as expected for a pheromone receptor, PsimOR14 is very narrowly tuned. While the authors first make use of an ectopic expression system, the empty neuron of *Drosophila melanogaster*, to characterise the receptor's responses, they next perform single sensillum recordings with different sensilla types on the termite antenna. By that, they are able to identify a sensillum that houses three neurons, of which the B neuron exhibits the narrow responses described for PsimOR14. Hence the authors do not only identify the first pheromone receptor in a termite but can even localize its expression on the antenna. The authors in addition perform a structural analysis to explain the binding properties of the receptor and its major and minor ligands (as this is beyond my expertise, I cannot judge this part of the manuscript). Finally, they compare expression patterns of ORs in different castes and find that PsimOR14 is more strongly expressed in workers than in soldier termites, which corresponds well with stronger antennal responses in the worker caste.*

Strengths:

The manuscript is well-written and a pleasure to read. The figures are beautiful and clear. I actually had a hard time coming up with suggestions.

We thank the reviewer for the positive comments.

Weaknesses:

Whenever it comes to the deorphanization of a receptor and its potential role in behaviour (in the case of the manuscript it would be trail-following of the termite) one thinks immediately of knocking out the receptor to check whether it is necessary for the behaviour. However, I definitely do not want to ask for this (especially as the establishment of CRISPR Cas-9 in eusocial insects usually turns out to be a nightmare). I

also do not know either, whether knockdowns via RNAi have been established in termites, but maybe the authors could consider some speculation on this in the discussion.

We agree that a functional proof of the PsimOR14 function using reverse genetics would be a valuable addition to the study to firmly establish its role in trail pheromone sensing. Nevertheless, such a functional proof is difficult to obtain. Due to the very slow ontogenetic development inherent to termites (several months from an egg to the worker stage) the CRISPR Cas-9 is not a useful technique for this taxon. By contrast, termites are quite responsive to RNAimediated silencing and RNAi has previously been used for the silencing of the ORCo co-receptor in termites resulting in impairment of the trail-following behavior (DOI: 10.1093/jee/toaa248). Likewise, our previous experiments showed a decreased ORCo transcript abundance, lower sensitivity to neocembrene and reduced neocembrene trail following upon dsPsimORCo administration to *P. simplex* workers, while we did not succeed in reducing the transcript abundance of PsimOR14 upon dsPsimOR14 injection. We do not report these negative results in the present manuscript so as not to dilute the main message. In parallel, we are currently developing an alternative way of dsRNA delivery using nanoparticle coating, which may improve the RNAi experiments with ORs in termites.

Reviewer #2 (Public review):

Summary:

*In this manuscript, the authors performed the functional analysis of odorant receptors (ORs) of the termite *Prorhinotermes simplex* to identify the receptor of trail-following pheromone. The authors performed single-sensillum recording (SSR) using the transgenic *Drosophila* flies expressing a candidate of the pheromone receptor and revealed that PsimOR14 strongly responds to neocembrene, the major component of the pheromone. Also, the authors found that one sensillum type (S I) detects neocembrene and also performed SSR for S I in wild termite workers. Furthermore, the authors revealed the gene, transcript, and protein structures of PsimOR14, predicted the 3D model and ligand docking of PsimOR14, and demonstrated that PsimOR14 is higher expressed in workers than soldiers using RNA-seq for heads of workers and soldiers of *P. simplex* and that EAG response to neocembrene is higher in workers than soldiers. I consider that this study will contribute to further understanding of the molecular and evolutionary mechanisms of the chemoreception system in termites.*

Strength:

The manuscript is well written. As far as I know, this study is the first study that identified a pheromone receptor in termites. The authors not only present a methodology for analyzing the function of termite pheromone receptors but also provide important insights in terms of the evolution of ligand selectivity of termite pheromone receptors.

We thank the reviewer for the overall positive evaluation of the manuscript.

Weakness:

As you can see in the "Recommendations to the Authors" section below, there are several things in this paper that are not fully explained about experimental methods. Except for this point, this paper appears to me to have no major weaknesses.

We address point by point the specific comments listed in the Recommendation to the authors chapter below.

Reviewer #3 (Public review):

Summary:

*Chemical communication is essential for the organization of eusocial insect societies. It is used in various important contexts, such as foraging and recruiting colony members to food sources. While such pheromones have been chemically identified and their function demonstrated in bioassays, little is known about their perception. Excellent candidates are the odorant receptors that have been shown to be involved in pheromone perception in other insects including ants and bees but not termites. The authors investigated the function of the odorant receptor PsimOR14, which was one of four target odorant receptors based on gene sequences and phylogenetic analyses. They used the *Drosophila* empty neuron system to demonstrate that the receptor was narrowly tuned to the trail pheromone neocembrene. Similar responses to the odor panel and neocembrene in antennal recordings suggested that one specific antennal sensillum expresses PsimOR14. Additional protein modeling approaches characterized the properties of the ligand binding pocket in the receptor. Finally, PsimOR14 transcripts were found to be significantly higher in worker antennae compared to soldier antennae, which corresponds to the worker's higher sensitivity to neocembrene.*

Strengths:

The study presents an excellent characterization of a trail pheromone receptor in a termite species. The integration of receptor phylogeny, receptor functional characterization, antennal sensilla responses, receptor structure modeling, and transcriptomic analysis is especially powerful. All parts build on each other and are well supported with a good sample size.

We thank the reviewer for these positive comments.

Weaknesses:

The manuscript would benefit from a more detailed explanation of the research advances this work provides. Stating that this is the first deorphanization of an odorant receptor in a clade is insufficient. The introduction primarily reviews termite chemical communication and deorphanization of olfactory receptors previously performed. Although this is essential background, it lacks a good integration into explaining what problem the current study solves.

We understand the comment about the lack of an intelligible cue to highlight the motivation and importance of the present study. In the current version of the manuscript the introduction has been reworked. As suggested by Reviewer 3 in the Recommendations section below, the introduction now integrates some parts of the original discussion, especially the part discussing the OR evolution and emergence of eusociality in hymenopteran social insects and in termites, while underscoring the need of data from termites to compare the commonalities and idiosyncrasies in neurophysiological (pre)adaptations potentially linked with the independent eusociality evolution in the two main social insect clades.

Selecting target ORs for deorphanization is an essential step in the approach. Unfortunately, the process of choosing these ORs has not been described. Were the authors just lucky that they found the correct OR out of the 50, or was there a specific selection process that increased the probability of success?

Indeed, we were extremely lucky. Our strategy was to first select a modest set of ORs to confirm the feasibility of the Empty Neuron *Drosophila* system and newly established SSR setup, while taking advantage of having a set of termite pheromones, including those previously identified in the *P. simplex* model, some of them *de novo* synthesized for this project. The selection criteria for the first set of four receptors were (i) to have full-length ORF and at least 6 unambiguously predicted transmembrane regions, and (ii) to be represented on different branches (subbranches) of the phylogenetic tree. Then it was a matter of a good luck to hit the PsimOR14 selectively responding to the genuine *P. simplex* trail-following pheromone main component. In the revised version, we state these selection criteria in the results section (Phylogenetic reconstruction and candidate OR selection).

The deorphanization attempts of additional *P. simplex* ORs are currently running.

The authors assigned antennal sensilla into five categories. Unfortunately, they did not support their categories well. It is not clear how they were able to differentiate SI and SII in their antennal recordings.

We agree that the classification of multiporous sensilla into five categories lacks robust discrimination cues. The identification of the neocembrene-responding sensillum was initially carried out by SSR measurements on individual olfactory sensilla of *P. simplex* workers one-by-one and the topology of each tested sensillum was recorded on optical microscope photographs taken during the SSR experiment. Subsequently, the SEM and HR-SEM were performed in which we localized the neocembrene sensillum and tried to find distinguishing characters. We admit that these are not robust. Therefore, in the revised version of the manuscript we decided to abandon the attempt of sensilla classification and only report the observations about the specific sensillum in which we consistently recorded the response to neocembrene (and geranylgeraniol). The modifications affect Fig. 4, its legend and the corresponding part of the results section (Identification of *P. simplex* olfactory sensillum responding to neocembrene).

The authors used a large odorant panel to determine receptor tuning. The panel included volatile polar compounds and non-volatile non-polar hydrocarbons. Usually, some heat is applied to such non-volatile odorants to increase volatility for receptor testing. It is unclear how it is possible that these non-volatile compounds can reach the tested sensilla without heat application.

The reviewer points at an important methodological error we made while designing the experiments. Indeed, the inclusion of long-chain hydrocarbons into Panel 1 without additional heat applied to the odor cartridges was inappropriate, even though the experiments were performed at 25–26 °C. We carefully considered the best solution to correct the mistake and finally decided to remove all tested ligands beyond C22 from Panel 1, i.e. altogether five compounds. These changes did not affect the remaining Panels 2–4 (containing compounds with sufficient volatility), nor did they affect the message of the manuscript on highly selective response of PsimOR14 to neocembrene (and geranylgeraniol). In consequence, Figures 2, 3 and 5 were updated, along with the supplementary tables containing the raw data on SSR measurements. In addition, the tuning curve for PsimOR14 was re-built and receptor lifetime sparseness value re-calculated (without any important change). We also exchanged squalene for limonene in the docking and molecular dynamics analysis and made new calculations.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) L 208: "than" instead of "that"

Corrected.

(2) L 527+527 strange squares (•) before dimensions

Apparently an error upon file conversion, corrected.

(3) L553 "reconstructing" instead of "reconstruct"

Corrected.

(4) Two references (Chahda et al. and Chang et al. appear too late in the alphabet.

Corrected. Thank you for spotting this mistake. Due to our mistake the author list was ordered according to the alphabet in Czech language, which ranks CH after H.

Reviewer #2 (Recommendations for the authors):

(1) L148: Why did the authors select only four ORs (PsimOR9, 14, 30, and 31) though there are 50 ORs in *P. simplex*? I would like you to explain why you chose them.

Our strategy was to first select a modest set of ORs to confirm the feasibility of the Empty Neuron *Drosophila* system and newly established SSR setup, while taking advantage of having a set of termite pheromones, including those previously identified in the *P. simplex* model, some of them *de novo* synthesized for this project. Then, it was a matter of a good luck to hit the PsimOR14 selectively responding to the genuine *P. simplex* trail-following pheromone main component, while the deorphanization attempts of a set of additional *P. simplex* ORs is currently running. In the revised version of the manuscript, we state the selection criteria for the four ORs studied in the Results section (Phylogenetic reconstruction and candidate OR selection).

(2) L149: Where is Figure 1A? Does this mean Figure 1?

Thank you for spotting this mistake. Fig. 1 is now properly labelled as Fig. 1A and 1B in the figure itself and in the legend. Also the text now either refers to either 1A or 1B.

(3) Figure 1: The authors also showed the transcription abundance of all 50 ORs of *P. simplex* in the right bottom of Figure 1, but there is no explanation about it in the main text.

The heatmap reporting the transcript abundances is now labelled as Fig. 1B and is referred to in the discussion section (in the original manuscript it was referred to on the same place as Fig. 1).

(4) L260-265: The authors confirmed higher expression of PsimOR14 in workers than soldiers by using RNA-seq data and stronger EAG responses of PsimOR14 to neocembrene in workers than soldiers, but I think that confirming the expression levels of PsimOR14 in workers and soldiers by RT-qPCR would strengthen the authors' argument (it is optional).

qPCR validation is a suitable complement to read count comparison of RNA Seq data, especially when the data comes from one-sample transcriptomes and/or low coverage sequencing. Yet, our RNA Seq analysis is based on sequencing of three independent biological replicates per phenotype (worker heads vs. soldier heads) with ~20 millions of reads per sample. Thus, the resulting differential gene expression analysis is a sufficient and powerful technique in terms of detection limit and dynamic range.

We admit that the replicate numbers and origin of the RNA seq data should be better specified since the Methods section only referred to the GenBank accession numbers in the original manuscript. Therefore, we added more information in the Methods section (Bioinformatics) and make clear in the Methods that this data comes from our previous research and related bioproject.

| (5) L491: I think that "The synthetic processes of these fatty alcohols are ..." is better.

We replaced the sentence with "The *de novo* organic synthesis of these fatty alcohols is described ..."

| (6) L525 and 527: There are white squares between the number and the unit. Perhaps some characters have been garbled.

Apparently an error upon file conversion, corrected.

| (7) L795: ORCo?

Corrected.

| (8) L829-830 & Figure 4: Where is Figure 4D?

Thank you for spotting this mistake from the older version of Figure 4. The SSR traces referred to in the legend are in fact a part of Figure 5. Moreover, Figure 4 is now reworked based on the comments by Reviewer 3.

| (9) L860-864: Why did the authors select the result of edgeR for the volcano plot in Figure 7 although the authors use both DESeq2 and edgeR? An explanation would be needed.

Both algorithms, DESeq2 and EdgeR, are routinely used for differential gene expression analysis. Since they differ in read count normalization method and statistical testing we decided to use both of them independently in order to reduce false positives. Because the resulting fold changes were practically identical in both algorithms (results for both analyses are listed in Supplementary table S15), we only reported in Fig. 7 the outputs for edgeR to avoid redundancies. We added in the Results section the information that both techniques listed PsimOR14 among the most upregulated in workers.

Reviewer #3 (Recommendations for the authors):

The discussion contains many descriptions that would fit better into the introduction, where they could be used to hint at the study's importance (e.g., 292-311, 381-412). The remaining parts often lack a detailed discussion of the results that integrates details from other insect studies. Although references were provided, no details were usually outlined. It would be helpful to see a stronger emphasis on what we learn from this study.

Along with rewriting the introduction, we also modified the discussion. As suggested, the lines 292-311 were rewritten and placed in the introduction. By contrast, we preferred to keep the two paragraphs 381-412 in the discussion, since both of them outline the potential future interesting targets of research on termite ORs.

As suggested, the discussion has been enriched and now includes comparative examples and relevant references about the broad/narrow selectivity of insect ORs, about the expected breadth of tuning of pheromone receptors vs. ORs detecting environmental cues, about the potential role of additional neurons housed in the neocembrene-detecting sensillum of *P. simplex* workers, etc. From both introduction and discussion the redundant details on the chemistry of termite communication have been removed.

This includes explanations of the advantages of the specific methodologies the authors used and how they helped solve the manuscript's problem. What does the phylogeny solve? Was it used to select the ORs tested? It would be helpful to discuss what the phylogeny shows in comparison to other well-studied OR phylogenies, like those from the social Hymenoptera.

We understand the comment. In fact, our motivation to include the phylogenetic tree of termite ORs was essentially to demonstrate (i) the orthologous nature of OR diversity with few expansions on low taxonomic levels, and (ii) to demonstrate graphically the relationship among the four selected sequences. We do not attempt here for a comprehensive phylogenetic analysis, because it would be redundant given that we recently published a large OR phylogeny which includes all sequences used in the present manuscript and analysed them in the proper context of related (cockroaches) and unrelated insect taxa (Johny et al., 2023). This paper also discusses the termite phylogenetic pattern with those observed in other Insecta. This paper is repeatedly cited on appropriate places of the present manuscript and its main observations are provided in the Introduction section. Therefore, we feel that thorough discussion on termite phylogeny would be redundant in the present paper.

The authors categorized the sensilla types. Potential problems in the categorization aside, it would be helpful to know if it is expected that you have sensilla specialized in perceiving one specific pheromone. What is known about sensilla in other insects?

We understand. In the discussion of the revised version, we develop more about the features typical/expected for a pheromone receptor and the sensillum housing this receptor together with two other olfactory sensory neurons, including examples from other insects.

As the manuscript currently stands, specialist readers with their respective background knowledge would find this study very interesting. In contrast, the general reader would probably fail to appreciate the importance of the results.

We hope that the re-organized and simplified introduction may now be more intelligible even for non-specialist readers.

(1) L35: Should "workers" be replaced with "worker antennae"?

Corrected.

(2) L62: Should "conservativeness" be replaced by "conservation"?

Replaced with "parsimony".

(3) L129: *How and why did the authors choose four candidate ORs? I could not find any information about this in the manuscript. I wondered why they did not pick the more highly expressed PsimOr20 and 26 (Figure 7).*

As already replied above in the Weaknesses section, we selected for the first deorphanization attempts only a modest set of four ORs, while an additional set is currently being tested. We also explained above the inclusion criteria, i.e. (i) full-length ORF and at least 6 unambiguously predicted transmembrane regions, and (ii) presence on different branches (subbranches) of the OR phylogeny. For these reasons, we did not primarily consider the expression patterns of different ORs. As for Fig. 7, it shows differential expression between soldiers and workers, which was not the primary guideline either and the data was obtained only after having the ORs tested by SSR. Yet, even though we had data on *P. simplex* ORs expression (Fig. 1B), we did not presume that pheromone receptors should be among the most expressed ORs, given the richness of chemical cues detected by worker termites and unlike, e.g., male moths, where ORs for sex pheromones are intuitively highly expressed.

The strategy of OR selection is specified in the results section of the revised manuscript under “Phylogenetic reconstruction and candidate OR selection”.

(4) 198 to 200: *SI, II, and III look very similar. Additional measurements rather than qualitative descriptions are required to consider them distinct sensilla. The bending of SIII could be an artifact of preparation. I do not see how the authors could distinguish between SI and SII under the optical microscope for recordings. A detailed explanation is required.*

As we responded above in “Weaknesses” chapter, we admit that the sensilla classification is not intelligible. Therefore, we decided in the revised version to abandon the classification of sensilla types and only focus on the observations made on the neocembreneresponding sensillum. To recognize the specific sensillum, we used its topology on the last antennal segment. Because termite antennae are not densely populated with sensilla, it is relatively easy to distinguish individual sensilla based on their topology on the antenna, both in optical microscope and SEM photographs. The modifications affect Fig. 4, its legend and the corresponding part of the results section (Identification of *P. simplex* olfactory sensillum responding to neocembrene).

(5) 208: *“Than” instead of “that”*

Corrected.

(6) 280: *I suggest replacing “demand” with “capabilities”*

Corrected.

(7) 312: *Why “nevertheless? It sounds as if the authors suggest that there is evidence that ORs are not important for communication. This should be reworded.*

We removed “Nevertheless” from the beginning of the sentence.

(8) 321 to 323: *This sentence sounds as if something is missing. I suggest rewriting it.*

This sentence simply says that empty neuron *Drosophila* is a good tool for termite OR deorphanization and that termite ORs work well *Drosophila* ORCo. We reworded the sentence.

| (9) 323: *I suggest starting a new paragraph.*

Corrected.

| (10) 421: *How many colonies were used for each of the analyses?*

The data for this manuscript were collected from three different colonies collected in Cuba. We now describe in the Materials and Methods section which analyses were conducted with each of the colonies.

| (11) 430: *Did the termites originate from one or multiple colonies and did the authors sample from the Florida and Cuba population?*

The data for this manuscript were collected from three different colonies collected in Cuba. We now describe in the Materials and Methods section which analyses were conducted with each of the colonies.

| (12) 501: *How was the termite antenna fixated? The authors refer to the Drosophila methods, but given the large antennal differences between these species, more specific information would be helpful.*

Understood. We added the following information into the Methods section under “Electrophysiology”: “The grounding electrode was carefully inserted into the clypeus and the antenna was fixed on a microscope slide using a glass electrode. To avoid the antennal movement, the microscope slide was covered with double-sided tape and the three distal antennal segments were attached to the slide.”

| (13) 509: *I want to confirm that the authors indicate that the outlet of the glass tube with the airstream and odorant is 4 cm away from the Drosophila or termite antenna. The distance seems to be very large.*

Thank you for spotting this obvious mistake. The 4 cm distance applies for the distance between the opening for Pasteur pipette insertion into the delivery tube, the outlet itself is situated approx. 1 cm from the antenna. This information is now corrected.

| (14) 510/527: *It looks like all odor panels were equally applied onto the filter paper despite the difference in solvent (hexane and paraffin oil). How was the solvent difference addressed?*

In our study we combine two types of odorant panels. First, we test on all four studied receptors a panel containing several compounds relevant for termite chemical communication including the C12 unsaturated alcohols, the diterpene neocembrene, the sesquiterpene (3R,6E)-nerolidol and other compounds. These compounds are stored in the laboratory as hexane solutions to prevent the oxidation/polymerization and it is not advisable to transfer them to another solvent. In the second step we used three additional panels of frequently occurring insect semiochemicals, which are stored as paraffin oil solutions, so as to address the breadth of PsimOR14 tuning. We are aware that the evaporation dynamics differ between the two solvents but we did not have any suitable option how to solve this problem. We believe that the use of the two solvents does not compromise the general message on the receptor specificity. For each panel, the corresponding solvent is used as a control. Similarly, the use of two different solvents for SSR can be encountered in other studies, e.g. 10.1016/j.celrep.2015.07.031.

(15) 518: delta spikes/sec works for all tables except for the wild type in Table S5. I could not figure out how the authors get to delta spikes/sec in that table.

Thank you for your sharp eye. Due to our mistake, the values of Δ spikes per second reported in Table S5 for W1118 were erroneously calculated using the formula for 0.5 sec stimulation instead of 1 sec. We corrected this mistake which does not impact the results interpretation in Table S5 and Fig. 2.

522: Did the workers and soldiers originate from different colonies or different populations?

We now clearly describe in the Material and Methods section the origin of termites for different experiments. EAG measurements were made using individuals (workers, soldiers) from one Cuban colony.

(16) Figure 6C/D: I suggest matching colors between the two figures. For example, instead of using an orange circle in C and a green coloration of the intracellular flap in D, I recommend using blue, which is not used for something else. In addition, the binding pocket could be separated better from anything else in a different color.

We agree that the color match for the intracellular flap was missing. This figure is now reworked and the colors should have a better match and the binding region is better delineated.

(17) Figure 7/Table S15: It is unclear where the transcriptome data originate and what they are based on. Are these antennal transcriptomes or head transcriptomes? Do these data come from previous data sets or data generated in this study? Figure 7 refers to heads, Table S15 to workers and soldiers, and the methods only refer to antennal extractions. This should be clarified in the text, the figure, and the table.

We admit that the replicate numbers and origin of the RNA seq data should be better specified and that the information that the RNASeq originated from samples of heads+antennae of workers and soldiers should be provided at appropriate places. Therefore, we added more information on replicates and origin of the data in the Methods section (Bioinformatics) and make clear that this data comes from our previous research and refer to the corresponding bioproject. Likewise, the Figure 7 legend and Table S15 heading have been updated.

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